

SingleRAN

Monitoring Management Feature Parameter Description

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 SRAN22.1 Draft B (2026-01-30)

This issue includes the following changes.

Technical Changes

None

Editorial Changes

Added descriptions of monitoring board configurations for the APM30H in a BTS3900A (AC) and that in a DBS3900 (AC). For details, see [Example of Configurations in a BTS3900A \(AC\)](#) and [Example of Configurations in a DBS3900 \(AC\)](#).

1.2 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 02 (2025-08-30).

Technical Changes

None

Editorial Changes

Added descriptions of data configuration requirements when adding a monitoring device. For details, see [4.4.3 Data Configuration](#).

Added descriptions of precautions for dual-side configuration in multimode base stations. For details, see [4.6.2 Precautions for Dual-Side Configuration](#).

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

Trial Features

Trial features are features that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these features can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial

features shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial features. Trial features are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial features may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial features but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial features are free, Huawei is not liable for any trial feature malfunctions or any losses incurred by using the trial features. Huawei does not promise that problems with trial features will be resolved in the current version. Huawei reserves the rights to convert trial features into commercial features in later R/C versions. If trial features are converted into commercial features in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial features. If a customer fails to purchase such a license, the trial feature(s) will be invalidated automatically when the product is upgraded.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, NB-IoT, and NR. This feature works the same way with each of these RATs.

2.3 Features in This Document

This document applies to macro base stations (including BTS3900, BTS3900L, BTS3900A, BTS3900AL, BTS3900C, and DBS3900 as well as BTS5900, BTS5900A, BTS5900L, and DBS5900), DBS3900 LampSite base stations, and DBS5900 LampSite base stations.

Any parameters, alarms, counters, or managed objects (MOs) described herein apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

Unless otherwise specified, in this document, LTE and eNodeB always include FDD, TDD, and NB-IoT. In scenarios where they need to be distinguished, LTE FDD, LTE TDD, and LTE NB-IoT are used. The same rules apply to eNodeB. The "G", "U", "L", "T", "N", and "M" in RAT acronyms refer to GSM, UMTS, LTE FDD, LTE TDD, NR, and LTE NB-IoT, respectively.

GBTS and eGBTS are GSM base stations. NodeB is a UMTS base station. eNodeB is an LTE base station. gNodeB is an NR base station.

The following table lists the features for different RATs.

RAT	Feature ID	Feature Name	Chapter/Section
GSM	GBFD-11230 1	Remote EAC Maintenance	4.5 Operation and Maintenance (GBTS)

RAT	Feature ID	Feature Name	Chapter/Section
GSM	GBFD-11121 1	Reporting the Temperature List of the BTS Equipment Room	4.5 Operation and Maintenance (GBTS)
LTE FDD	LBFD-00401 2	Environment Monitoring	4.4 Operation and Maintenance (eGBTS/NodeB/eNodeB/gNodeB)
LTE TDD	TDLBFD-004012	Environment Monitoring	4.4 Operation and Maintenance (eGBTS/NodeB/eNodeB/gNodeB)
LTE NB-IoT	MLBFD-12000412	Environment Monitoring	4.4 Operation and Maintenance (eGBTS/NodeB/eNodeB/gNodeB)

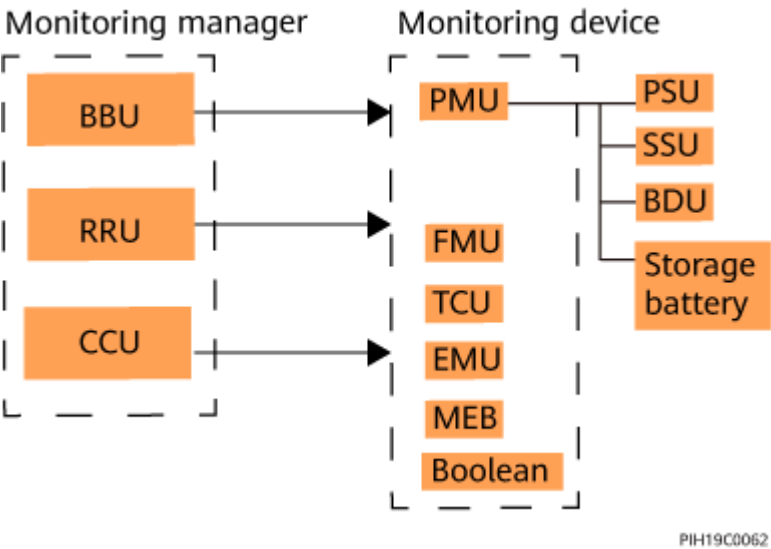
3 Overview

The monitoring system is an important part of a base station. It provides the following monitoring functions:

- Monitors the power supply, fans and environment in the cabinet, and automatically reports an alarm once an exception is detected.
- Collects alarm signals from outside the cabinet to implement user-defined alarm monitoring, such as monitoring the environment, diesel generator, and access control system (ACS) of the equipment room.

The monitoring system consists of monitoring devices and monitoring managers. Monitoring devices (such as the FMU, TCU, and PMU) monitor the base station and connect to the monitoring managers (BBU, RRU, or CCU) through the RS485 bus. Monitoring managers summarize the monitoring information from connected monitoring devices to the BBU for unified management.

Figure 3-1 Monitoring system



4 Monitoring Management

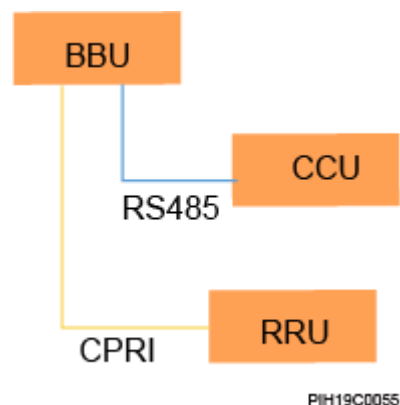
4.1 Principles

4.1.1 Monitoring Managers

4.1.1.1 Introduction to Monitoring Managers

There are three types of monitoring managers. The following figure shows their relationships.

Figure 4-1 Relationships of monitoring managers



- **Baseband unit (BBU)**

The BBU is the monitoring management center for most base stations. Both internal and external alarms are directly reported to the BBU and then forwarded to the alarm system.

BBUs are further classified into indoor BBUs (BBU3900/BBU3910/BBU5900/BBU5910) and outdoor BBUs (BBU3910A/BBU3910C/BBU5900A/BBU5900C/BookBBU5901/BookBBU5901a).

- **Cabinet control unit (CCU)**

When a base station has many devices to be monitored in the cabinets, and the BBU monitoring ports cannot meet the monitoring requirements, the CCU of each cabinet monitors devices housed in the cabinet.

- The power cabinet TP48600A-H17B1 (TP48600A for short) and battery cabinets IBBS700D, IBBS700T, IBBS300D, and IBBS300T use CCUs to implement monitoring management. The CCUs are cascaded, and the CCU directly connected to the BBU reports alarms to the BBU.
- APM30H (Ver.E) and TMC11H (Ver.E) cabinets can be monitored and managed by the CCUs which can centrally manage all cabinets in a base station but do not support cascading.

• **Remote radio unit (RRU)**

For some DBS3900 and DBS5900 base stations, when RRUs are installed far away from the BBU and the alarms of devices near the RRUs need to be collected, RRUs can directly manage these alarms. RRUs report alarms to the BBU through common public radio interface (CPRI) optical fibers and the BBU then forwards the alarms to the alarm console. This scenario is called RRU remote monitoring.

A BBU3910C/BBU5900C/BookBBU5901/BookBBU5901a does not have any monitoring ports and does not support RS485 or dry contacts. External equipment can be monitored only through RRU monitoring ports.

For details about BBU configurations, see *Base Station Cabinets and Subracks (Including the BBU Subrack) Configuration in 3900 & 5900 Series Base Station Product Documentation*. For details about RRU configurations, see *RF Unit and Topology Management in 3900 & 5900 Series Base Station Product Documentation*. For details about CCU configurations, see the following sections about operation and maintenance.

The following table describes the relationships between the monitoring managers and hardware devices.

Base Station Type	Cabinet Configuration/Power Input	Physical Board	Logical Name
-	-	BBU	BBU
DBS3900	IBBS700D/IBBS700T	CCU	CCU
	TP48600A cabinet	CCU	CCU
BTS3900AL (Ver.A)	BTS3900AL AC/ BTS3900AL DC	CCU	CCU
	IBBS700D/IBBS700T	CCU	CCU
	IBBS300D/IBBS300T	CCU01D-03	CCU
-	-	RRU	RRU

LampSite base stations do not support monitoring by RRUs, but they are the same as macro base stations in supporting other monitoring managers.

4.1.1.2 BBU Monitoring Ports

4.1.1.2.1 BBU3900/BBU3910/BBU5900/BBU5910 Monitoring Ports

In a BBU3900/BBU3910/BBU5900/BBU5910, the universal power and environment interface unit (UPEU) and universal environment interface unit (UEIU) boards provide monitoring ports. Each board provides two Boolean signal input ports and two RS485 signal input ports. Each Boolean signal input port supports four Boolean signal inputs.

- When a BBU is configured with one UPEU, the BBU provides eight Boolean signal inputs and two RS485 signal inputs.
- When a BBU is configured with two UPEUs or with one UPEU and one UEIU, the BBU provides 16 Boolean signal inputs and four RS485 signal inputs. On the UPEUs or on the UPEU and UEIU, the two MON0 ports are interconnected through RS485 bus 0, and the two MON1 ports are interconnected through RS485 bus 1.

Figure 4-2 shows the monitoring ports on the UPEU and UEIU.

Figure 4-2 Monitoring ports on the UPEU and UEIU

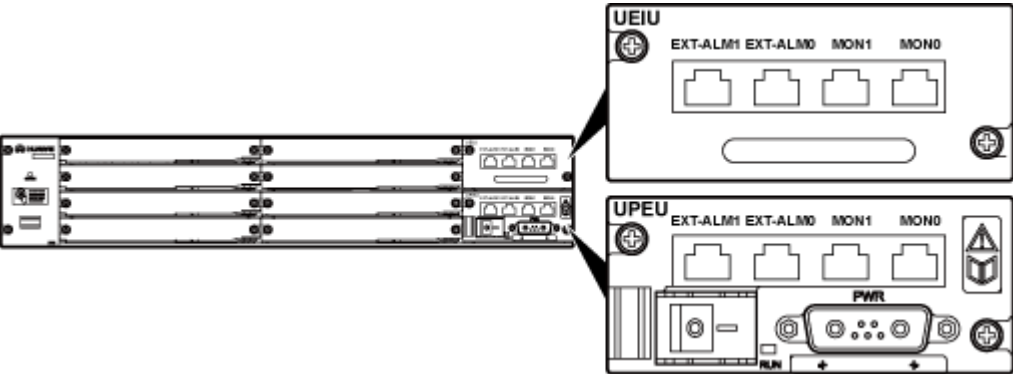


Table 4-1 describes the monitoring ports on the UPEU and UEIU.

Table 4-1 Monitoring ports on the UPEU and UEIU

Board	Silkscreen	Connector	Note
UPEU	EXT-ALM0	RJ45	Port for Boolean signal inputs 0 to 3
	EXT-ALM1	RJ45	Port for Boolean signal inputs 4 to 7
	MON0	RJ45	Port for RS485 signal input 0
	MON1	RJ45	Port for RS485 signal input 1
UEIU	EXT-ALM0	RJ45	Port for Boolean signal inputs 0 to 3

Board	Silkscreen	Connector	Note
	EXT-ALM1	RJ45	Port for Boolean signal inputs 4 to 7
	MON0	RJ45	Port for RS485 signal input 0
	MON1	RJ45	Port for RS485 signal input 1

4.1.1.2.2 BBU3910A Monitoring Port

The BBU3910A has one monitoring port, which provides one RS485 monitoring input, or six dry contact inputs and one relay output.

Figure 4-3 shows the monitoring port on the BBU3910A.

Figure 4-3 Monitoring port on the BBU3910A

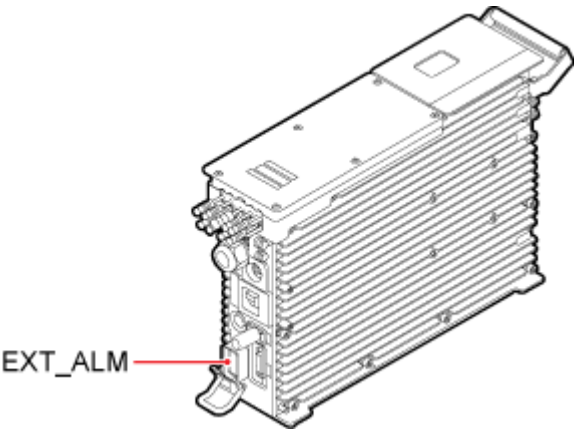


Table 4-2 describes the monitoring port on the BBU3910A.

Table 4-2 Monitoring port on the BBU3910A

Module	Silkscreen	Connector	Note
BBU3910A	EXT_ALM	DB15	- Port for Boolean signal inputs 0 to 5 - Port for Boolean signal output 0 or Port for RS485 signal input 0

4.1.1.2.3 BBU5900A Monitoring Ports

In a BBU5900A, the UEIUd board provides monitoring ports, including two Boolean signal input ports and two RS485 signal input ports. The two Boolean signal input ports support six Boolean signal inputs and one Boolean signal output.

Figure 4-4 UEIUd slot

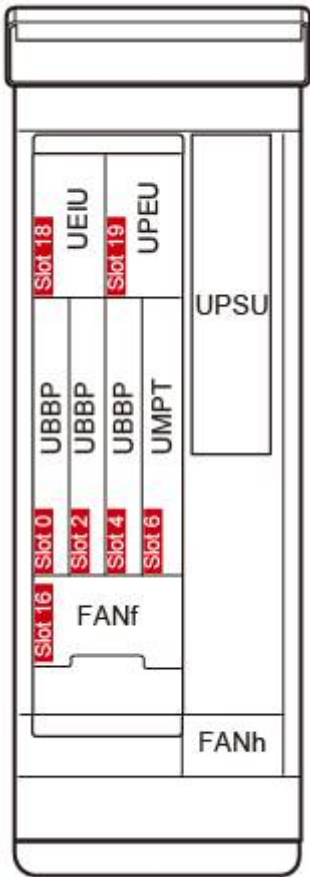


Figure 4-5 UEIUd ports

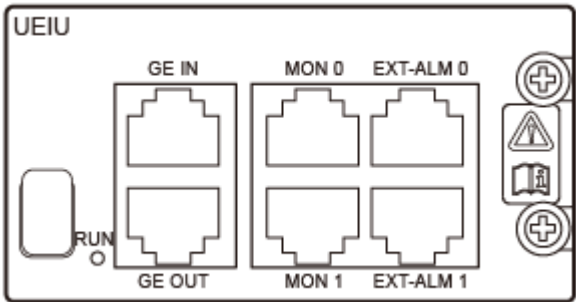


Table 4-3 describes the monitoring ports on the UEIUd.

Table 4-3 Monitoring ports on the UEIUd

Board	Silkscreen	Connector	Note
UEIUd	EXT-ALM0	RJ45	Port for Boolean signal inputs 0 to 3

Board	Silkscreen	Connector	Note
	EXT-ALM1	RJ45	Port for Boolean signal inputs 4 to 5 Port for Boolean signal output 6
	MON0	RJ45	Port for RS485 signal input 0
	MON1	RJ45	Port for RS485 signal input 1

4.1.1.3 CCU Monitoring Ports

A CCU monitors the operating environment of the cabinet and manages devices in the cabinet. Available CCU models are as follows:

- CCU
- CCU01D-03
- CCUB

For details about the CCU monitoring ports, see *3900 & 5900 Series Base Station Component Description*.

4.1.1.4 RRU Monitoring Ports

RRU monitoring ports vary with RRU models. Some RRUs provide one monitoring port which supports two or four Boolean inputs and one RS485 input. Some RRUs do not provide monitoring ports but can transmit monitoring information through power cables. Some RRUs neither provide monitoring ports nor transmit monitoring information through power cables. For details, see section "RRU Ports" in the RRU-specific hardware description.

NOTE

The following types of RRUs cannot be remotely connected to and monitored by the CCUs:

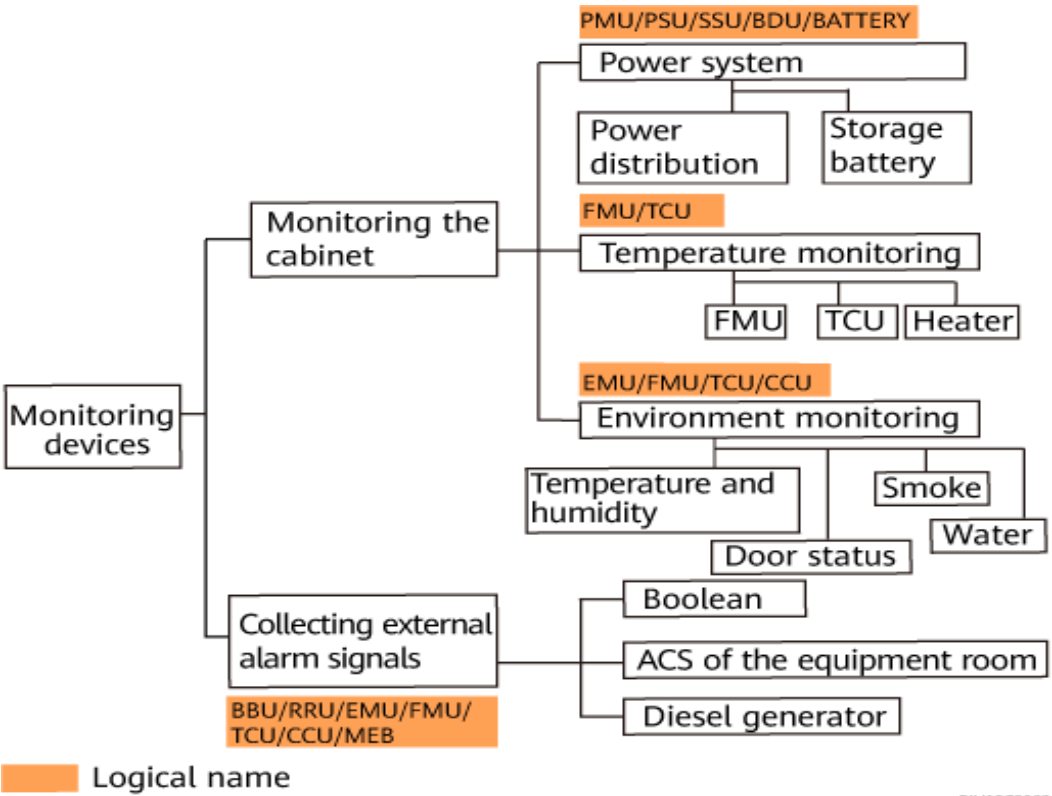
- Internal functions of the following RRUs do not support this type of monitoring: RRU3908, RRU3008, RRU3004(DRRU), RRU3804, RRU3806, RRU3801E, RRU3805, RRU3801C, RRU3808, RRU3201, RRU3220, RRU3203, RRU3222, RRU3260, RRU3221, and RRU3606
- RRUs without monitoring ports, such as Book RRUs, do not support this type of monitoring.

4.1.2 Monitoring Devices

4.1.2.1 Introduction to Monitoring Devices

Monitoring devices in a base station report monitoring information. For example, the FMU reports fan monitoring information, the PMU reports power supply monitoring information, and the EMU/EMUB reports environment monitoring alarms and customized alarms. All monitoring signals are transmitted to the BBU/CCU/RRU monitoring ports through an RS485 bus. The following figure shows the monitoring devices in a base station.

Figure 4-6 Monitoring devices



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For details about monitoring devices, see *3900 & 5900 Series Base Station Component Description*. For details about cable connections between monitoring devices and monitoring managers, see **Cable Connections > Monitoring Signal Cable Connections** in *3900 & 5900 Series Base Station Cables*.

4.1.2.2 Mapping Between Physical Boards and Logical Names of Monitoring Devices

The logical names in the following tables are the names of devices displayed on the configuration tool interface.

Table 4-4 Index table

Base Station Model	Table of Mapping Between Physical Boards and Logical Names
BTS3900	Table 4-5
BTS3900L	Table 4-6
BTS3900A	Table 4-7 Table 4-8
BTS3900C	Table 4-9
DBS3900	Table 4-10

Base Station Model	Table of Mapping Between Physical Boards and Logical Names
BTS3900AL (Ver.A)	Table 4-11
BTS5900/BTS5900A/BTS5900L	Table 4-12
DBS5900	Table 4-13

Mapping Table

Table 4-5 BTS3900 boards required in different scenarios

Base Station Type	Cabinet Configuration/ Power Input	Physical Board	Logical Name
BTS3900 (Ver.C)	–48 V DC power input	FMUE	FMU
	+24 V DC power input	FMUE	FMU
	110 V AC/220 V AC power input	FMUE	FMU
		PMU	PMU
BTS3900 (Ver.D) BTS3900 (Ver.D_A) BTS3900 (Ver.D_B)	–48 V DC power input	FMUEA	FMU
	+24 V DC power input	FMUEA	FMU
	220 V AC power input	FMUEA	FMU
		PMU ¹	PMU

Table 4-6 BTS3900L boards required in different scenarios

Base Station Type	Cabinet Configuration/ Power Input	Physical Board	Logical Name
BTS3900L (Ver.C)	–48 V DC power input	FMUE	FMU
BTS3900L (Ver.D) BTS3900L (Ver.D_B)	–48 V DC power input	FMUEA	FMU
	220 V AC power input	FMUEA	FMU
		PMU ²	PMU

NOTE

¹: The PMU 12A is supported from SRAN12.1 onwards, and it is supported only by the BTS3900 (Ver.D) cabinet.

²: The PMU 12A is supported from SRAN12.1 onwards, and it is supported only by the BTS3900L (Ver.D) DC cabinet.

Table 4-7 BTS3900A boards required in different scenarios

Base Station Type	Cabinet Configuration/ Power Input	Physical Board	Logical Name
BTS3900A (Ver.C)	APM30H (Ver.C)	PMU	PMU
		CMUE	TCU
	RFC (Ver.C)	CMUE	FMU
	TMC11H (Ver.C)	CMUE	TCU
	IBBS200D (Ver.C)/IBBS200T (Ver.C)	CMUE	TCU
		PMU 15A ¹	PMU
BTS3900A (Ver.D)	APM30H (Ver.D)	PMU 11A	PMU
		CMUEA	TCU
		PMU 12A	PMU
	RFC (Ver.D)	CMUEA	FMU
	TMC11H (Ver.D)	CMUEA	TCU
		PMU 12A	PMU
		PMU 15A ¹	PMU
	IBBS200D (Ver.D)/IBBS200T (Ver.D)	CMUEA	TCU
	IBBS700D/IBBS700T	CMUF/CMUEA	TCU
	IBBS300D/IBBS300T	CCU01D-03	TCU
BTS3900A (Ver.E)	APM30H (Ver.E)	CCUB ² /CCUC	PMU
		CCUB ² /CCUC	TCU
		PMU 12A	PMU
	RFC (Ver.E)	CMUH	FMU
	TMC11H (Ver.E)	CCUB/CCUC	TCU
		PMU 12A	PMU
	IBBS200D (Ver.E)/IBBS200T (Ver.E)	CMUH	TCU
		PMU 15A ³	PMU
	IBBS300D/IBBS300T	CCU01D-03	TCU
	BBC5200D-L (Ver.A)	CMUH	TCU
		PMU 15A ⁴	PMU

 NOTE

- 1: When lithium batteries are installed in an IBBS200D, the PMU 15A is required.
- 2: The PMU and TCU in an APM30H share the CCUB/CCUC whose E-Label can be queried through the TCU.
- 3: When the IBBS200D cabinet is configured with lithium batteries and the required PMU 15A, data configuration for the PMU 15A needs to be performed.
- 4: When the battery cabinet BBC5200D-L is used with the power cabinet APM30H, the PMU 15A is required.

Table 4-8 Boards required for an enhanced BTS3900A in different scenarios

Base Station Type	Cabinet Configuration/ Power Input	Physical Board	Logical Name
BTS3900A (Ver.D_B)	APM30H (Ver.D_B)	CMUEA	TCU
		PMU 12A	PMU
	RFC (Ver.D_B)	CMUEA	TCU
	TMC11H (Ver.D_B)	CMUEA	TCU
		PMU 12A	PMU
	IBBS200D (Ver.D_B)/ IBBS200T (Ver.D_B)	CMUEA	TCU
BTS3900A (Ver.D_A1)	APM30H (Ver.D_A1)	CMUEA	TCU
	RFC (Ver.D_A1)	CMUEA	TCU
	TMC11H (Ver.D_A1)	CMUEA	TCU
BTS3900A (Ver.D_A2)	APM30H (Ver.D_A2)	CMUEA	TCU
	RFC (Ver.D_A2)	CMUEA	TCU
	TMC11H (Ver.D_A2)	CMUEA	TCU

Table 4-9 BTS3900C boards required in different scenarios

Base Station Type	Cabinet Configuration/ Power Input	Physical Board	Logical Name
BTS3900C (Ver.C)	DC power input	HEUB	TCU
	AC power input	PMU 11A	PMU
		HEUB	TCU

Table 4-10 DBS3900 boards required in different scenarios

Base Station Type	Cabinet Configuration/Power Input	Physical Board	Logical Name
DBS3900	IBBS700D/IBBS700T	CCU	CCU
		CMUF/CMUEA	TCU
		HAU	TCU
	IBBS300D/IBBS300T	CCU01D-03	TCU
	APM30H (Ver.C)	PMU	PMU
DBS3900	TMC11H (Ver.C)	CMUE	TCU
		PMU 12A	PMU
	IBBS200D (Ver.C)/IBBS200T (Ver.C)	CMUE	TCU
		PMU 15A ¹	PMU
	APM30H (Ver.D_A1)/APM30H (Ver.D_A2)	PMU 11A	PMU
		CMUEA	TCU
	APM30H (Ver.D)/APM30H (Ver.D_B)	PMU 11A	PMU
		CMUEA	TCU
		PMU 12A	PMU
	TMC11H (Ver.D_A1)/TMC11H (Ver.D_A2)	CMUEA	TCU
	TMC11H (Ver.D)/TMC11H (Ver.D_B)	CMUEA	TCU
		PMU 12A	PMU
	APM30H (Ver.E)	CCUB/CCUC	PMU
			TCU
		PMU 12A	PMU
	TMC11H (Ver.E)	CCUB/CCUC	TCU
		PMU 12A	PMU
	IBBS200D (Ver.E)/IBBS200T (Ver.E)	CMUH	TCU
		PMU 15A ¹	PMU
	BBC5200D-L (Ver.A)	CMUH	TCU
		PMU 15A	PMU

Base Station Type	Cabinet Configuration/Power Input	Physical Board	Logical Name
DBS3900	IBBS300D/IBBS300T	CCU01D-03	TCU
	OMB (Ver.C)	HEUB	TCU
		PMU 11A/PMU 11B	PMU
		PMU 12A	PMU
DBS3900	IMB	PMU	PMU
	IMB (ICR scenario)	PMU	PMU
	TP48600A cabinet	CCU	CCU
		PMU	PMU
		CMUF	TCU
		HAU	TCU
	OPM50M (Ver.A)	OPM50M	PMU
	OPM50M (Ver.B)	OPM50M	PMU
	RFC5906	CMUH	FMU
	IFS5906	FMUF	FMU

 NOTE

- ¹: When lithium batteries are installed in an IBBS200D, the PMU 15A is required.
- Due to the version compatibility design, the hardware version of the CMUH/FMUF/PMU 11A/PMU 11B/PMU 12A/PMU 15A/CCUB/CCUC board queried by running the **DSP BRDVER** command may be different before and after the board software is activated (by running **ACT SOFTWARE**). The functions supported by the hardware, however, remain unchanged.

Table 4-11 BTS3900AL (Ver.A) boards required in different scenarios

Base Station Type	Cabinet Configuration/Power Input	Physical Board	Logical Name
BTS3900AL (Ver.A)	BTS3900AL AC	CCU	CCU
		CMUF	TCU
		PMU	PMU
		FAU	FMU
		HAU	TCU

Base Station Type	Cabinet Configuration/ Power Input	Physical Board	Logical Name
	IBBS700D/IBBS700T	PMU 12A	PMU
		CCU	CCU
		CMUF/CMUEA	TCU
		HAU	TCU
	BTS3900AL DC	CCU	CCU
		CMUF	TCU
		FAU	FMU
		HAU	TCU
		PMU 12A	PMU
	TMC11H (Ver.C)	CMUE	TCU
	IBBS300D/IBBS300T	CCU01D-03	CCU
		CCU01D-03	TCU

Table 4-12 BTS5900/BTS5900A/BTS5900L boards required in different scenarios

Base Station Type	Cabinet Configuration/ Power Input	Physical Board	Logical Name
BTS5900A (Ver.A)	APM5930(AC)	CCUC	PMU
			TCU
		PMU 12A	PMU
	BBC5200D (Ver.A)/ BBC5200T (Ver.A)	CMUH	TCU
	BBC5300D (Ver.A)/ BBC5300T (Ver.A)	CMUH (replacing the CCU01D-03)	TCU
	BBC5200D-L (Ver.A)	CMUH	TCU
BTS5900A (Ver.A)	RFC5906	CMUH	FMU
BTS5900A (Ver.A)	BBC5600D(Ver.A)/ BBC5600A(Ver.A)	CMUH	TCU
	BBC5600D-V(Ver.A)/ BBC5600A-V(Ver.A)	CMUH	TCU
BTS5900A (Ver.A)	APM5950H(Ver.A)/ APM5950H-L(Ver.A)	PMU 11C	PMU

Base Station Type	Cabinet Configuration/ Power Input	Physical Board	Logical Name
BTS5900A (Ver.A)	APM5900H (Ver.A)	CCUC	TCU
BTS5900A (Ver.A)	APM5900H-L(Ver.A)	CMUH	TCU
BTS5900 (Ver.A)	-48 V DC power input	FMUF	FMU
		PMU 12A	PMU
BTS5900L (Ver.A)	-48 V DC power input	FMUF	FMU
		PMU 12A	PMU

 NOTE

The monitoring boards need to be configured only when a TP48600A cabinet is used.

Table 4-13 DBS5900 boards required in different scenarios

Base Station Type	Cabinet Configuration/ Power Input	Physical Board	Logical Name
DBS5900	IMB05	-	-
	ILC29 (Ver.E)	FMUF	FMU
	APM5930(DC)	CCUC	TCU
	ICR: The BBU and PMU are installed in the upper and lower racks respectively in the IMB05 cabinet.	PMU 11B	PMU
	APM5930(AC)/BBC5200D/ BBC5200T/BBC5300D/ BBC5300T	CCUC	PMU
			TCU
		CMUH	TCU
	OPM200/OPM200 (Ver.B)	PMU 16A	PMU
DBS5900	APM5900H	CCUC	TCU
DBS5900	APM5900H-L	CMUH	TCU

4.1.3 Monitoring Networking for a Single-mode/Dual-mode Base Station

The monitoring boards of a base station report monitoring information. For example, the FMU reports fan monitoring information, the PMU reports power supply monitoring information, and the EMU/EMUB reports environment

monitoring alarms and customized alarms. All the monitoring signals are transmitted to the BBU, CCU, or RRU monitoring ports through an RS485 bus. The following describes the connections between boards and monitoring components.

 NOTE

For details about how to connect the monitoring signal cables of the boards to relevant ports, see sections "Cable Connections" > "Monitoring Signal Cable Connections" in *3900 & 5900 Series Base Station Cables*.

4.1.3.1 BTS3900/BTS5900 Monitoring Networking

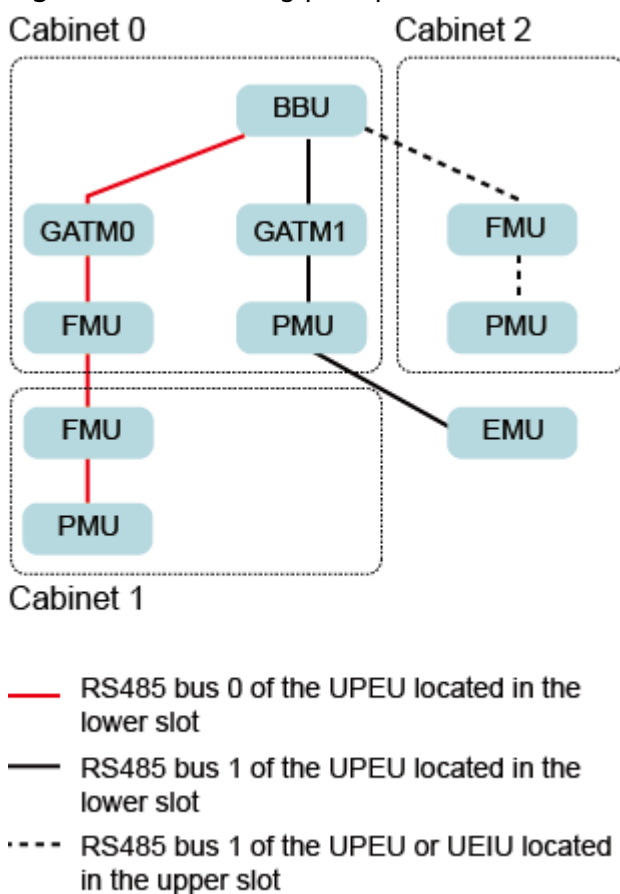
Figure 4-7 shows the monitoring principles of the BTS3900 cabinet (AC). The BTS3900 (Ver.C), BTS3900 (Ver.D), BTS3900 (Ver.D_A), and BTS3900 (Ver.D_B) cabinets use the same monitoring principles. The FMU, PMU, and GATM (short for GSM antenna and TMA control module) communicate with the BBU's MON ports through an RS485 serial bus. The GATM is GBTS-specific and is not used for the eGBTS/NodeB/eNodeB/gNodeB.

A BTS3900 (Ver.E_B~D) or BTS5900 (Ver.A) cabinet used for a BTS5900 applies the same monitoring principles as those used for a BTS3900.

The monitored boards are cascaded in sequence. If an upper-level board is not configured, the monitoring signal cable is directly connected to the lower-level board. The bottom-level board does not need to be connected if it is not configured. For example:

- If the GATM is not configured, the FMU and PMU are directly connected to the BBU's MON ports.
- If cabinet 1 and cabinet 2 are not configured, pay attention to components in cabinet 0 only.

Figure 4-7 Monitoring principles of the BTS3900 cabinet (AC)



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NOTE

Cabinet 0 is the first cabinet of this type configured at a site. The same rule also applies to numbering cabinets of other types.

The monitoring principles of the BTS3900 cabinet (–48 V DC) are similar to those of the BTS3900 cabinet (AC). In these cabinets, the PMU is not configured and can be ignored in [Figure 4-7](#).

4.1.3.2 BTS3900L/BTS5900L Monitoring Networking

The BTS3900L (Ver.D) cabinet stacked with the IMS06 cabinet supports AC power supply. [Figure 4-8](#) shows the monitoring principles of the BTS3900L (AC). The FMU, PMU, GATM, and EMU communicate with the BBU's MON ports through an RS485 serial bus. The GATM is GBTS-specific and is not used for the eGBTS/NodeB/eNodeB/gNodeB.

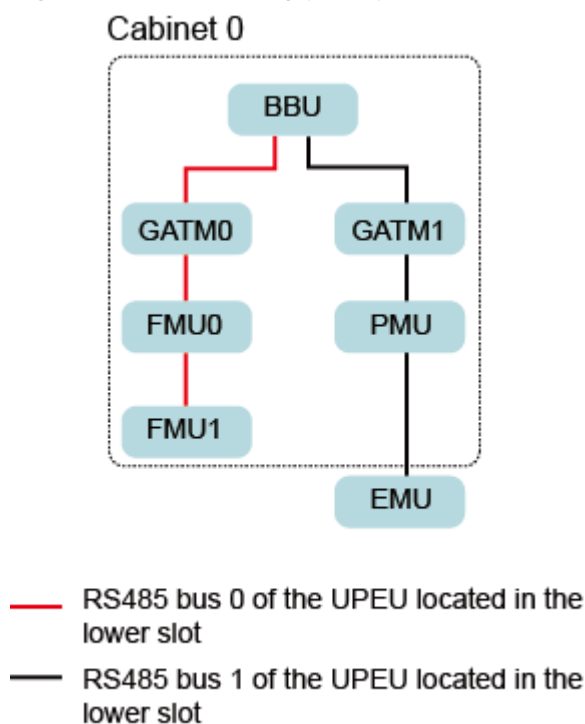
The monitored boards are cascaded in sequence. If an upper-level board is not configured, the monitoring signal cable is directly connected to the lower-level board. The bottom-level board does not need to be connected if it is not configured. For example, if the GATM is not configured, the FMU is directly connected to the BBU's MON port.

A BTS3900L (Ver.E_B~D) or BTS5900L (Ver.A) cabinet used for a BTS5900L applies the same monitoring principles as those used for a BTS3900L.

NOTICE

In the following figure, FMU0 indicates the FMU in the lower-level subrack and FMU1 indicates the FMU in the upper-level subrack in the BTS3900L cabinet. Cascade them following the shown sequence.

Figure 4-8 Monitoring principles of the BTS3900L (AC)



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The monitoring principles of the BTS3900L (–48 V DC) are similar to those of the BTS3900L (AC). In the BTS3900L (–48 V DC) cabinet, the PMU in [Figure 4-8](#) is not configured and can be ignored. The BTS3900L (Ver.C), BTS3900L (Ver.D), and BTS3900L (Ver.D_B) cabinets use the same monitoring principles.

4.1.3.3 BTS3900A/BTS5900A Monitoring Networking

BTS3900A (Ver.C)/(Ver.D)

Figures "Monitoring principles of the BTS3900A (AC)" and "Monitoring principles of the BTS3900A (–48 V DC)" show the monitoring principles of these cabinets. The central monitoring unit type E (CMUE)/central monitoring unit type EA (CMUEA), PMU, GATM, and EMU communicate with the BBU's MON ports through an RS485 serial bus. The GATM is GBTS-specific and is not used for the eGBTS/NodeB/eNodeB/gNodeB. The following uses BTS3900A (Ver.C) cabinets as an example.

The monitored boards are cascaded in sequence. If an upper-level board is not configured, the monitoring signal cable is directly connected to the lower-level board. The bottom-level board does not need to be connected if it is not configured. For example:

- If the TMC cabinet is not configured, ignore the connections related to the TMC cabinet. The CMUE in cabinet RFC0 is directly connected to the CMUE in cabinet RFC1.
- If the battery backup cabinet (BBC) is not configured, ignore the connections to the CMUE in the BBC.

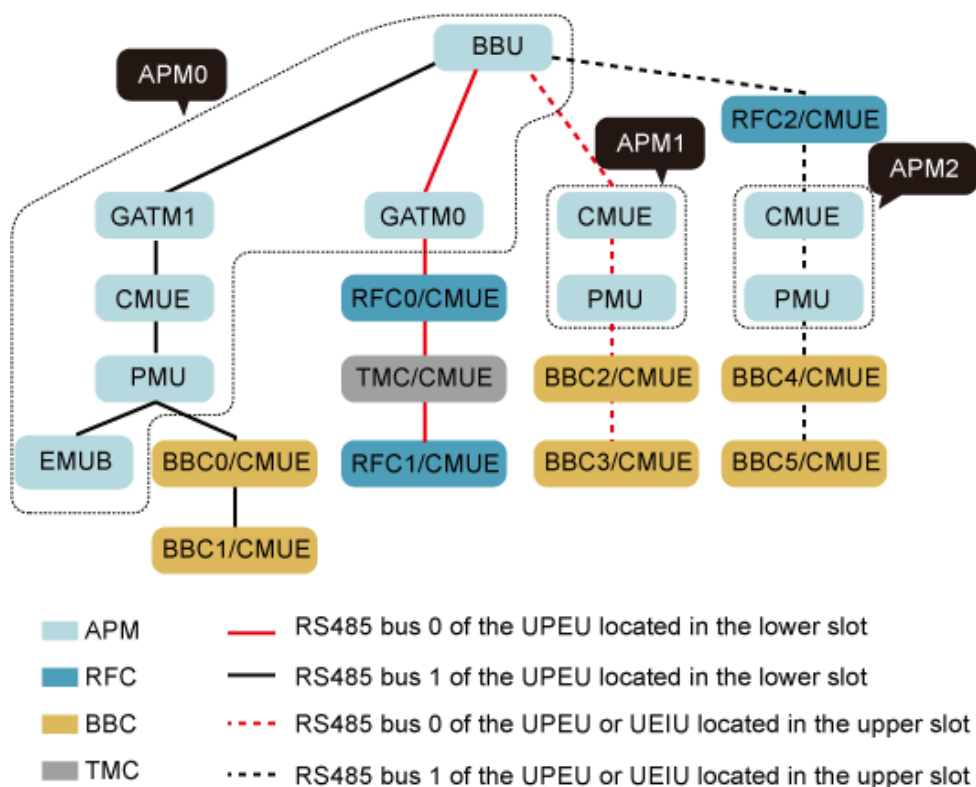
The monitoring principles of the BTS3900A (Ver.D), BTS3900A (Ver.D_B), BTS3900A (Ver.D_A1), and BTS3900A (Ver.D_A2) cabinets are the same as those of the BTS3900A (Ver.C) cabinet, except that no HERT power monitoring interface unit (HPMI) is configured. The HPMI does not monitor signals but transfers signals for the PMU. It is not shown in the figures. Figures "Monitoring principles of the BTS3900A (AC)" and "Monitoring principles of the BTS3900A (-48 V DC)" use the BTS3900A (Ver.C) cabinet as the example to illustrate the monitoring principles. For the BTS3900A (Ver.D), BTS3900A (Ver.D_B), BTS3900A (Ver.D_A1), and BTS3900A (Ver.D_A2), the CMUE is replaced with the CMUEA.

A BTS5900A applies the same monitoring principles as a BTS3900A.

NOTICE

Two BBCs are cascaded in sequence and cannot be connected to the same PMU. For example, if BBC0 is connected to the PMU's COM_485 port, BBC1 is connected to the CMUE's COM_OUT port in BBC0 and cannot be connected to the other RS485 ports on the PMU.

Figure 4-9 Monitoring principles of the BTS3900A (AC)

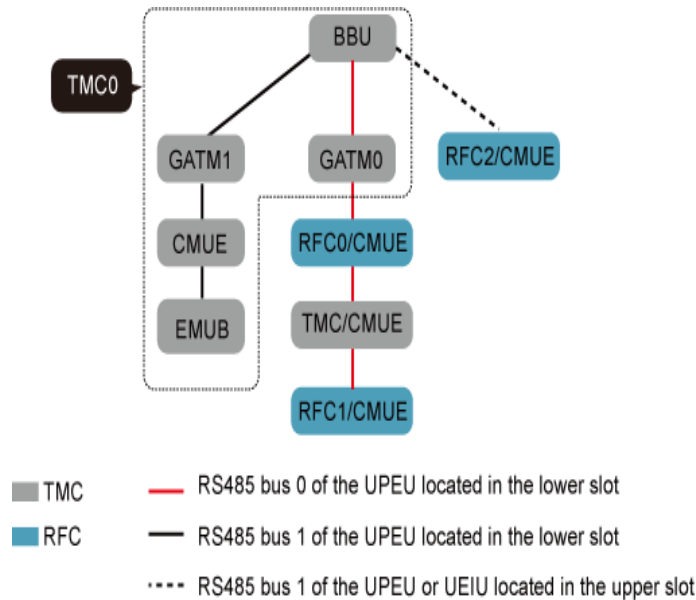


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NOTE

BBCs are IBBS200D, IBBS200T, IBBS700D, IBBS700T, IBBS300D, and IBBS300T cabinets. (IBBS is short for integrated backup battery system.) The monitoring board in an IBBS700D cabinet is CMUEA and the monitoring board in an IBBS700T cabinet is CMUF. The monitoring board in an IBBS300D or IBBS300T cabinet is CCU01D-03.

Figure 4-10 Monitoring principles of the BTS3900A (–48 V DC)

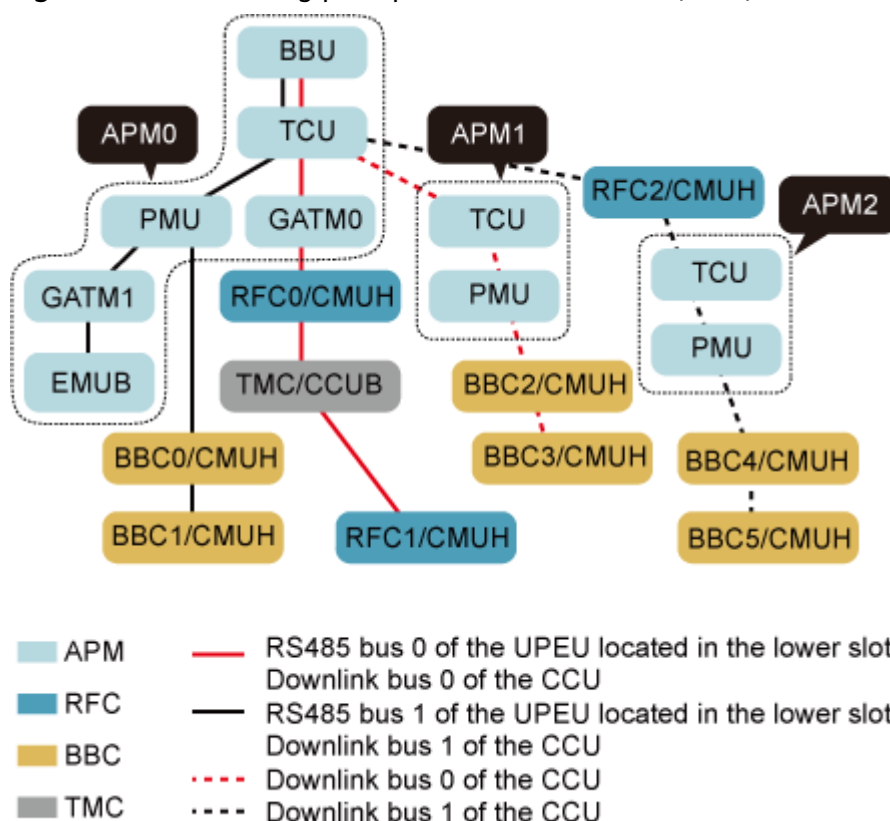


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BTS3900A (Ver.E)

Figure 4-11 and **Figure 4-12** show the monitoring principles of a BTS3900A (Ver.E) cabinet.

Figure 4-11 Monitoring principles of the BTS3900A (Ver.E) AC cabinet

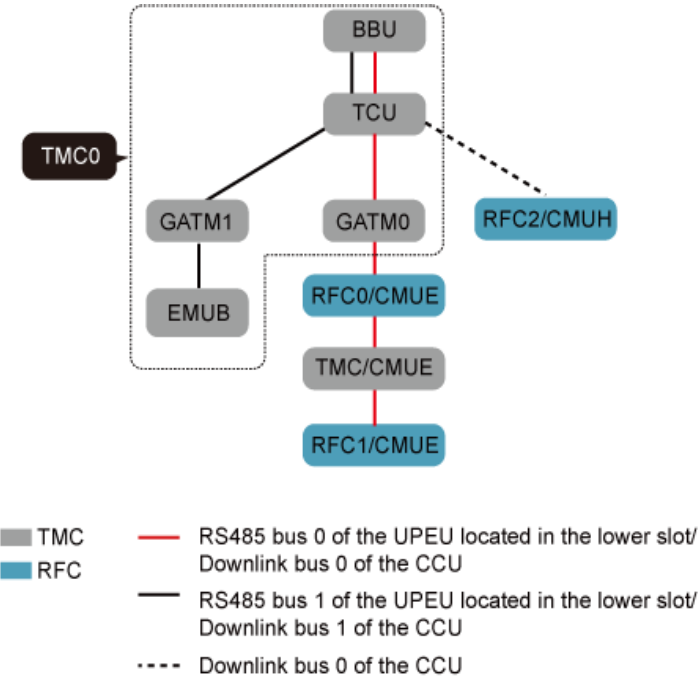


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NOTE

- BBCs are IBBS200D, IBBS200T, IBBS300D, and IBBS300T cabinets. The monitoring board in an IBBS300D or IBBS300T cabinet is CCU01D-03.
- The TCU and PMU in an APM30H cabinet share the CCUB, whose E-Label can be queried through the TCU. Port U_COM0 on the CCUB in the cabinet where the BBU is installed is connected to port MON0 of the BBU, port U_COM1 is connected to port MON1 of the BBU, lower-level electromechanical components are connected to BBU communication ports MON0-0, MON0-1, MON1-0, and MON1-1 to which the CCUB is mapped, respectively. The BBC is connected to COM_485 port of the CCUB.
- If the eBat battery management function is used for the Ver.E cabinet with RRUs remotely installed, the CCU monitoring is implemented by connecting the CCU to remote RRUs, which cannot be the following RRUs: RRU3908, RRU3008, RRU3004 (DRRU), RRU3804, RRU3806, RRU3801E, RRU3805, RRU3801C, RRU3808, RRU3201, RRU3220, RRU3203, RRU3222, RRU3260, RRU3221, and RRU3606. The intelligent battery monitoring function has been disused and is no longer supported. Therefore, the board does not need to be configured as CCU.

Figure 4-12 Monitoring principles of the BTS3900A (Ver.E) (-48 V)



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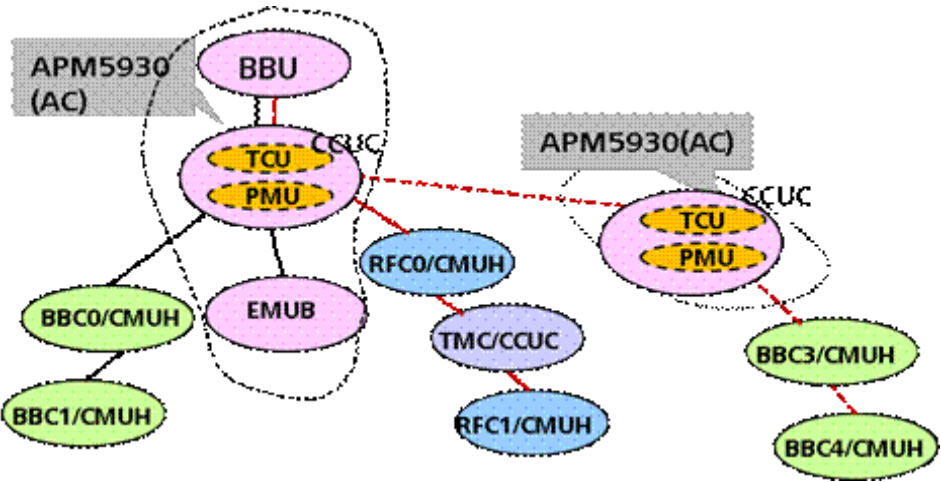
NOTE

The TCU hardware board in a TMC cabinet is the CCUB. Port U_COM0 on the CCUB in the cabinet where the BBU is installed is connected to port MON0 of the BBU, port U_COM1 is connected to port MON1 of the BBU, lower-level electromechanical components are connected to BBU communication ports MON0-0, MON0-1, MON1-0, and MON1-1 to which the CCUB is mapped, respectively.

BTS5900A (Ver.A)

Figure 4-13 and Figure 4-15 show the monitoring principles of the BTS5900A (Ver.A) cabinet.

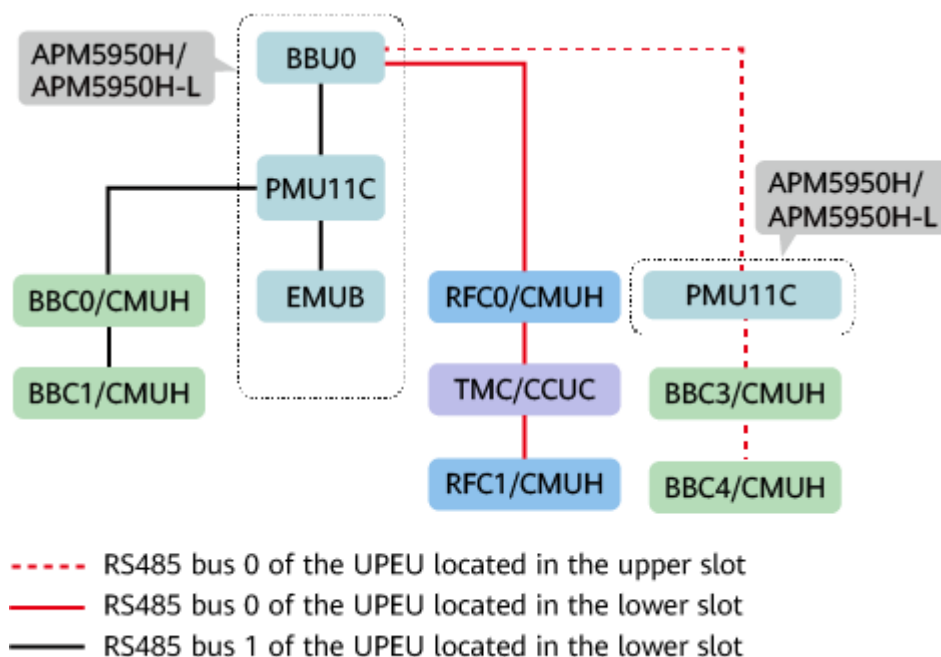
Figure 4-13 Monitoring principles of the BTS5900A (Ver.A) AC



NOTE

- BBCs are BBC5200D, BBC5200D-L, BBC5200T, BBC5300D, or BBC5300T cabinets. The monitoring board in a BBC5200D, BBC5200D-L, or BBC5200T cabinet is CMUH. The monitoring board CCU01D-03 in a BBC5300D or BBC5300T cabinet is replaced by the CMUH.
- The TCU and PMU in an APM5930 cabinet share the CCUC, whose E-Label can be queried through the TCU. Port U_COM0 on the CCUC in the cabinet where the BBU is installed is connected to port MON0 of the BBU, port U_COM1 is connected to port MON1 of the BBU, lower-level electromechanical components are connected to BBU communication ports MON0-0, MON0-1, MON1-0, and MON1-1 to which the CCUC is mapped, respectively. The BBC is connected to COM_485 port of the CCUC.
- The CCUC board in the APM5930 cannot be configured as CCU.

Figure 4-14 Monitoring principles of BTS5900A(Ver.A)-APM5950H/APM5950H-L

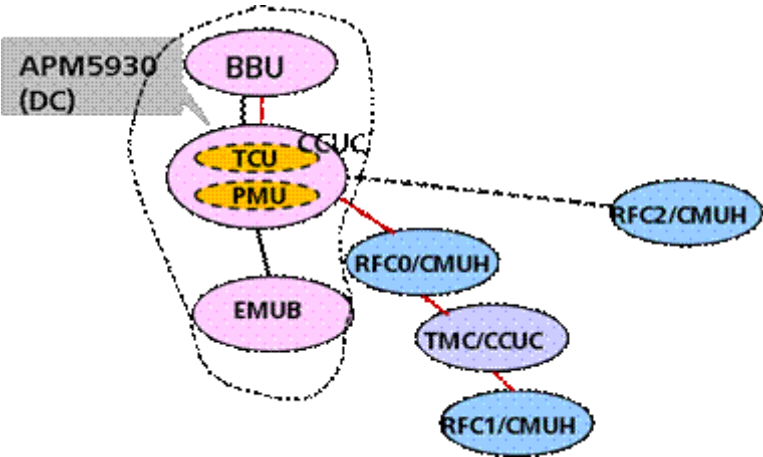


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NOTE

- BBCs are BBC5600D/BBC5600D-V/BBC5600A/BBC5600A-V cabinets. The monitoring board in a BBC5600D/BBC5600D-V/BBC5600A/BBC5600A-V cabinet is CMUH.
- Only the PMU 11C is installed in the APM5950H/APM5950H-L cabinet. You can query the electronic label of the cabinet through the PMU. The COM_IN port on the PMU 11C in the cabinet where the BBU is installed is connected to the MON1 port on the BBU.
- The monitored boards are cascaded in sequence. If an upper-level board is not configured, the monitoring signal cable is directly connected to the lower-level board. The bottom-level board does not need to be connected if it is not configured. For example, if the TMC cabinet is not configured, ignore the connections related to the TMC cabinet. The CMUH in cabinet RFC0 is directly connected to the CMUH in cabinet RFC1. If the BBC is not configured, ignore the connections to the CMUH in the BBC.

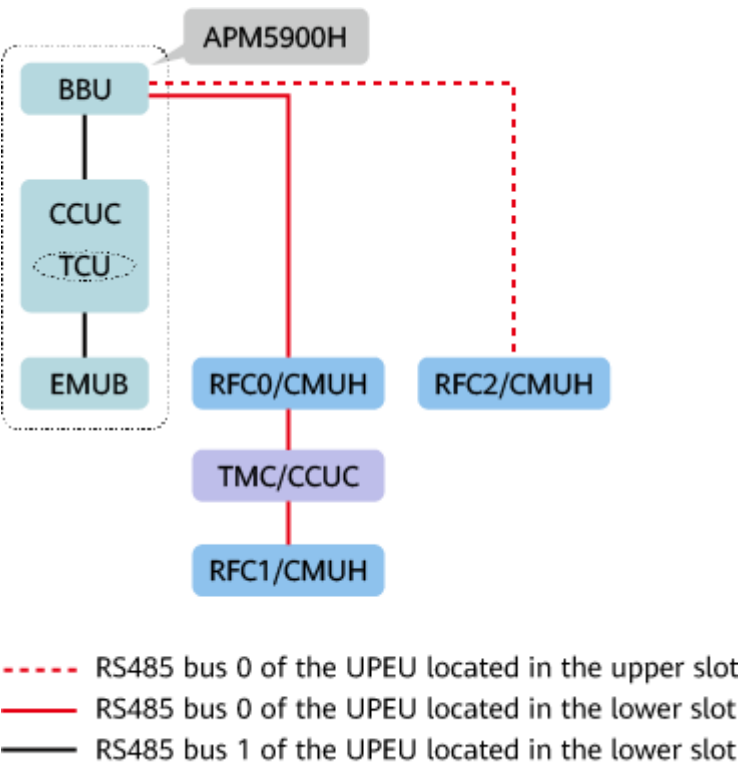
Figure 4-15 Monitoring principles of the BTS5900A (Ver.A) (-48 V)



NOTE

The TCU hardware board in an APM5930(DC) cabinet is the CCUC. Port U_COM0 on the CCUC in the cabinet where the BBU is installed is connected to port MON0 of the BBU, port U_COM1 is connected to port MON1 of the BBU, lower-level electromechanical components are connected to BBU communication ports MON0-0, MON0-1, MON1-0, and MON1-1 to which the CCUC is mapped.

Figure 4-16 Monitoring principles of BTS5900A(Ver.A)-APM5900H

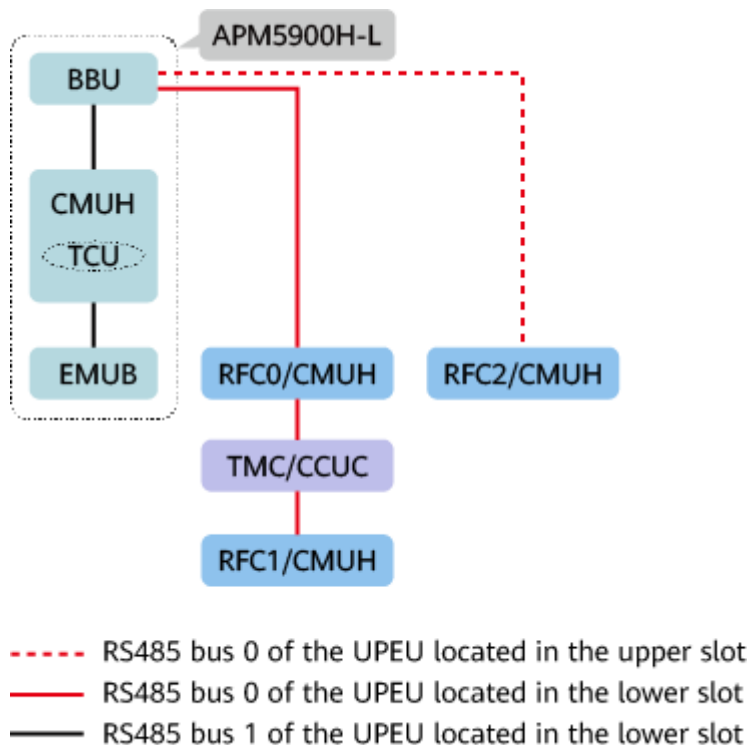


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NOTE

The monitored boards are cascaded in sequence. If an upper-level board is not configured, the monitoring signal cable is directly connected to the lower-level board. The bottom-level board does not need to be connected if it is not configured. For example, if the TMC cabinet is not configured, ignore the connections related to the TMC cabinet. The CMUH in cabinet RFC0 is directly connected to the CMUH in cabinet RFC1. If the BBC is not configured, ignore the connections to the CMUH in the BBC.

Figure 4-17 Monitoring principles of BTS5900A(Ver.A)-APM5900H-L



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NOTE

The monitored boards are cascaded in sequence. If an upper-level board is not configured, the monitoring signal cable is directly connected to the lower-level board. The bottom-level board does not need to be connected if it is not configured. For example, if the TMC cabinet is not configured, ignore the connections related to the TMC cabinet. The CMUH in cabinet RFC0 is directly connected to the CMUH in cabinet RFC1. If the BBC is not configured, ignore the connections to the CMUH in the BBC.

4.1.3.4 DBS3900/DBS5900 Monitoring Networking

The DBS3900 has multiple application scenarios. The monitoring principles vary with hardware configurations, as described in [Table 4-14](#).

Table 4-14 Index table of DBS3900 monitoring principles

Input Power	Cabinet Configuration	Setting Notes
110 V AC/220 V AC	• APM30H (Ver.C) • TMC11H (Ver.C) • IBBS200D (Ver.C)/IBBS200T (Ver.C)	The monitoring principles are the same as those of the BTS3900A AC cabinet. For details, see Figure 4-9 .
110 V AC/220 V AC	• APM30H (Ver.D)/APM30H (Ver.D_B)/APM30H (Ver.D_A1)/APM30H (Ver.D_A2) • TMC11H (Ver.D)/TMC11H (Ver.D_B)/TMC11H (Ver.D_A1)/TMC11H (Ver.D_A2) • IBBS200D (Ver.D)/IBBS200T (Ver.D) • IBBS700D/IBBS700T • IBBS300D/IBBS300T	The monitoring principles are the same as those of the BTS3900A AC cabinet. For details, see Figure 4-9 .
	• APM30H (Ver.E) • TMC11H (Ver.E) • IBBS200D (Ver.E)/IBBS200T (Ver.E) • MRE1000	
110 V AC/220 V AC	• APM30H (Ver.C) • APM30H (Ver.D)/APM30H (Ver.D_B)/APM30H (Ver.D_A1)/APM30H (Ver.D_A2) • IBBS200D/IBBS200T • IBBS700D/IBBS700T • IBBS300D/IBBS300T	For details, see Figure 4-18 .
110 V AC/220 V AC	TP48600A	For details, see Figure 4-19 .
-48 V DC	TMC11H (Ver.C)	For details, see (1) in Figure 4-20 .
	APM30H (Ver.D)/APM30H (Ver.D_B)/APM30H (Ver.D_A1)/APM30H (Ver.D_A2)	
	TMC11H (Ver.E)	For details, see Figure 4-22 .
110 V AC/220 V AC	OMB (Ver.C)	For details, see (3) in Figure 4-20 .
-48 V DC	OMB (Ver.C)	For details, see (3) in Figure 4-20 . The monitoring principles are similar to those of the AC scenario except that the PMU is not used in this scenario.

Input Power	Cabinet Configuration	Setting Notes
110 V AC/220 V AC	Indoor centralized rack (ICR): The BBU and PMU are installed in the upper and lower racks respectively in the indoor mini box (IMB03/IMB05).	For details, see (4) in Figure 4-20 .
+24 V DC	ICR: The BBU is installed in the IMB03/IMB05 cabinet.	For details, see (5) in Figure 4-20 .
110 V AC/220 V AC	IMB03/IMB05	For details, see (4) in Figure 4-20 .
-48 V DC	IMB03/IMB05	Monitoring signal cables are not required.
110 V AC/220 V AC	BBU3910A+outdoor power module 50M (OPM50M)	For details, see (6) in Figure 4-20 .
110 V AC/220 V AC	OPM30M+RRU	For details, see (7) in Figure 4-20 .
-48 V DC	BBU3910A	Monitoring signal cables are not required.
-48 V DC	BBU3910C	For details, see Example of Configurations in a DBS3900 (-48 V DC) .

 NOTE

- The DBS3900 is not configured with the GATM or the RFC cabinet, so the cable connections to the GATM or to the RFC cabinet can be ignored. The BBU in the APM30H cabinet is directly connected to the CMUE in the TMC11H cabinet.
- The MRE1000 cabinet does not support the eBat battery management function.
- When the BBU3910A functions as the monitoring manager, it can monitor a maximum of three cascaded OPM50M/PMU modules. If only one or two OPM50M/PMU modules are configured at a site, you only need to connect the cables for the configured OPM50M/PMU modules.

The DBS5900 has multiple application scenarios. The monitoring principles vary with hardware configurations, as described in [Table 4-15](#).

Table 4-15 Index table of DBS5900 monitoring principles

Input Power	Cabinet Configuration	Setting Notes
-48 V DC	IMB05	Monitoring signal cables are not required.

Input Power	Cabinet Configuration	Setting Notes
-48 V DC	ILC29 (Ver.E)	The working principles of an ILC29 cabinet are the same as those of a BTS3900L. The FMU, PMU, GATM, and EMU communicate with the BBU's MON ports through an RS485 serial bus. For details, see Figure 4-8 .
-48 V DC	APM5930(DC)	For details, see (1) in Figure 4-20 .
-48 V DC	APM5900H/APM5900H-L	For details, see Figure 4-16 and Figure 4-17 .
110 V AC/220 V AC	ICR: The BBU and PMU are installed in the upper and lower racks respectively in the IMB05 cabinet.	The monitoring principles are the same as those of a DBS3900. For details, see (4) in Figure 4-20 .
110 V AC/220 V AC	APM5930(AC) BBC5200D/BBC5200T BBC5300D/BBC5300T BBC5200D-L	The monitoring principles are the same as those of the BTS5900A AC cabinet. For details, see 4.1.3.3 BTS3900A/BTS5900A Monitoring Networking .
110 V AC/220 V AC	OPM200/OPM200 (ver.B)	For details, see Figure 4-25 .
-	Indoor 19-inch cabinet/subrack (including the GDU)	Figure 4-26
110 V AC/220 V AC	APM5950H/APM5950H-L BBC5600D/BBC5600D-V/BBC5600A/ BBC5600A-V	The monitoring principles are the same as those of the BTS5900A AC cabinet. For details, see Figure 4-14 .

 NOTE

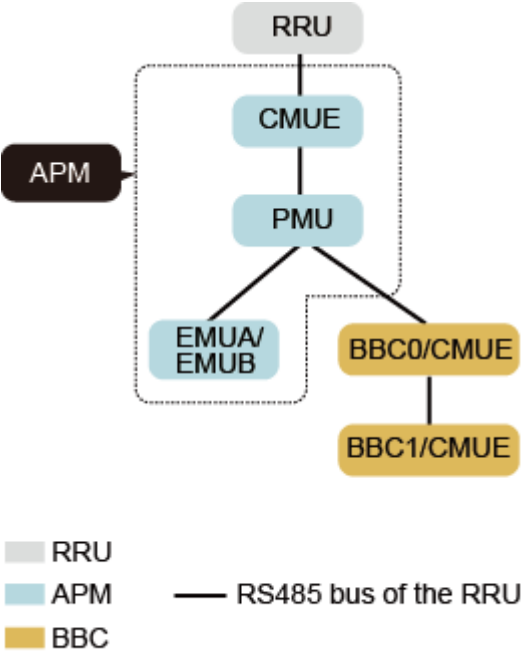
If the DBS5900 is not configured with the GATM and RFC, ignore the cable connections for the GATM and RFC.

For the APM/TMC series cabinets used in the DBS3900 in RRU remote monitoring mode, the monitoring principles of the Ver.D/Ver.D_B/Ver.D_A1/Ver.D_A2 cabinets are the same as those of the Ver.C cabinets, except that the CMUE is replaced with the CMUEA. [Figure 4-18](#) shows the monitoring principles of the Ver.C cabinets.

[Figure 4-21](#) shows the monitoring principles of APM/TMC (Ver.E) cabinets of the DBS3900 when RRU remote monitoring is used.

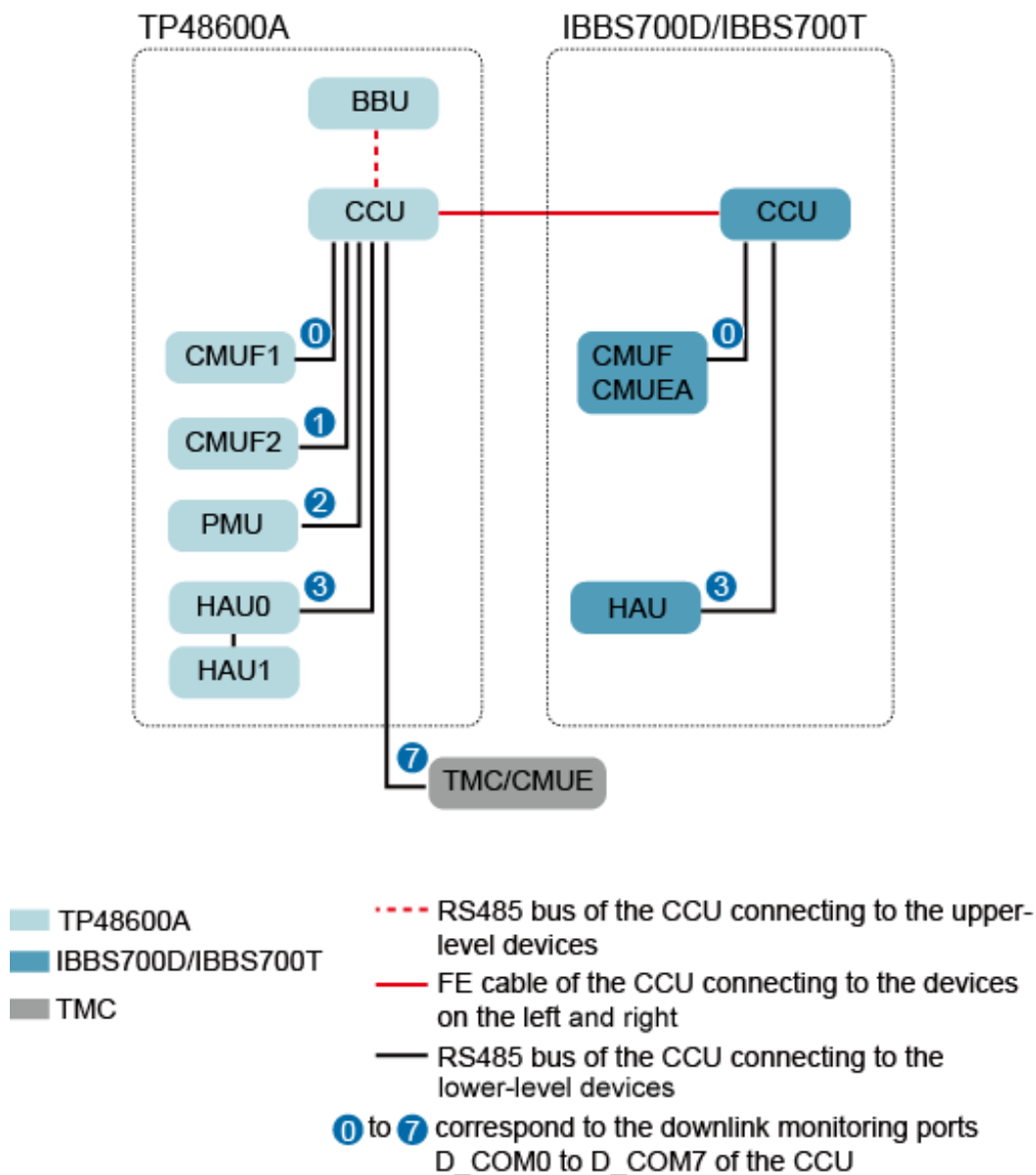
[Figure 4-21](#) shows the monitoring principles of APM5930 series cabinets of the DBS5900 when RRU remote monitoring is used.

Figure 4-18 RRU remote monitoring principles of the DBS3900



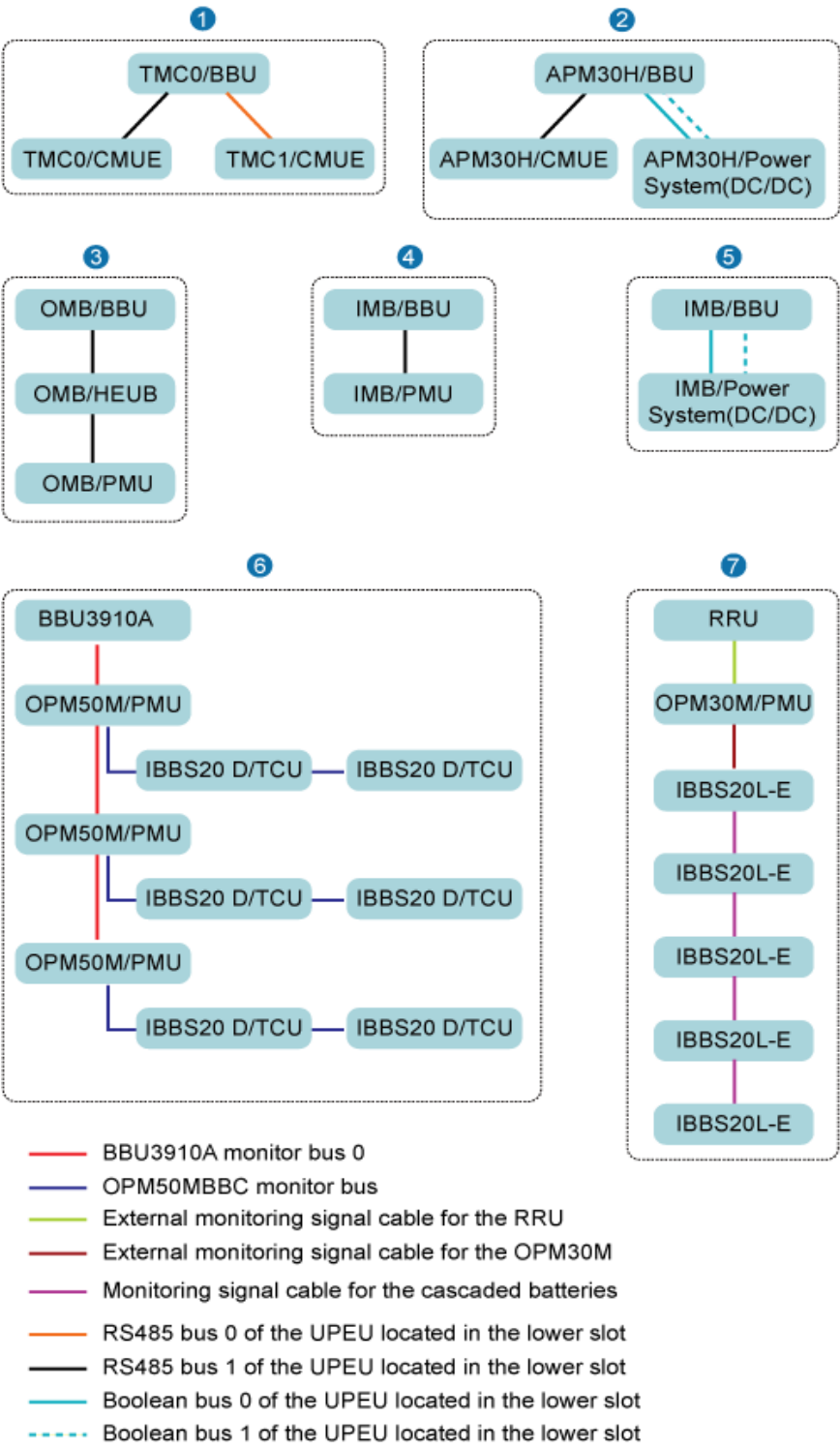
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Figure 4-19 Monitoring principles of the DBS3900 configured with the TP48600A



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Figure 4-20 Monitoring principles of the DBS3900 in different scenarios



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Figure 4-21 RRU remote monitoring principles of the DBS3900 (Ver.E)

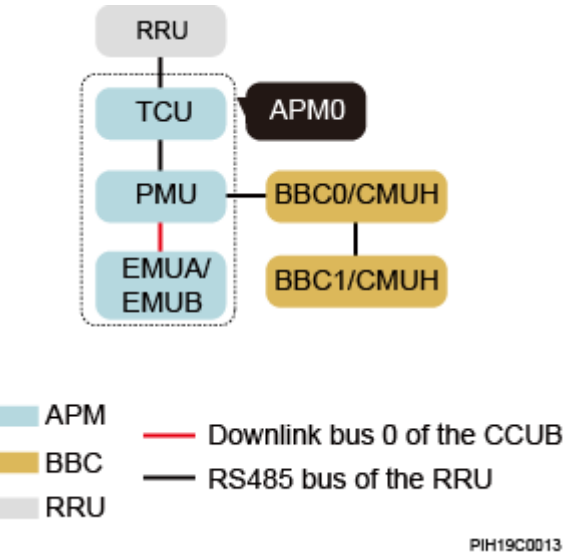
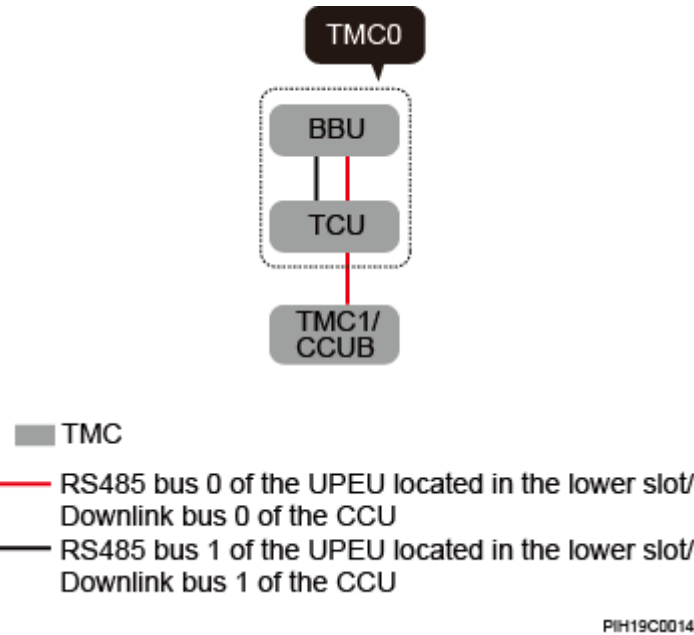


Figure 4-22 Monitoring principles of the DBS3900 (Ver.E) DC



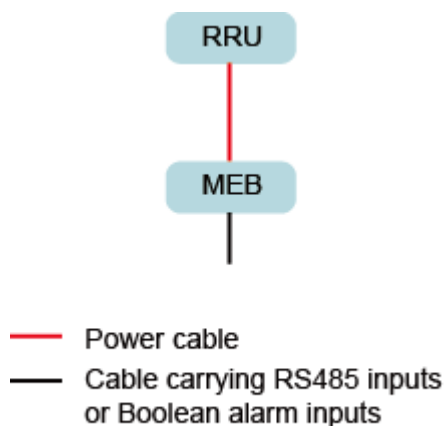
NOTE

The hardware board of the TCU in a TMC is the CCUB/CCUC.

The following provides the monitoring principles when an RRU transmits monitoring information through power cables in two scenarios:

If a device to be monitored can transmit monitoring information through RS485 ports, the RRU connects to a monitoring extension box (MEB) using a power cable and the MEB provides a monitoring port to connect to the device. In this case, the RRU and MEB serve as an entity externally.

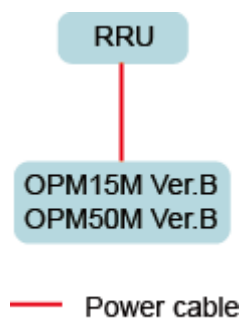
Figure 4-23 Monitoring principles when an RRU connects to a device to be monitored using a power cable (1)



PIH19C0046

If a device such as the OPM15M (Ver.B) or the OPM50M (Ver.B) to be monitored can transmit monitoring information through a power cable, the RRU connects to the device using a power cable and no MEB is required.

Figure 4-24 Monitoring principles when an RRU connects to a device to be monitored using a power cable (2)



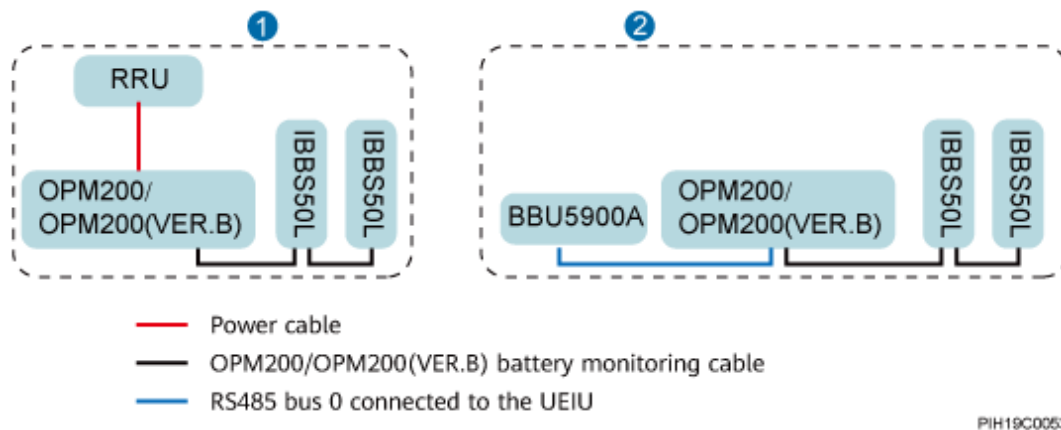
PIH19C0050

In SRAN10.1 and SRAN11.1, the software of the base station does not support MEB configuration. The MEB provides physical links only and does not require configuration. The RS485 and dry contact functions are configured on the RRU connecting to the MEB using a power cable. In this case, the MEB does not support the alarm, log, inventory, and upgrade functions.

In SRAN12.1 and later versions, the MEB must be configured on a newly deployed base station in order to support the alarm, log, inventory, and upgrade functions. You are advised to configure the dry contact alarms on the MEB. The MEB can be configured for only one RAT. When an MEB is configured, the dry contacts configured on the manager RRU do not take effect.

During a base station upgrade from SRAN10.1 or SRAN11.1 to SRAN12.1, MEB configuration does not require modification.

Figure 4-25 Monitoring principles of the DBS5900A powered by the OPM200/OPM200 (Ver.B)



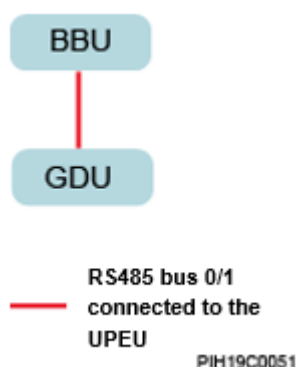
PIH19C0053

NOTE

- The OPM200/OPM200 (Ver.B) is preferentially monitored by the remotely installed RRU that is powered by the OPM200/OPM200 (Ver.B). When the BBU5900A is used for monitoring, the monitoring cable for the OPM200/OPM200 (Ver.B) is connected to the MON0/MON1 monitoring port on the UEIUd in the BBU5900A.
- The RS485 bus 0 is required for the RRU that does not support the power cable connection shown in the preceding figure.

The GDU cannot be cascaded and must be directly connected to the BBU. The GDU in the cabinet can be connected to the BBU for management through the RS485 communication link.

Figure 4-26 GDU monitoring principle



PIH19C0051

In SRAN17.0 and later versions, the GDU can be configured but not on the GBTS.

4.1.3.5 DBS5900E Monitoring Networking

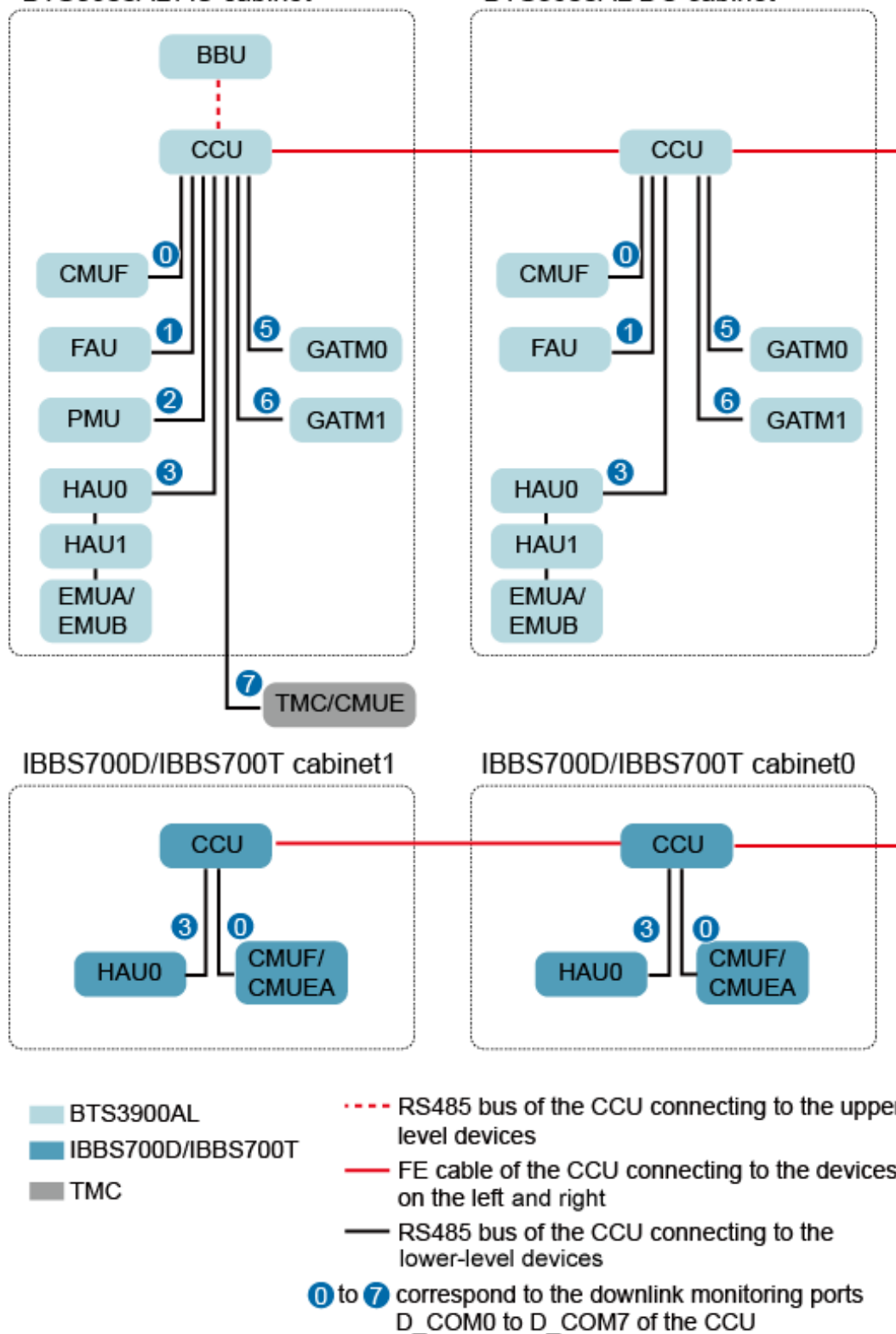
The DBS5900E is an outdoor base station and is used together with the BBU5900C/BookBBU5901/BookBBU5901a. As there is no monitoring port on the BBU5900C/BookBBU5901/BookBBU5901a, the DBS5900E supports only RRU remote monitoring. For details, see [Example of Configurations in a DBS5900E](#).

4.1.3.6 BTS3900AL Monitoring Networking

Figure 4-27 shows the monitoring principles of the BTS3900AL cabinet. To monitor boards in cabinets, the BTS3900AL uses CCUs. The CMUF, FAU, and PMU communicate with the CCUs' downlink RS485 ports through an RS485 bus. Each CCU reports alarms to the BBU through the uplink RS485 port. The GATM is GBTS-specific and is not used for the eGBTS/NodeB/eNodeB/gNodeB.

If TMC, IBBS700D/IBBS700T, and IBBS300D/IBBS300T cabinets are not configured, ignore the boards in these cabinets.

Figure 4-27 Monitoring principles of the single-mode or dual-mode BTS3900AL
BTS3900AL AC cabinet



PIH19C0016

 NOTE

Unlike that in an IBBS700D/IBBS700T cabinet, the CCU and CMUF/CMUEA are integrated into a CCU01D-03 in an IBBS300D/IBBS300T cabinet. However, the CCU and TCU must be both configured, and only the CCU reports inventory information. Ignore the cable connections between the CCU and the CMUF/CMUEA. In addition, connect the HAU to the D_COM0 port on the CCU01D-03.

4.1.3.7 BTS3900C Monitoring Networking

The monitoring principles of the BTS3900C are the same as those of the DBS3900 configured with an OMB. For details, see [Table 4-14](#).

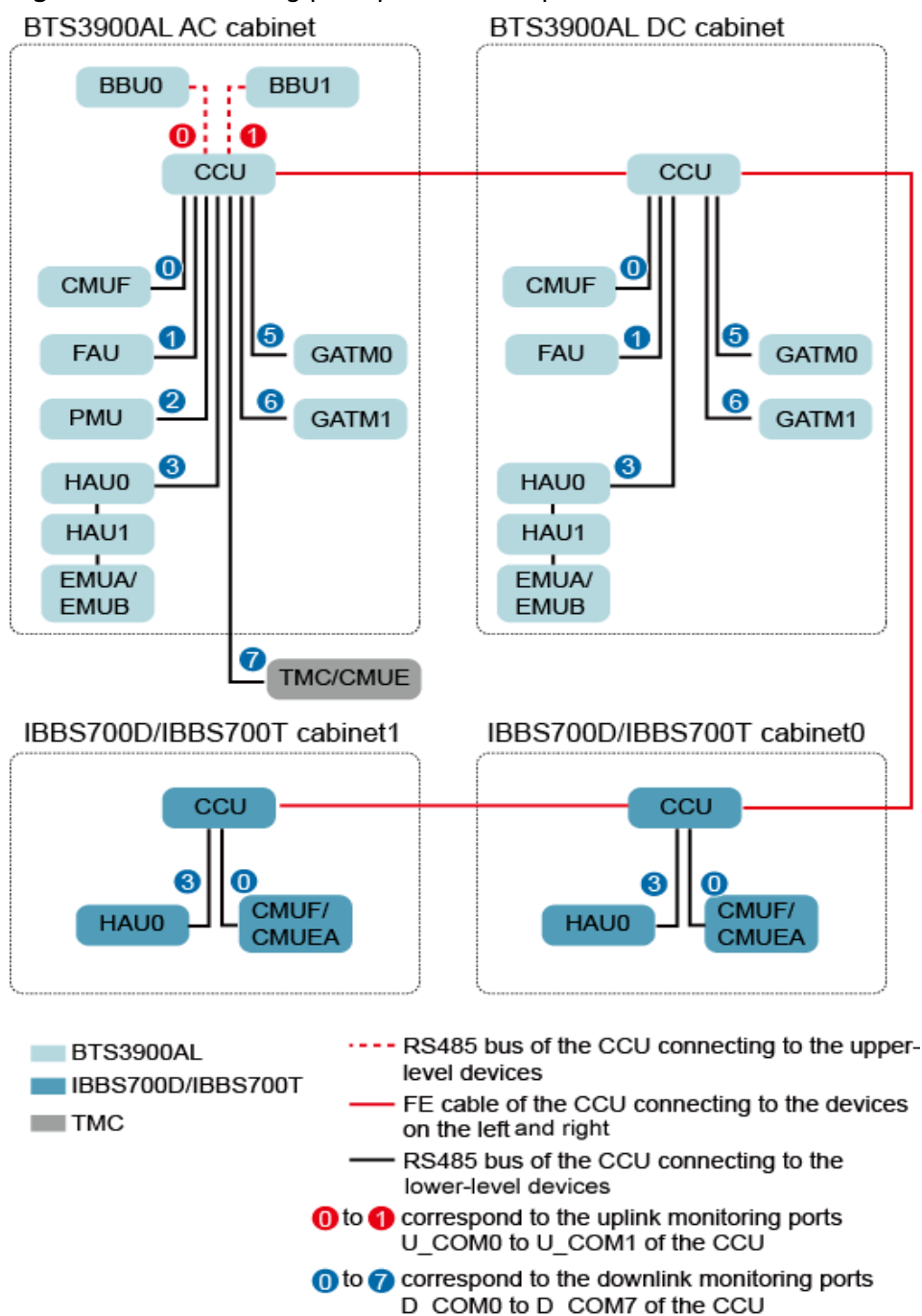
4.1.4 Monitoring Networking for a Triple-mode or Quadruple-mode Base Station

For a newly deployed triple-mode base station with only one BBU, co-MPT G*U*L or separate-MPT G[U*L] can be implemented in the BBU to support GSM, UMTS, and LTE. The monitoring principles in this scenario are the same as those of single-mode or dual-mode base stations. For details, see [4.1.3 Monitoring Networking for a Single-mode/Dual-mode Base Station](#).

For a newly deployed triple-mode base station or a base station evolving from single-mode or dual-mode to triple-mode, if two BBUs are configured to support GSM, UMTS, and LTE simultaneously, the monitoring principles are as follows:

- All the monitored devices of the BTS3900, BTS3900L, BTS3900A, BTS3012 (Ver.D_Z), BTS3012AE (Ver.D_Z), and DBS3900 (not configured with a TP48600A) are connected only to BBU0 (root BBU). The extension BBU does not provide monitoring management.
- [Figure 4-28](#) shows the monitoring principles of the BTS3900AL where two BBUs are configured. BBU0 and BBU1 are connected to the uplink monitoring ports U_COM0 and U_COM1 of the CCU, respectively. The monitoring principles of the DBS3900 configured with the TP48600A are similar.
- The monitoring board in an IBBS700D cabinet is CMUEA and the monitoring board in an IBBS700T cabinet is CMUF.

Figure 4-28 Monitoring principles of the triple-mode BTS3900AL



PIH19C0020

For a newly deployed quadruple-mode base station, two BBUs must be configured to support the four RATs simultaneously. In this scenario, the monitoring principles are the same as those of triple-mode base stations with two BBUs.

4.1.5 Networking for Customized Alarms

You can customize alarms to monitor external devices.

- Boolean signal input
Connect the monitored devices to the BBU/RRU/EMU/EMUB that has Boolean alarm signal input ports.
- Analog signal input
Connect the monitored devices to the EMU/EMUB that has analog alarm signal input ports.
- Boolean signal output
Connect the monitored devices to the EMU/EMUB that has Boolean signal output ports.

Environment monitoring alarms are common customized alarms. They are generated after sensors installed outdoors or in the equipment room receive alarm signals. Some base stations have sensors that are preinstalled before delivery, as shown in [Table 4-16](#), [Table 4-17](#), [Table 4-18](#), and [Table 4-19](#).

Table 4-16 Sensors preinstalled on base stations before delivery (1)

Base Station Type	Cabinet	Preinstalled Sensor
BTS3900A/ DBS3900 ⁽¹⁾ (2)	APM30H (Ver.C)	Door status sensor, which is connected to the GATE port on an HPMI
	APM30H (Ver.D)/APM30H (Ver.D_B)/APM30H (Ver.D_A1)/APM30H (Ver.D_A2)	Door status sensor, which is connected to the GATE port on a PMU
	APM30H (Ver.E)	Door status sensor, which is connected to the GATE port on a CCUB/CCUC
	RFC	• Door status sensor, which is connected to the GATE port on a CMUE/CMUEA/CMUH • Temperature sensor, which is connected to the TEM port on a CMUE/CMUEA/CMUH
	TMC11H	Door status sensor, which is connected to the GATE port on a CMUE/CMUEA/CCUB/CCUC
	IBBS200D/IBBS200T	• Door status sensor, which is connected to the GATE port on a CMUE/CMUEA/CMUH • Temperature sensor for the battery, which is connected to the TEM_BAT port on a CMUE/CMUEA/CMUH • BAU02D, which is connected to the IN0, IN1, and IN2 ports on a CMUE/CMUEA/CMUH

Base Station Type	Cabinet	Preinstalled Sensor
	IBBS700D/IBBS700T	- Door status sensor, which is connected to the GATE port on a CMUEA or CMUF - Temperature sensor for the battery, which is connected to the TEM_BAT port on a CMUEA or CMUF
	IBBS300D/IBBS300T	- Door status sensor, which is connected to the GATE port on a CCU01D-03 - Temperature sensor for the battery, which is connected to the TEMP1 port on a CCU01D-03
	IBBS20D	Door status sensor, which is connected to the GATE port on a CMUG

Table 4-17 Sensors preinstalled on base stations before delivery (2)

Base Station Type	Cabinet	Preinstalled Sensor
BTS3900/ BTS3900L	BTS3900/BTS3900L	-
BTS3900AL ⁽⁵⁾)	BTS3900AL (AC)/ BTS3900AL (DC)	- Door status sensor, which is connected to the GATE port on a CCU - Return air temperature sensor for HEX inner circulation, which is connected to the TEM0 port on a CMUF - Intake air temperature sensor for the RF compartment, which is connected to the TEM0 port on an FAU - Supply air temperature sensor for HEX inner circulation, which is connected to the TEMP1 port on a CMUF
	TMC11H	Door status sensor, which is connected to the GATE port on a CMUE/CMUEA
	IBBS700D/IBBS700T	- Door status sensor, which is connected to the GATE port on a CMUF - Temperature sensor for the battery, which is connected to the TEM_BAT port on a CMUEA or CMUF

Base Station Type	Cabinet	Preinstalled Sensor
	IBBS300D/IBBS300T	- Door status sensor, which is connected to the GATE port on a CCU01D-03 - Temperature sensor for the battery, which is connected to the TEMP1 port on a CCU01D-03
BTS3900C	BTS3900C	-
BTS3012 (Ver.D_Z)	BTS3012 (Ver.D_Z)	-
BTS3012AE (Ver.D_Z)	BTS3012AE (Ver.D_Z) AC	- Door status sensor, which is connected to the GATE port on a PMU - Temperature sensor for the battery, which is connected to the BAT_TEMP port on a PMU
	BTS3012AE (Ver.D_Z) DC	Door status sensor, which is connected to the GATE port on a CMUEA
BTS5900	BTS3900 (Ver.E_A~D)/ BTS5900 (Ver.A) DC	-

Table 4-18 Sensors preinstalled on base stations before delivery (3)

Base Station Type	Cabinet	Preinstalled Sensor
BTS5900L	BTS3900L (Ver.E_B~D)/ BTS5900L (Ver.A) DC	-
DBS5900	ILC29 (Ver.E)	-
	IMB05	-
	APM30H (Ver.E)	Door status sensor, which is connected to the GATE port on a CCUB/CCUC
	RFC (Ver.E)	- Door status sensor, which is connected to the GATE port on a CMUH - Temperature sensor, which is connected to the TEM port on a CMUH
	TMC11H (Ver.E)	Door status sensor, which is connected to the GATE port on a CCUB/CCUC

Base Station Type	Cabinet	Preinstalled Sensor
	IBBS200D (Ver.E)/ IBBS200T (Ver.E) BBC5200D (Ver.A)/ BBC5200T (Ver.A) BBC5300D (Ver.A)/ BBC5300T (Ver.A) BBC5200D-L(Ver.A)	- Door status sensor, which is connected to the GATE port on a CMUH - Temperature sensor for the battery, which is connected to the TEM_BAT port on a CMUH - BAU02D, which is connected to the IN0, IN1, and IN2 ports on a CMUH
	IBBS300D/IBBS300T	- Door status sensor, which is connected to the GATE port on a CCU01D-03 - Temperature sensor for the battery, which is connected to the TEMP1 port on a CCU01D-03
	APM5930(DC) (Ver.A)	Door status sensor, which is connected to the GATE port on a CCUC
	RFC5906 (Ver.A) ⁽¹⁾	Door status sensor⁽¹⁾, which is connected to the GATE port on a CMUH
DBS5900	BBC5600D(Ver.A)/ BBC5600D-V(Ver.A) BBC5600A(Ver.A)/ BBC5600A-V(Ver.A)	- Door status sensor, which is connected to the GATE port on a CMUH - Temperature sensor for the battery, which is connected to the TEM port on a CMUH
DBS5900	APM5900H(Ver.A)	Door status sensor, which is connected to the GATE port on a CCUC
DBS5900	APM5900H-L(Ver.A)	Door status sensor, which is connected to the GATE port on a CMUH

 NOTE

⁽¹⁾: The door status sensor is installed only in the RFC5906 complying with UK standards.

Table 4-19 Sensors preinstalled on base stations before delivery (4)

Base Station Type	Cabinet	Preinstalled Sensor
BTS5900A	APM30H (Ver.E)	Door status sensor, which is connected to the GATE port on a CCUB/CCUC
	RFC (Ver.E)	- Door status sensor, which is connected to the GATE port on a CMUH - Temperature sensor, which is connected to the TEM port on a CMUH

Base Station Type	Cabinet	Preinstalled Sensor
	TMC11H (Ver.E)	Door status sensor, which is connected to the GATE port on a CCUB/CCUC
	IBBS200D (Ver.E)/ IBBS200T (Ver.E) BBC5200D (Ver.A)/ BBC5200T (Ver.A) BBC5300D (Ver.A)/ BBC5300T (Ver.A) BBC5200D-L(Ver.A)	- Door status sensor, which is connected to the GATE port on a CMUH - Temperature sensor for the battery, which is connected to the TEM_BAT port on a CMUH - BAU02D, which is connected to the IN0, IN1, and IN2 ports on a CMUH
	IBBS300D/IBBS300T	- Door status sensor, which is connected to the GATE port on a CCU01D-03 - Temperature sensor for the battery, which is connected to the TEMP1 port on a CCU01D-03
	APM5930(AC) (Ver.A)	Door status sensor, which is connected to the GATE port on a CCUC
	RFC5906 (Ver.A) ⁽⁷⁾	Door status sensor, which is connected to the GATE port on a CMUH
BTS5900A	BBC5600D(Ver.A)/ BBC5600D-V(Ver.A) BBC5600A(Ver.A)/ BBC5600A-V(Ver.A)	- Door status sensor, which is connected to the GATE port on a CMUH - Temperature sensor for the battery, which is connected to the TEM port on a CMUH
BTS5900A	APM5950H(Ver.A)	Door status sensor, which is connected to the GATE port on a PMIE
BTS5900A	APM5950H-L(Ver.A)	Door status sensor, which is connected to the Door port on a PMIE

 NOTE

1. The DBS3900 configured with the OMB or IMB does not have preinstalled sensors. The BBU3910A configured with the IBBS20D cabinet has a preinstalled door status sensor and does not have one when the BBU3910A is not configured with the IBBS20D cabinet.
2. The DBS3900 may also be configured with the TP48600A cabinet. For details about the sensor configuration in this cabinet, see *TP48600A-H17B1 User Manual*.
3. In the BTS3900A (Ver.C), the sensors are connected to the CMUE. In the BTS3900A (Ver.D), the sensors are connected to the CMUEA. In the BTS3900A (Ver.E), the sensors are connected to the CMUH.
4. In the TMC11H (Ver.C), the sensors are connected to the CMUE. In the TMC11H (Ver.D), the sensors are connected to the CMUEA. In the TMC11H (Ver.E), the sensors are connected to the CCUB/CCUC.
5. The BTS3900AL site may also be configured with the TP48600A, IBBS700D, IBBS700T, IBBS300D, and IBBS300T. For details about the sensor configurations of these cabinets, see the *TP48600A-H17B1 User Manual* and *3900 & 5900 Series Base Station Cabinet Description*.
6. Most small cell products use dry contact inputs and customized alarms for monitoring.
7. The door status sensor is installed only in the RFC5906 complying with UK standards.

4.2 Network Analysis

4.2.1 Benefits

None

4.2.2 Impacts

None

4.3 Requirements

4.3.1 Licenses

There are no license requirements for basic functions.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

None

4.3.3 Hardware

Software configuration varies with hardware configuration and monitoring cable connections among boards. Before you configure software, collect information about the following devices based on the actual hardware configuration or hardware configuration plan:

- Monitoring managers: BBU, CCU, and RRU. Plan the manager of each board according to [4.1 Principles](#).
- Monitoring device: see [Mapping Table](#). The boards that are not listed in the tables are configured by default. If the GATM board is installed in a GBTS, the board must be configured in any scenario listed in the corresponding table. The logical name of GATM is GATM.

4.3.4 Others

None

4.4 Operation and Maintenance (eGBTS/NodeB/eNodeB/gNodeB)

4.4.1 When to Use

It is recommended that the monitoring management feature be enabled when you intend to monitor power and fan units in a base station. If auxiliary devices with monitoring functions are configured for a base station, it is recommended that customized alarm function be enabled.

4.4.2 Planning

RF Planning

None

Network Planning

None

Hardware Planning

Configuration Planning for Monitoring Devices

The EMU/EMUB is required for configuring customized alarms on base stations. [Table 4-20](#) and [Table 4-21](#) describe the hardware planning of equipment related to customized alarms. The logical names in tables are the names of devices displayed on the configuration tool interface.

Table 4-20 Hardware planning for customized alarm configuration (1)

Base Station Type	Usage Scenario	Required Device	Logical Name
BTS3900 (Ver.C) BTS3900 (Ver.D) BTS3900 (Ver.D_A) BTS3900 (Ver.D_B) BTS3900L (Ver.C) BTS3900L (Ver.D) BTS3900L (Ver.D_B) BTS5900 BTS5900L	1 to 16 Boolean alarm inputs	_(1)	-
BTS3900 (Ver.C) BTS3900 (Ver.D) BTS3900 (Ver.D_A) BTS3900 (Ver.D_B) BTS3900L (Ver.C) BTS3900L (Ver.D) BTS3900L (Ver.D_B) BTS5900 BTS5900L	17 to 32 Boolean alarm inputs	EMU ⁽²⁾	EMU
	1 to 6 Boolean alarm outputs	EMU	EMU
	RRU remote monitoring	MEB or not required ⁽³⁾	(1)
BTS3900 (Ver.C) BTS3900 (Ver.D) BTS3900 (Ver.D_A) BTS3900 (Ver.D_B) BTS3900L (Ver.C) BTS3900L (Ver.D) BTS3900L (Ver.D_B) BTS5900 BTS5900L	BBU3910A, 0 to 5 ⁽⁵⁾ Boolean alarm inputs	_(5)	_(5)
BTS3900 (Ver.C) BTS3900 (Ver.D) BTS3900 (Ver.D_A) BTS3900 (Ver.D_B) BTS3900L (Ver.C) BTS3900L (Ver.D) BTS3900L (Ver.D_B) BTS5900 BTS5900L	BBU5900A, 0 to 6 Boolean alarm inputs	-	-
	1 to 4 analog alarm inputs	EMU	EMU

Base Station Type	Usage Scenario	Required Device	Logical Name
BTS3900C (Ver.C) ⁽⁴⁾	1 to 16 Boolean alarm inputs	_(1)	-

Table 4-21 Hardware planning for customized alarm configuration (2)

Base Station Type	Usage Scenario	Required Device	Logical Name
BTS3900A (Ver.C) BTS3900A (Ver.D) BTS3900A (Ver.E) DBS3900 BTS3900AL (Ver.A) BTS3900A (Ver.D_B) BTS3900A (Ver.D_A1) BTS3900A (Ver.D_A2) DBS5900 BTS5900A	1 to 16 Boolean alarm inputs	_(1)	-
BTS3900A (Ver.C) BTS3900A (Ver.D)	17 to 32 Boolean alarm inputs	EMUB ⁽²⁾	EMU
BTS3900A (Ver.E) DBS3900 BTS3900AL (Ver.A) BTS3900A (Ver.D_B) BTS3900A (Ver.D_A1) BTS3900A (Ver.D_A2) DBS5900 BTS5900A	1 to 4 remote Boolean alarm inputs	MEB or not required ⁽³⁾	-
BTS3900A (Ver.C) BTS3900A (Ver.D) BTS3900A (Ver.E) DBS3900 BTS3900AL (Ver.A) BTS3900A (Ver.D_B) BTS3900A (Ver.D_A1) BTS3900A (Ver.D_A2) DBS5900 BTS5900A	1 to 6 Boolean alarm outputs	EMUB	EMU

Base Station Type	Usage Scenario	Required Device	Logical Name
BTS3900A (Ver.C) BTS3900A (Ver.D)	1 to 4 analog alarm inputs	EMUB	EMU
BTS3900A (Ver.E) DBS3900 BTS3900AL (Ver.A) BTS3900A (Ver.D_B) BTS3900A (Ver.D_A1) BTS3900A (Ver.D_A2) DBS5900 BTS5900A	BBU5900A, 0 to 6 Boolean alarm inputs	-	-

NOTE

1. When 1 to 16 customized Boolean inputs are required, the UPEU and UEIU boards can be configured in the BBU to meet the requirements. You are not advised to configure customized alarms for the Boolean alarm ports on the FMU, PMU, CMUE, CMUF, CMUEA, CMUH, and CCU. If you need to use these ports, contact Huawei technical support engineers.
2. The EMU and EMUB support 32 Boolean inputs and 4 analog inputs. For details about the monitoring ports on them, see *EMU User Guide* and *EMUB User Guide*. You are advised to preferentially use the EMUB.
3. Some RRUs support 2 or 4 Boolean inputs. Some RRUs can connect to an MEB and the MEB supports 6 Boolean inputs. For details, see section "RRU Ports" in the related RRU hardware description.
4. The EMU and EMUB cannot be installed on a BTS3900C (Ver.C).
5. If a BBU3910A configured in a DBS3900 uses AC power supplies and an OPM50M is connected to the EXT_ALM port, the EXT_ALM_A port on the OPM50M can provide 4 Boolean inputs for the BBU3910A when no battery cabinet can provide backup power.

Requirements for Boolean Alarm Inputs

Consider the following requirements while planning Boolean alarm inputs:

- Base station's Boolean alarm inputs must be dry contact alarm inputs.
- It is recommended that optical coupling isolation or relay isolation Boolean inputs be used. When either of the inputs is used, open circuit indicates alarm reporting, and short circuit indicates no alarm.
- It is not recommended that the voltage signals be used as the BBU Boolean inputs. If voltage signal alarms are used, ensure that alarm devices and alarm input devices (BBU, EMU, or EMUB) are properly grounded. When the BBU input alarms are of the voltage type, 0 V to 0.7 V indicate low voltage, and 3 V to 5 V indicate high voltage. When the EMU/EMUB input alarms are of the voltage type, 0 V to 0.7 V indicate low voltage, and 5 V to 12 V indicate high voltage.

4.4.3 Data Configuration

The scenarios for configuring monitoring management can be used together. If they are used together, data for all these scenarios must be configured. There is no

fixed sequence between the configuration scenarios. When the EMU/EMUB is installed for configuring customized alarms, configure the EMU, and then the **ALMPORT** or **OUTPORT** MO.

 NOTE

After adding a monitoring device, it must be configured. Otherwise, its monitoring function may be affected.

4.4.3.1 Data Preparation

4.4.3.1.1 MO Setting Notes

The following scenarios can be used together. If they are used together, data for all these scenarios must be prepared.

Scenario 1: Configuring the PMU

For the parameter settings for the PMU, see *Power Supply Management*.

Scenario 2: Configuring the FMU

The following table describes the parameter settings for the FMU.

Table 4-22 Key parameters to be set in the **FMU** MO

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	FMU.CN (LTE eNodeB, 5G gNodeB)	This parameter specifies the number of the cabinet housing the FMU. If the number of the RFC housing the FMU is 1, set this parameter to 1 .
Subrack No.	FMU.SRN (LTE eNodeB, 5G gNodeB)	This parameter specifies the number of the subrack housing the FMU. Set this parameter to 11 or 12 .
Slot No.	FMU.SN (LTE eNodeB, 5G gNodeB)	This parameter specifies the number of the slot housing the FMU. Set this parameter to the default value 0 .
Manager Cabinet No.	FMU.MCN (LTE eNodeB, 5G gNodeB)	The manager of the board is generally a BBU or CCU that directly communicates with the board. Set this parameter to the cabinet number of the BBU or CCU.
Manager Subrack No.	FMU.MSRN (LTE eNodeB, 5G gNodeB)	The manager of the board is generally a BBU or CCU. Set this parameter to the subrack number of the BBU or CCU.

Parameter Name	Parameter ID	Setting Notes
Manager Port No.	FMU.MPN (LTE eNodeB, 5G gNodeB)	This parameter specifies the number of the monitoring port on the manager to which the board is connected. If the FMU is connected to the BBU: • UPEU/UEIU (MON1): 1 • UPEU/UEIU (MON0): 0 If the FMU is connected to the CCU, it must be connected to port D_COM1. Set this parameter to 1.
Address	FMU.ADDR (LTE eNodeB, 5G gNodeB)	This parameter specifies the communication address of the FMU. A maximum of two FMUs can be configured on an RS485 bus. Set this parameter as follows: • Upper-level FMU in two cascaded FMUs: 14 • Lower-level FMU in two cascaded FMUs: 15
Special Boolean Alarm Flag	FMU.SBAF (LTE eNodeB, 5G gNodeB)	This parameter specifies whether to report a dedicated Boolean alarm. If the shield flag for a Boolean alarm is selected, the Boolean alarm cannot be reported. It is recommended that the FMU not be connected to external sensors and all sensors be disabled. If you want to use the sensors, contact Huawei technical support. Retain the default settings, which mean that all sensors are disabled.
Smart Temperature Control	FMU.STC (LTE eNodeB, 5G gNodeB)	This parameter specifies whether to enable smart temperature control. When this parameter is set to ENABLE, the cabinet adjusts the fan speed according to the carrier temperature. In this mode, the fan noise is low. When this parameter is set to DISABLE, the cabinet adjusts the fan speed according to the temperature of the air intake vent. In this mode, the fan noise is high. ENABLE(Enable) is recommended.

Parameter Name	Parameter ID	Setting Notes
Temperature Control Mode	FMU.TCMODE (LTE eNodeB, 5G gNodeB)	This parameter specifies which mode is adopted to adjust the fan speed. The value range is 0 to 65535. 0: default mode. This mode achieves a balance among system power consumption, noise, and temperature control. It applies to most scenarios. 1: low noise mode. This mode brings low noise and applies to scenarios with strict requirements for noise control. 2: low temperature rise mode. This mode applies to high-temperature areas. When this mode is used, power consumption and noise are higher than those in the default mode but heat dissipation effect is better than that in the default mode. Other values: Before using other modes, run the DSP BTSTCMODE command to query the temperature control modes and features that the base station supports. If the query result does not contain a certain mode, do not configure this mode. This parameter is valid only when CFGFLAG is set to YES.
Temperature Alarm Detection Enh Switch	FMU.TEMPERATUREALMDETENHSW	This parameter specifies whether to enable the alarm detection enhancement function for the temperature exception of the air exhaust vent and air intake vent of an FMU. If this switch is turned on, the alarm threshold of the temperature difference between the air exhaust vent and air intake vent and the upper limit of the temperature alarm for the air intake vent can be configured. The system detects alarms based on the configured thresholds. Recommended value: OFF(OFF)
Temperature Delta Alarm Threshold	FMU.TEMPERATUREDELTAALMTHLD	This parameter specifies the alarm threshold of the temperature difference between the air exhaust vent and air intake vent. When the temperature difference between the air exhaust vent and air intake vent is higher than this threshold, ALM-25655 Cabinet Air Outlet Temperature Unacceptable is reported. This parameter is valid when the temperature alarm detection enhancement switch is turned on.

Parameter Name	Parameter ID	Setting Notes
Inlet Temperature Alarm Threshold	FMU.INLETTEMPERATUREALMTHLD	This parameter specifies the upper limit of the temperature alarm for the air intake vent. When the temperature of the air intake vent is higher than this threshold, ALM-25656 Cabinet Air Inlet Temperature Unacceptable is reported. This parameter is valid when the temperature alarm detection enhancement switch is turned on.

Scenario 3: Configuring the TCU

The following table describes the parameter settings for the TCU.

Table 4-23 Key parameters to be set in the TCU MO

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	TCU.CN (LTE eNodeB, 5G gNodeB)	This parameter specifies the number of the cabinet housing the TCU. If the number of the APM30H housing the TCU is 0, set this parameter to 0 .
Subrack No.	TCU.SRN (LTE eNodeB, 5G gNodeB)	This parameter specifies the number of the subrack housing the TCU. Set this parameter to 8 or 14 .
Slot No.	TCU.SN (LTE eNodeB, 5G gNodeB)	This parameter specifies the number of the slot housing the TCU. Value range: 0 to 3 If the TCU is installed in the BTS3900AL AC, BTS3900AL DC, IBBS700D, or IBBS700T cabinet, set this parameter as follows: • Set the slot number of the CMUF to 0. • Set the slot number of HAU0 to 1. • Set the slot number of HAU1 to 2. If the TCU is installed in the TP48600A cabinet, set this parameter as follows: • Set the slot number of the CMUF for cabinet temperature control to 0. • Set the slot number of the CMUF for battery cabinet temperature control to 1. • Set the slot number of HAU0 to 2. • Set the slot number of HAU1 to 3. If the TCU is installed in other cabinets, set this parameter to 0 .

Parameter Name	Parameter ID	Setting Notes
Manager Cabinet No.	<i>TCU.MCN (LTE eNodeB, 5G gNodeB)</i>	The manager is generally a BBU, CCU, or RRU. Set this parameter to the cabinet number of the BBU, CCU, or RRU.
Manager Subrack No.	<i>TCU.MSRN (LTE eNodeB, 5G gNodeB)</i>	The manager is generally a BBU, CCU, or RRU. Set this parameter to the subrack number of the BBU, CCU, or RRU.
Manager Port No.	<i>TCU.MPN (LTE eNodeB, 5G gNodeB)</i>	This parameter specifies the number of the monitoring port on the manager to which the board is connected. If the TCU is connected to the BBU: • UPEU/UEIU (MON1): 1 • UPEU/UEIU (MON0): 0 If the TCU is connected to the CCU: • The CMUF must be connected to port D_COM0 on the CCU. In this case, set this parameter to 0. • The FAU0 or FAU1 must be connected to port D_COM3 on the CCU. In this case, set this parameter to 3. If the TCU is connected to the RRU, set this parameter to 0. If the TCU is connected to the BBU3910A, set this parameter to 0.

Parameter Name	Parameter ID	Setting Notes
Address	TCU.ADDR (LTE eNodeB, 5G gNodeB)	In the APM30H or TMC11H cabinet, a maximum of two TCUs can be configured on an RS485 bus. Set this parameter as follows: • Upper-level TCU in two cascaded TCUs: 7 • Lower-level TCU in two cascaded TCUs: 6 In the IBBS200D/IBBS200T cabinet, a maximum of four TCUs can be configured on an RS485 bus. • If the upper-level TCU is managed by the PMU whose communication address is 3, set this parameter to 23. If the lower-level TCU is managed by this PMU, set this parameter to 24. • If the upper-level TCU is managed by the PMU whose communication address is 4, set this parameter to 25. If the lower-level TCU is managed by this PMU, set this parameter to 26. If the HAU is installed, a maximum of two TCUs can be configured on an RS485 bus. Set this parameter as follows: • Upper-level TCU in two cascaded TCUs: 28 • Lower-level TCU in two cascaded TCUs: 29 If the CMUF is installed, only one TCU can be configured on an RS485 bus. Set this parameter to 7. In the IBBS20D configured together with the BBU3910A, a maximum of six TCUs can be configured on an RS485 bus. – If the upper-level TCU is managed by the OPM50M whose communication address is 3, set this parameter to 23. If the lower-level TCU is managed by this OPM50M, set this parameter to 24. – If the upper-level TCU is managed by the OPM50M whose communication address is 4, set this parameter to 25. If the lower-level TCU is managed by this OPM50M, set this parameter to 26. If the upper-level TCU is managed by the OPM50M whose communication address is 9, set this parameter to 28. If the lower-level TCU is managed by this OPM50M, set this parameter to 29.

Parameter Name	Parameter ID	Setting Notes
Temperature Alarm Lower Threshold	TCU.TLTHD (LTE eNodeB, 5G gNodeB)	This parameter specifies the lower temperature limit for the cabinet. If the temperature is below the lower limit, the base station reports ALM-25652 Cabinet Temperature Unacceptable. It is recommended that this parameter be set to -33 for the TCU in the OMB and -20 for the TCU in other cabinets.
Temperature Alarm Upper Threshold	TCU.TUTHD (LTE eNodeB, 5G gNodeB)	This parameter specifies the upper temperature limit for the cabinet. If the temperature exceeds the upper limit, the base station reports ALM-25652 Cabinet Temperature Unacceptable. - Set this parameter to 50 for the TCU in the BBC. - Set this parameter to 68 for the TCU in other cabinets.
Special Boolean Alarm Flag	TCU.SBAF (LTE eNodeB, 5G gNodeB)	This parameter specifies whether to report a dedicated Boolean alarm. If the shield flag for a Boolean alarm is selected, the Boolean alarm cannot be reported. It is recommended that the TCU not be connected to external sensors and all sensors be disabled. If you want to use the sensors, contact Huawei technical support. Retain the default settings, which mean that all sensors are disabled.

Parameter Name	Parameter ID	Setting Notes
Temperature Control Mode	TCU.TCMODE (LTE eNodeB, 5G gNodeB)	This parameter specifies which mode is adopted to adjust the fan speed. The value range is 0 to 65535. 0: default mode. This mode achieves a balance among system power consumption, noise, and temperature control. It applies to most scenarios. 1: low noise mode. This mode brings low noise and applies to scenarios with strict requirements for noise control. 2: low temperature rise mode. This mode applies to high-temperature areas. When this mode is used, power consumption and noise are higher than those in the default mode but heat dissipation effect is better than that in the default mode. Other values: Before using other modes, run the DSP BTSTCMODE command to query the temperature control modes and features that the base station supports. If the query result does not contain a certain mode, do not configure this mode. This parameter is valid only when CFGFLAG is set to YES.

Scenario 4: Configuring the CCU

The following table describes the parameter settings for the CCU.

Table 4-24 Key parameters to be set in the CCU MO

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	CCU.CN (LTE eNodeB, 5G gNodeB)	This parameter specifies the number of the cabinet housing the CCU. If the number of the BTS3900AL cabinet housing the CCU is 0, set this parameter to 0.
Subrack No.	CCU.SRN (LTE eNodeB, 5G gNodeB)	This parameter specifies the number of the subrack housing the CCU. Set this parameter to the default value 15.
Slot No.	CCU.SN (LTE eNodeB, 5G gNodeB)	This parameter specifies the number of the slot housing the CCU. Set this parameter to the default value 0.

Parameter Name	Parameter ID	Setting Notes
Manager Cabinet No.	CCU.MCN (LTE eNodeB, 5G gNodeB)	The manager of the board is generally a BBU or RRU. Set this parameter to the cabinet number of the BBU or RRU.
Manager Subrack No.	CCU.MSRN (LTE eNodeB, 5G gNodeB)	The manager of the board is generally a BBU or RRU. Set this parameter to the subrack number of the BBU or RRU.
Manager Port No.	CCU.MPN (LTE eNodeB, 5G gNodeB)	This parameter specifies the number of the serial port on the manager to which the board is connected. The CCU must be connected to port MON0 on the BBU. In this case, this parameter must be set to 0 . The CCU must be connected to port EXT_ALM on the RRU. In this case, this parameter must be set to 0 .
DirectConnectFlag	CCU.DCF (LTE eNodeB, 5G gNodeB)	This parameter specifies whether the CCU is directly connected to its manager. If the CCU is directly connected to its manager, set this parameter to YES . If the CCU is connected to its manager over a cascaded CCU, set this parameter to NO .
Cascade No.	CCU.CCN (LTE eNodeB, 5G gNodeB)	This parameter specifies the position of the CCU on a CCU chain. Set this parameter to 1 for the leftmost CCU on the chain (the CCU whose port FE_L is not connected). Set this parameter to an integer in ascending order for the other CCUs on the chain.
Connected RAT	CCU.CS (LTE eNodeB, 5G gNodeB)	This parameter specifies the working standard of the BBU that is directly connected to the CCU. Set this parameter based on the actual situation. The parameter is valid only when DCF is set to Yes .
Special Boolean Alarm Flag	CCU.SBAF (LTE eNodeB, 5G gNodeB)	This parameter specifies whether to report a dedicated Boolean alarm. If the shield flag for a Boolean alarm is selected, the Boolean alarm cannot be reported. It is recommended that the CCU not be connected to external sensors and all sensors be disabled. If you want to use the sensors, contact Huawei technical support. Retain the default settings, which mean that the water sensor and smoke sensor are disabled.

Scenario 5: Configuring Customized Alarms

To prepare data for customized alarm configuration, perform the following steps:

Step 1 Prepare EMU/MEB configuration data. Before data preparation, check whether the EMU/MEB needs to be configured by referring to [Table 4-25](#). If the EMU/MEB is required, prepare configuration data by referring to the following table. Otherwise, skip this step.

Step 2 Prepare configuration data of the alarm signal input and output ports based on the input and output requirements of customized alarms.

----End

- Prepare the configuration data for the **ALMPORT** MO by referring to [Table 4-26](#).
- Prepare the configuration data for the **OUTPORT** MO by referring to [Table 4-27](#).

Table 4-25 Key parameter to be set in the **EMU** MO

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	EMU.CN (LTE eNodeB, 5G gNodeB)	Set this parameter to the actual number of the cabinet where the EMU is located.
Subrack No.	EMU.SRN (LTE eNodeB, 5G gNodeB)	This parameter specifies the number of the subrack housing the EMU. Set this parameter to the default value.
Slot No.	EMU.SN (LTE eNodeB, 5G gNodeB)	This parameter specifies the number of the slot housing the EMU. Set this parameter to the default value.
Manager Cabinet No.	EMU.MCN (LTE eNodeB, 5G gNodeB)	The manager is generally a BBU, CCU, or RRU. Set this parameter to the cabinet number of the BBU, CCU, or RRU.
Manager Subrack No.	EMU.MSRN (LTE eNodeB, 5G gNodeB)	The manager is generally a BBU, CCU, or RRU. Set this parameter to the subrack number of the BBU, CCU, or RRU.
Manager Port No.	EMU.MPN (LTE eNodeB, 5G gNodeB)	This parameter specifies the number of the monitoring port on the manager to which the board is connected. If the EMU is connected to the BBU: • UPEU/UEIU (MON1): 1 • UPEU/UEIU (MON0): 0 If the EMU is connected to the CCU, it must be connected to port D_COM3. Set this parameter to 3 . If the module is connected to the RRU, set this parameter to 0 .

Parameter Name	Parameter ID	Setting Notes
Address	EMU.ADDR (LTE eNodeB, 5G gNodeB)	This parameter specifies the communication address of the EMU. Set this parameter to 2 .
Temperature Alarm Lower Threshold	EMU.TLTHD (LTE eNodeB, 5G gNodeB)	This parameter needs to be configured only when TS_DISABLE in SAAF is cleared. The default value is recommended for this parameter.
Temperature Alarm Upper Threshold	EMU.TUTHD (LTE eNodeB, 5G gNodeB)	This parameter needs to be configured only when TS_DISABLE in SAAF is cleared. The default value is recommended for this parameter.
Humidity Alarm Lower Threshold	EMU.HLTHD (LTE eNodeB, 5G gNodeB)	This parameter needs to be configured only when HS_DISABLE in SAAF is cleared. The default value is recommended for this parameter.
Humidity Alarm Upper Threshold	EMU.HUTHD (LTE eNodeB, 5G gNodeB)	This parameter needs to be configured only when HS_DISABLE in SAAF is cleared. The default value is recommended for this parameter.
Special Analog Alarm Flag	EMU.SAAF (LTE eNodeB, 5G gNodeB)	Set this parameter as follows: - The -48 V alarm is used to check whether the input voltage for the cabinet is normal. When the input voltage for the cabinet is -48 V, deselect the 48V_DISABLE(-48 Voltage Disabled) option. - Set this parameter based on the types of sensors connected to the EMU.
Special Boolean Alarm Flag	EMU.SBAF (LTE eNodeB, 5G gNodeB)	It is recommended that this parameter be set based on the types of sensors connected to the EMU.

Table 4-26 Key parameters to be set in the **ALMPORT** MO

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	ALMPORT.CN (LTE eNodeB, 5G gNodeB)	Set this parameter to the actual cabinet number.
Subrack No.	ALMPORT.SRN (LTE eNodeB, 5G gNodeB)	Set this parameter to the actual subrack number.
Slot No.	ALMPORT.SN (LTE eNodeB, 5G gNodeB)	Set this parameter to the actual slot number.
Port No.	ALMPORT.PN (LTE eNodeB, 5G gNodeB)	This parameter specifies the number of the port on each board as follows: UPEU/UEIU: 0 to 7 (for Boolean alarms) RRU: 0 to 3 (for Boolean alarms) EMU/EMUB: 0 to 31 (for Boolean alarms) EMU/EMUB: 32 to 35 (for analog alarms) BBU3910A: 0 to 5 (for Boolean alarms)
Switch	ALMPORT.SW (LTE eNodeB, 5G gNodeB)	Turn on this switch for the port configured with a customized alarm. OFF(Off) : indicates that the port is disabled and alarms cannot be detected. This is the default value for this parameter.
Alarm ID	ALMPORT.AID (LTE eNodeB, 5G gNodeB)	The value range of this parameter is 65033 to 65233. The alarm name, alarm severity, and OSS type corresponding to the alarm ID can be configured and saved on the MAE client.
Port Type	ALMPORT.PT (LTE eNodeB, 5G gNodeB)	This parameter specifies the type of an alarm port. Set this parameter to BOOL or VALUE . An analog port, provided by the EMU, is used to monitor the external environment that changes continuously. A digital port, provided by the UPEU, UEIU, RRU, or EMU, is used to monitor the external environment with Boolean or discrete values.

Parameter Name	Parameter ID	Setting Notes
Alarm Voltage	ALMPORT.AVOL (LTE eNodeB, 5G gNodeB)	This parameter is valid only when PT is set to BOOL. This parameter specifies whether the sensor sends the port a high-level or low-level alarm. Set this parameter based on the alarm cable connection. For details about the pin assignment for the wires of alarm cables, see the following sections: "BBU Alarm Cable" in <i>3900 & 5900 Series Base Station Cables</i> "RRU Alarm Cable" in the RRU-specific hardware description - Section "RS485 Serial Port" in <i>EMU User Guide</i> and <i>EMUB User Guide</i>
Alarm Upper Limit	ALMPORT.UL (LTE eNodeB, 5G gNodeB)	If the environment measurement value is greater than the alarm upper limit configured for the sensor, the base station reports the alarm. Environment measurement value = Measurement lower limit + (Measurement upper limit - Measurement lower limit) x (Observed current or voltage - Output lower limit)/(Output upper limit - Output lower limit). UL must be less than or equal to SMUL and greater than LL.
Alarm Lower Limit	ALMPORT.LL (LTE eNodeB, 5G gNodeB)	If the environment measurement value is below the alarm lower limit configured for the sensor, the base station reports the alarm. Environment measurement value = Measurement lower limit + (Measurement upper limit - Measurement lower limit) x (Observed current or voltage - Output lower limit)/(Output upper limit - Output lower limit). LL must be greater than or equal to SMLL and less than UL.
Sensor Type	ALMPORT.ST (LTE eNodeB, 5G gNodeB)	Set this parameter to VOLTAGE(Voltage Type) or CURRENT(Current Type) based on the external sensor specifications.

Parameter Name	Parameter ID	Setting Notes
Sensor Measurement Upper Limit	ALMPORT.SMUL (LTE eNodeB, 5G gNodeB)	This parameter is valid only when PT is set to VALUE . SMUL must be greater than or equal to UL . SMUL must be greater than SMLL .
Sensor Measurement Lower Limit	ALMPORT.SMLL (LTE eNodeB, 5G gNodeB)	This parameter is valid only when PT is set to VALUE . SMLL must be less than or equal to LL . SMLL must be less than SMUL .
Sensor Output Upper Limit	ALMPORT.SOUL (LTE eNodeB, 5G gNodeB)	This parameter is valid only when PT is set to VALUE . SOUL must be greater than SOLL .
Sensor Output Lower Limit	ALMPORT.SOLL (LTE eNodeB, 5G gNodeB)	This parameter is valid only when PT is set to VALUE . SOLL must be less than SOUL .
Detection Period	ALMPORT.DTPRD (LTE eNodeB, 5G gNodeB)	When this parameter is set to 255 , the default alarm detecting period is used. When DTPRD is not set to 255, SRN must be set to a value within the range of 60 to 254 . The AARU, RRU3801C, RRU3821E, and RRU3004 modules do not support the configuration of the detection period.

Table 4-27 Key parameters to be set in the **OUTPORT MO**

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	OUTPORT.CN (LTE eNodeB, 5G gNodeB)	Set this parameter to the actual cabinet number.
Subrack No.	OUTPORT.SRN (LTE eNodeB, 5G gNodeB)	Set this parameter to the actual subrack number.
Slot No.	OUTPORT.SN (LTE eNodeB, 5G gNodeB)	Set this parameter to the actual slot number.
Port No.	OUTPORT.PN (LTE eNodeB, 5G gNodeB)	Value range: 0 to 5 for the EMU/EMUB; 0 for the BBU3910A

Parameter Name	Parameter ID	Setting Notes
Port Name	OUTPORTNAME (<i>LTE eNodeB, 5G gNodeB</i>)	-
Port Switch	OUTPORT.SW (<i>LTE eNodeB, 5G gNodeB</i>)	This switch is turned on based on the monitoring device configuration. To enable a port, set this parameter to ON . To disable a port, set this parameter to OFF .

Scenario 6: Configuring the MEB

If the software version of the base station does not support MEB configuration, the MEB provides physical links only and does not require configuration. The RS485 and dry contact functions are configured on the RRU connecting to the MEB. In this case, the MEB does not support the alarm, log, inventory, and upgrade functions.

If the software version of the base station supports MEB configuration, the MEB must be configured in order to support the alarm, log, inventory, and upgrade functions. The MEB can be configured for only one RAT. When an MEB is configured, the dry contacts configured on the manager RRU do not take effect.

Configure an MEB by referring to [Table 4-28](#).

NOTE

If devices to be monitored connect to an MEB, the manager of the boards is the RRU which also connects to the MEB.

Table 4-28 Key parameters to be set in the MEB MO

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	MEB.CN (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter specifies the number of the cabinet housing the MEB.
Subrack No.	MEB.SRN (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter specifies the number of the subrack housing the MEB.
Slot No.	MEB.SN (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter specifies the number of the slot housing the MEB.
Manager Cabinet No.	MEB.MCN (<i>LTE eNodeB, 5G gNodeB</i>)	The manager is the RRU which connects to the board using a power cable. Set this parameter to the cabinet number of the manager.
Manager Subrack No.	MEB.MSRN (<i>LTE eNodeB, 5G gNodeB</i>)	The manager is the RRU which connects to the board using a power cable. Set this parameter to the subrack number of the manager.

Scenario 7: Configuring the GDU

The GDU can be configured for only one RAT.

- The GDU does not need to be configured when the base station software version does not support GDU configuration. In this case, the OSS cannot manage the GDU, that is, the OSS cannot report alarms, logs, and inventory data of the GDU, and does not support the remote upgrade for the GDU +SLPA.
- The GDU must be configured when the base station software version supports the GDU configuration. If the GDU is not configured, it cannot be managed by the OSS.

The following table describes the parameter settings for the GDU.

Table 4-29 Key parameters to be set in the **GDU MO**

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	GDU.CN (LTE eNodeB, 5G gNodeB)	This parameter specifies the number of the cabinet where the GDU is located.
Subrack No.	GDU.SRN (LTE eNodeB, 5G gNodeB)	This parameter specifies the number of the subrack where the GDU is located.
Slot No.	GDU.SN (LTE eNodeB, 5G gNodeB)	This parameter indicates the number of the slot where the GDU is located.
Manager Cabinet No.	GDU.MCN (LTE eNodeB, 5G gNodeB)	This parameter specifies the cabinet number of the BBU that communicates with the GDU directly (through the RS485 monitoring cable).
Manager Subrack No.	GDU.MSRN (LTE eNodeB, 5G gNodeB)	The manager is the BBU which directly communicates with the GDU (through the RS485 monitoring cable). Set this parameter to the subrack number of the BBU.
Manager Port No.	GDU.MPN (LTE eNodeB, 5G gNodeB)	This parameter specifies the number of the monitoring port on the manager to which the board is connected. • UPEU/UEIU (MON1): 1 • UPEU/UEIU (MON0): 0
Address	GDU.ADDR (LTE eNodeB, 5G gNodeB)	The communication address of the GDU is always 1 .

Parameter Name	Parameter ID	Setting Notes
Port No.	GDU.PN (LTE eNodeB, 5G gNodeB)	The GDU has two GNSS input ports. If this parameter is set to AUTO(Auto) , the GDU automatically selects a port number. If it is set to 0 or 1 , the GDU forcibly switches to the corresponding port. The value AUTO is recommended.

Reference for Configuring the Monitoring Function

For details, see [4.4.3.1.2 Reference for Configuring the Monitoring Function](#).

4.4.3.1.2 Reference for Configuring the Monitoring Function

The monitoring function should be configured based on the hardware installed in a cabinet. This section describes how to configure the monitoring function in some typical base station cabinets. Monitoring configurations involve the settings of parameters such as cabinet number, subrack number, slot number, board type, cabinet number of a board manager, subrack number of a board manager, serial port number, and communication address. For details about the settings of other parameters, see [Data Preparation](#).

NOTE

1. This section describes the configurations in the Ver.C cabinets for the BTS3900 and BTS3900L. The configurations in the Ver.D/Ver.D_B/Ver.D_A series cabinets and BTS3012 (Ver.D_Z) cabinets are the same as those in the Ver.C cabinets, except that the FMUE is replaced with the FMUEA.
2. This section also describes the configurations in the Ver.C cabinets for the BTS3900A and DBS3900. The configurations in the Ver.D/Ver.D_B/Ver.D_A1/Ver.D_A2 cabinets are the same as those in the Ver.C cabinets, except that the CMUE is replaced with the CMUEA and no HPMI is configured.
3. This section describes the configurations of a single-, dual-, or triple-mode base station with only one BBU. If two BBUs are configured for a triple-mode base station, the second BBU does not perform the monitoring function and the monitoring configuration is the same as that when one BBU is configured.
4. This section describes the maximum configuration. If a cabinet is not configured at a site, the monitoring signal cable is directly connected to the lower-level cabinet. If a device is not configured at a site, the monitoring signal cable is directly connected to the lower-level device.

Example of Configurations in a BTS3900 (-48 V DC)

[Figure 4-29](#) shows the monitoring signal cable connections in a BTS3900 (-48 V DC) housed in two cabinets.

Figure 4-29 Monitoring signal cable connections in a BTS3900 (–48 V DC)

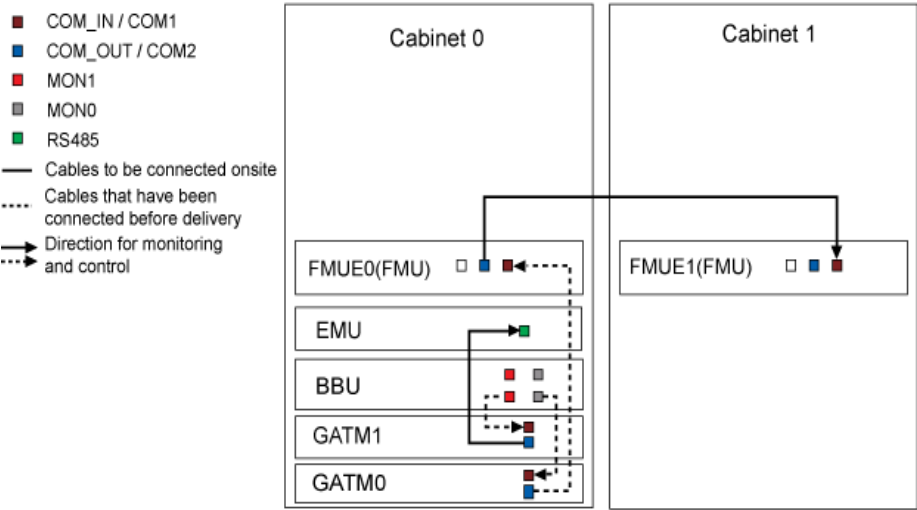


Table 4-30 describes the monitoring board configurations in this scenario.

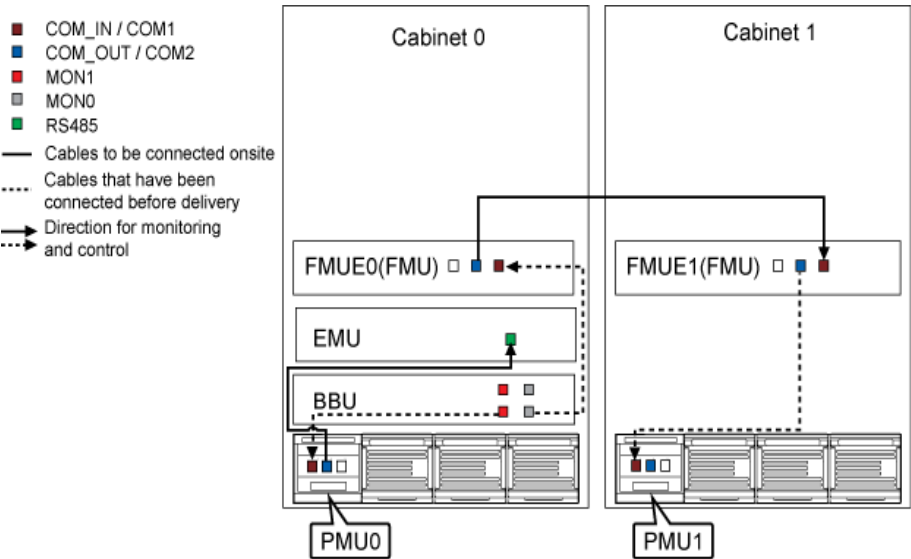
Table 4-30 Monitoring board configurations in a BTS3900 (–48 V DC)

Cabinet	Cabinet Type	CN	SRN	SN	BT	MCN	MSRN	MPN	ADDR	Note
Cabinet 0	BTS3900	0	50	0	GATM	0	0	0	22	For GBTS only
		0	51	0	GATM	0	0	1	22	For GBTS only
		0	11	0	FMU	0	0	0	14	FMUE
		0	40	0	EMU	0	0	1	2	The EMU is optional.
Cabinet 1	BTS3900	1	11	0	FMU	0	0	0	15	FMUE

Example of Configurations in a BTS3900 (AC)

Figure 4-30 shows the monitoring signal cable connections in the BTS3900 (AC) cabinets when two BTS3900 (AC) cabinets are installed.

Figure 4-30 Monitoring signal cable connections in BTS3900 (AC) cabinets



NOTE

When a BTS3900 (AC) is stacked with an IMS06, the PMU is installed in the IMS06, which makes no difference to connections of monitoring signal cables.

When two BTS3900L cabinets are deployed, each BTS3900L cabinet is stacked with an IMS06.

Table 4-31 describes the monitoring board configurations in this scenario.

Table 4-31 Monitoring board configurations in a BTS3900 (AC)

Cabinet	Cabinet Type	CN	SR N	SN	BT	MC N	MSRN	MPN	ADDR	Note
Cabinet 0	BTS3900	0	7	0	PMU	0	0	1	3	-
		0	11	0	FMU	0	0	0	14	-
		0	40	0	EMU	0	0	1	2	The EMU is optional.
Cabinet 1	BTS3900	1	11	0	FMU	0	0	0	15	-
		1	7	0	PMU	0	0	0	3	-

NOTE

The cabinet number and cabinet type of the IMS06 must be set to the same values as those of the BTS3900.

Example of Configurations in a BTS3900L (-48 V DC)

Figure 4-31 shows the monitoring signal cable connections in a BTS3900L (-48 V DC) cabinet.

Figure 4-31 Monitoring signal cable connections in a BTS3900L (-48 V DC)

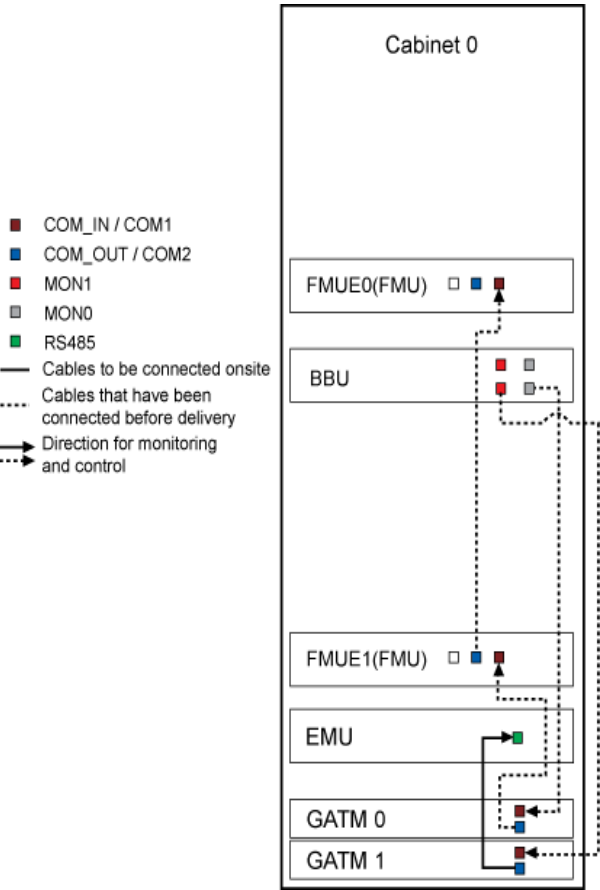


Table 4-32 describes the monitoring board configurations in this scenario.

Table 4-32 Configurations of monitoring boards in a BTS3900L (-48 V DC)

Cabinet	Cabinet Type	CN	SRN	SN	BT	MCN	MSRN	MPN	ADDR	Note
Cabinet 0	BTS3900L	0	40	0	EMU	0	0	1	2	The EMU is optional.
		0	50	0	GATM	0	0	0	22	For GBTS only

Cabin et	Cabin et Type	CN	SRN	SN	BT	MCN	MSR N	MPN	ADD R	Note
		0	51	0	GAT M	0	0	1	22	For GBTS only
		0	11	0	FMU	0	0	0	14	-
		0	12	0	FMU	0	0	0	15	-

Example of Configurations in a BTS3900L (AC)

Figure 4-32 shows monitoring signal cable connections in a BTS3900L (Ver.D) cabinet which is stacked with the IMS06 cabinet and supplied with AC power.

Figure 4-32 Monitoring signal cable connections in a BTS3900L (AC)

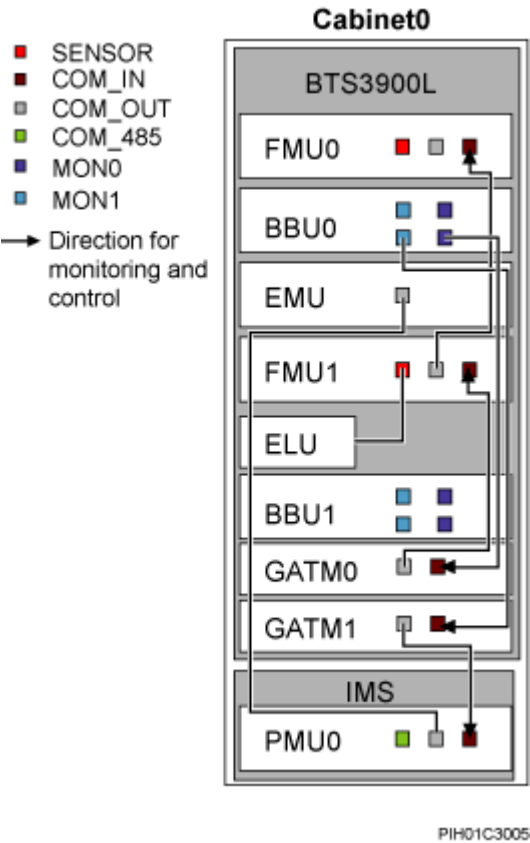


Table 4-33 describes the monitoring board configurations in this scenario.

Table 4-33 Monitoring board configurations in a BTS3900L (AC)

Cabin	Cabin Type	CN	SRN	SN	BT	MCN	MSRN	MPN	ADDR	Note
Cabin 0	BTS3900L	0	40	0	EMU	0	0	1	2	The EMU is optional.
		0	7	0	PMU	0	0	1	3	-
		0	50	0	GATM	0	0	0	22	For GBTS only
		0	51	0	GATM	0	0	1	22	For GBTS only
		0	11	0	FMU	0	0	0	14	-
		0	12	0	FMU	0	0	0	15	-

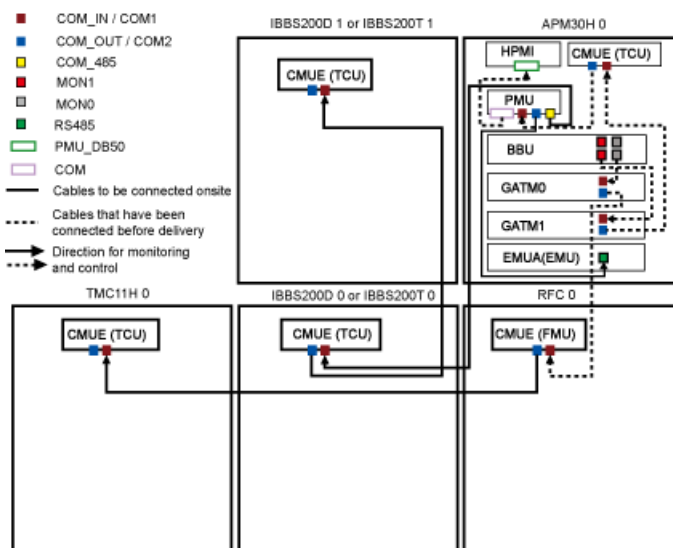
Example of Configurations in a BTS3900A (AC)

NOTICE

The HPMI board is configured in the APM30H (Ver.C), but is not configured in the APM30H (Ver.D). Therefore, the monitoring signal cable connections between the PMU and the HPMI do not need to be considered for the APM30H (Ver.D).

Figure 4-33 and Figure 4-34 show the monitoring signal cable connections in a BTS3900A (AC) (1 APM30H+1 RFC+2 IBBS200D/IBBS200T cabinets+1 TMC11H).

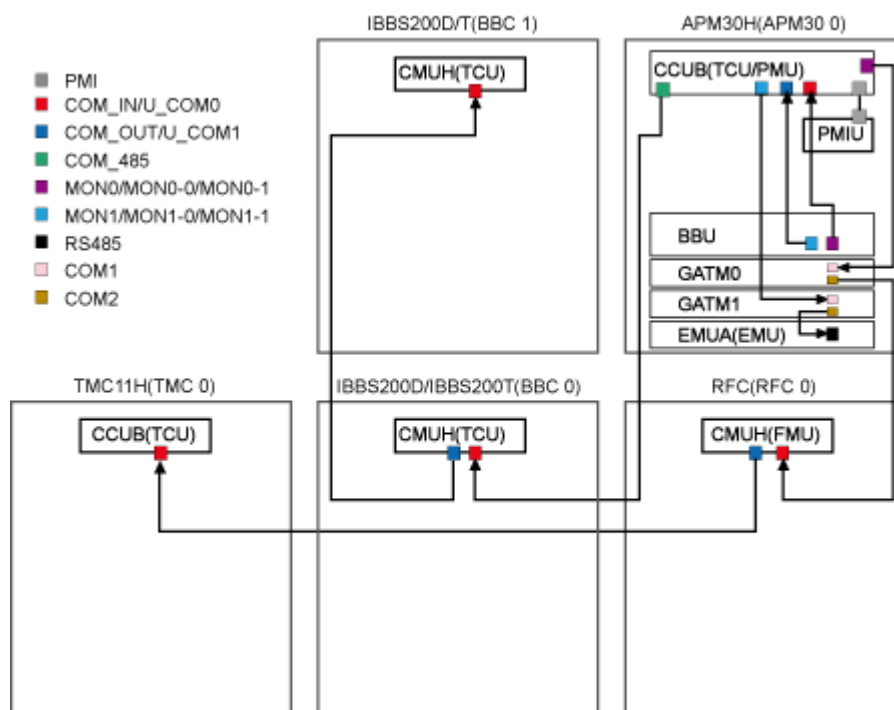
Figure 4-33 Monitoring signal cable connections in a BTS3900A (1 APM30H+1 RFC+2 IBBS200D/IBBS200T cabinets+1 TMC11H)



NOTE

The preceding figure uses the Ver.C cabinet as an example. The monitoring unit of a Ver.C cabinet is CMUE. The monitoring unit of a Ver.D cabinet is CMUEA.

Figure 4-34 Monitoring signal cable connections in a BTS3900A (Ver.E) (1 APM30H+1 RFC+2 IBBS200D/IBBS200T cabinets+1 TMC11H)



PIH19C0001

 NOTE

The preceding figure uses the Ver.E cabinet as an example. The TCU and PMU in an APM30H cabinet are connected to the same CCUB/CCUC. PMIU is the power signal transfer board and does not need to be configured.

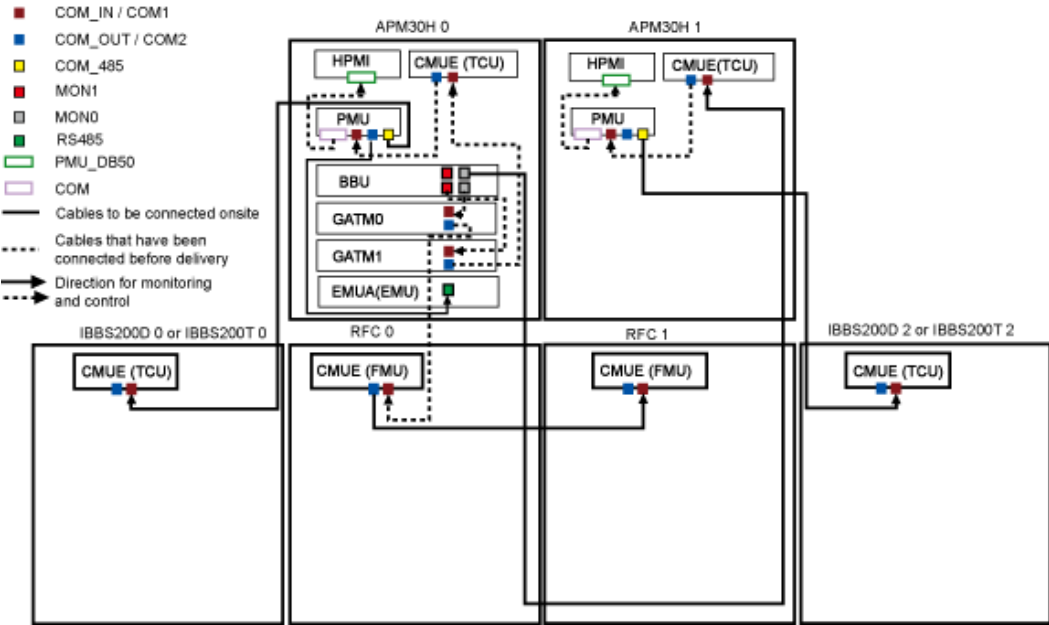
Table 4-34 describes the monitoring board configurations in this scenario.

Table 4-34 Monitoring board configurations in a BTS3900A (1 APM30H+1 RFC+2 IBBS200D/IBBS200T cabinets+1 TMC11H)

Cabin	Cabin Type	CN	SRN	SN	BT	MCN	MSRN	MPN	ADDR	Note
APM30H0	APM30	0	8	0	TCU	0	0	1	7	CMUE
APM30H0	APM30	0	7	0	PMU	0	0	1	3	-
APM30H0	APM30	0	50	0	GATM	0	0	0	22	For GBTS only
APM30H0	APM30	0	51	0	GATM	0	0	1	22	For GBTS only
APM30H0	APM30	0	40	0	EMU	0	0	1	2	-
RFC0	RFC	1	11	0	FMU	0	0	0	14	CMUE
TMC11H0	TMC	8	8	0	TCU	0	0	0	6	CMUE
IBBS200D / IBBS200T0	BBC	9	8	0	TCU	0	0	1	23	CMUE
IBBS200D / IBBS200T1	BBC	10	8	0	TCU	0	0	1	24	CMUE

Figure 4-35 and Figure 4-36 show the monitoring signal cable connections in a BTS3900A (AC) (2 APM30H cabinets+2 RFCs+2 IBBS200D/IBBS200T cabinets).

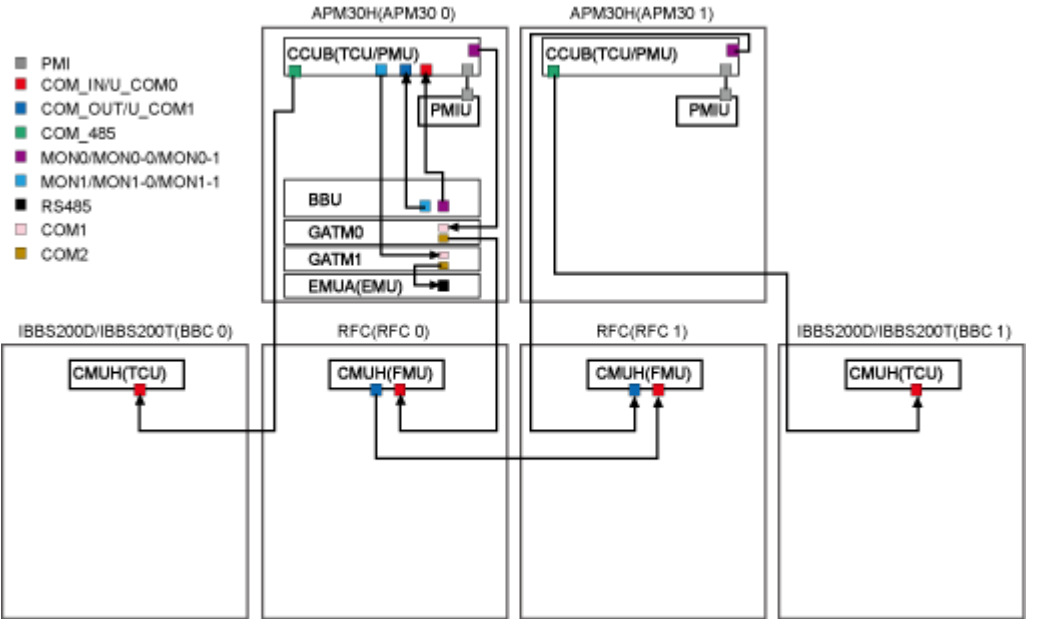
Figure 4-35 Monitoring signal cable connections in a BTS3900A (Ver.C) (2 APM30H cabinets+2 RFCs+ 2 IBBS200D/IBBS200T cabinets)



NOTE

The preceding figure uses the Ver.C cabinet as an example. The monitoring unit of a Ver.C cabinet is CMUE. The monitoring unit of a Ver.D cabinet is CMUEA.

Figure 4-36 Monitoring signal cable connections in a BTS3900A (Ver.E) (2 APM30H cabinets+2 RFCs+2 IBBS200D/IBBS200T cabinets)



PIH19C0002

 NOTE

The preceding figure uses the Ver.E cabinet as an example. The TCU and PMU in an APM30H cabinet are connected to the same CCUB/CCUC. PMIU is the power signal transfer board and does not need to be configured.

Table 4-35 describes the monitoring board configurations in this scenario.

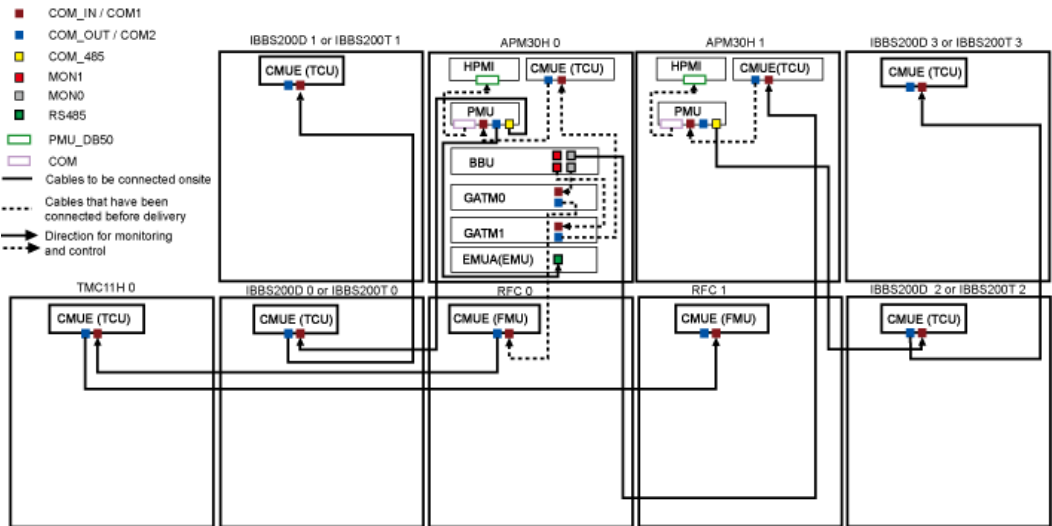
Table 4-35 Monitoring board configurations in a BTS3900A (2 APM30H cabinets +2 RFCs+2 IBBS200D/IBBS200T cabinets)

Cabin	Cabin Type	CN	SRN	SN	BT	MCN	MSRN	MPN	ADDR	Note
APM30H0	APM30	0	7	0	PMU	0	0	1	3	-
APM30H0	APM30	0	8	0	TCU	0	0	1	7	CMUE
APM30H0	APM30	0	50	0	GATM	0	0	0	22	For GBTS only
APM30H0	APM30	0	51	0	GATM	0	0	1	22	For GBTS only
APM30H0	APM30	0	40	0	EMU	0	0	1	2	-
RFC0	RFC	1	11	0	FMU	0	0	0	14	CMUE
RFC1	RFC	2	11	0	FMU	0	0	0	15	CMUE
APM30H1	APM30	5	7	0	PMU	0	0	0	3	-
APM30H1	APM30	5	8	0	TCU	0	0	0	7	CMUE
IBBS200D / IBBS200T0	BBC	9	8	0	TCU	0	0	1	23	CMUE

Cabin	Cabin Type	CN	SRN	SN	BT	MCN	MSRN	MPN	ADDR	Note
IBBS 200D / IBBS 200T 2	BBC	11	8	0	TCU	0	0	0	23	CMUE

Figure 4-37 shows the monitoring signal cable connections in a BTS3900A (AC) (2 APM30H cabinets+2 RFCs+4 IBBS200D/IBBS200T cabinets+1 TMC11H).

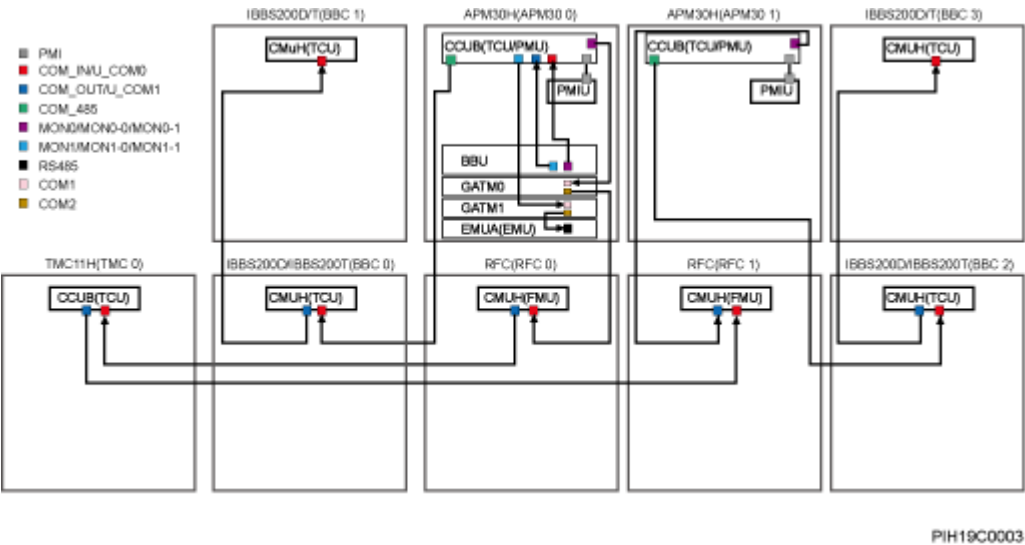
Figure 4-37 Monitoring signal cable connections in a BTS3900A (Ver.C) (2 APM30H cabinets+2 RFCs+4 IBBS200D/IBBS200T cabinets+1 TMC11H)



NOTE

The preceding figure uses the Ver.C cabinet as an example. The monitoring unit of a Ver.C cabinet is CMUE. The monitoring unit of a Ver.D cabinet is CMUEA.

Figure 4-38 Monitoring signal cable connections in a BTS3900A (Ver.E) (2 APM30H cabinets+2 RFCs+4 IBBS200D/IBBS200T cabinets+1 TMC11H)



NOTE

The preceding figure uses the Ver.E cabinet as an example. The TCU and PMU in an APM30H cabinet are connected to the same CCUB/CCUC. PMIU is the power signal transfer board and does not need to be configured.

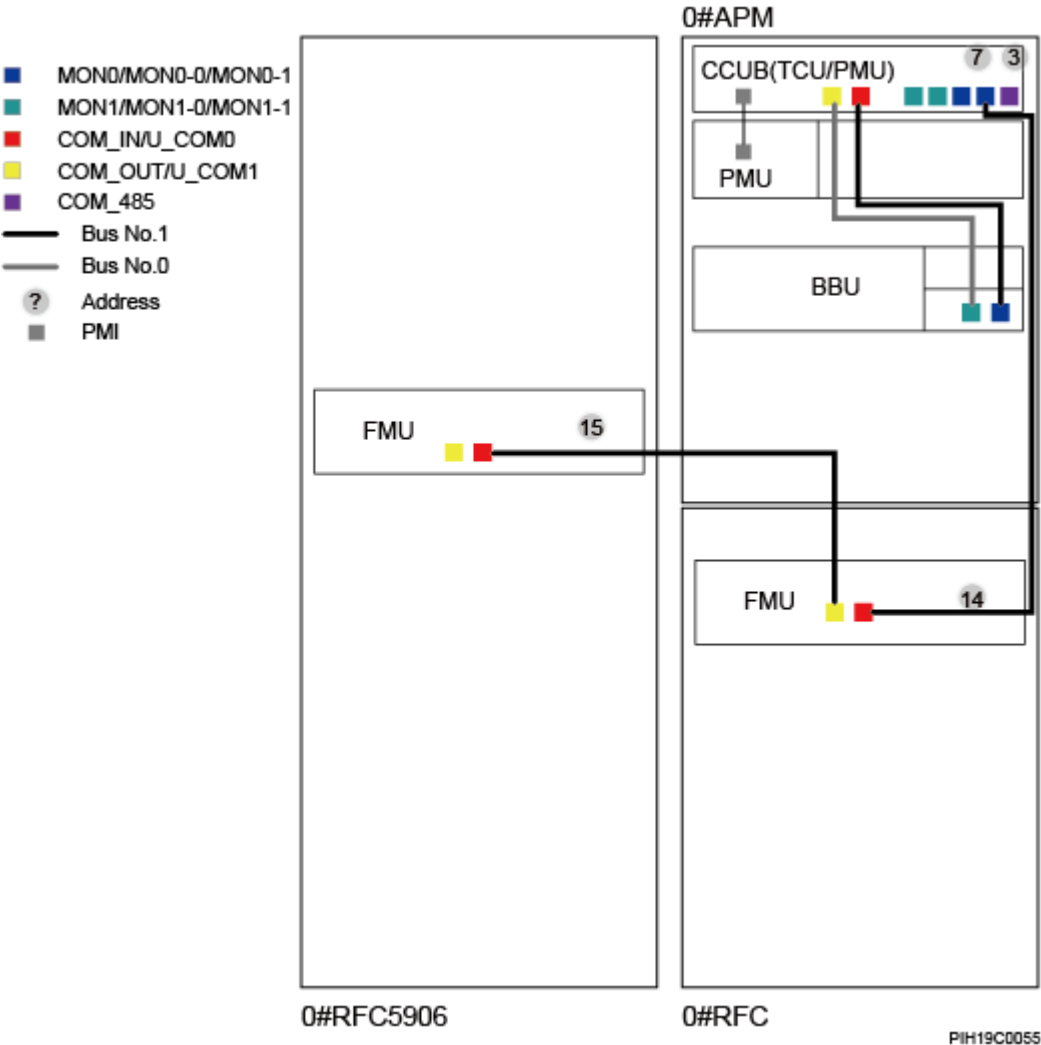
Table 4-36 describes the monitoring board configurations in this scenario.

Table 4-36 Monitoring board configurations in a BTS3900A (2 APM30H cabinets +2 RFCs+4 IBBS200D/IBBS200T cabinets+1 TMC11H)

Cabin	Cabin Type	CN	SRN	SN	BT	MCN	MSR N	MPN	ADD R	Note
APM30H0	APM30	0	7	0	PMU	0	0	1	3	-
APM30H0	APM30	0	8	0	TCU	0	0	1	7	CMUE
APM30H0	APM30	0	50	0	GATM	0	0	0	22	For GBTS only
APM30H0	APM30	0	51	0	GATM	0	0	1	22	For GBTS only
APM30H0	APM30	0	40	0	EMU	0	0	1	2	-

Cabinet	Cabinet Type	CN	SRN	SN	BT	MCN	MSRN	MPN	ADDR	Note
RFC0	RFC	1	11	0	FMU	0	0	0	14	CMUE
RFC1	RFC	2	11	0	FMU	0	0	0	15	CMUE
APM30H1	APM30	5	7	0	PMU	0	0	0	3	-
APM30H1	APM30	5	8	0	TCU	0	0	0	7	CMUE
TMC11H0	TMC	8	8	0	TCU	0	0	0	6	-
IBBS200D / IBBS200T0	BBC	9	8	0	TCU	0	0	1	23	CMUE
IBBS200D / IBBS200T1	BBC	10	8	0	TCU	0	0	1	24	CMUE
IBBS200D / IBBS200T2	BBC	11	8	0	TCU	0	0	0	23	CMUE
IBBS200D / IBBS200T3	BBC	12	8	0	TCU	0	0	0	24	CMUE

Figure 4-39 Monitoring signal cable connections in a BTS3900A (Ver.E) (2 APM30H cabinets+2 RFCs+4 IBBS200D/IBBS200T cabinets+1 TMC11H)



NOTE

The preceding figure uses the Ver.E cabinet as an example. The TCU and PMU in an APM30H cabinet are connected to the same CCUB/CCUC.

PMIU is the power signal transfer board and does not need to be configured with monitoring data.

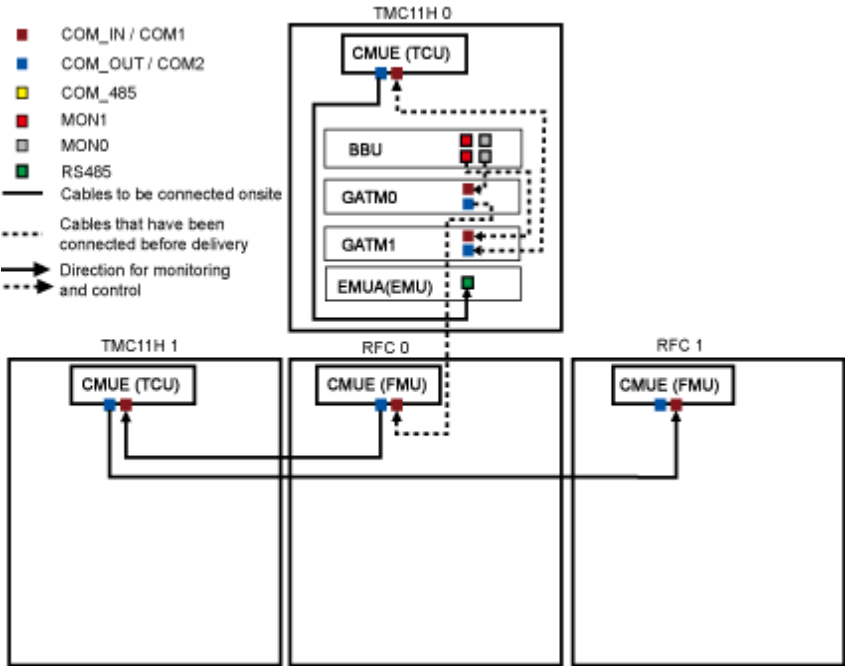
Cabin	Cabin Type	CN	SRN	SN	BT	MCN	MSR N	MPN	ADD R	Note
APM 30H 0	APM 30	0	7	0	PMU	0	0	1	3	CCU B
APM 30H 0	APM 30	0	8	0	TCU	0	0	1	7	CCU B

Cabin et	Cabin et Type	CN	SRN	SN	BT	MCN	MSR N	MPN	ADD R	Note
RFC 0	RFC	1	11	0	FMU	0	0	0	14	CMU H
RFC5 906 0	RFC5 906	2	11	0	FMU	0	0	0	15	CMU H

Example of Configurations in a BTS3900A (-48 V DC)

Figure 4-40 and Figure 4-41 show the connections of monitoring signal cables in a BTS3900A (-48 V DC) (2 TMC11H+2 RFC). If cabinet TMC11H1 is not configured, connect the CMUE/CMUH in cabinet RFC0 to the CMUE/CMUH in cabinet RFC1 and ignore the configuration data of cabinet TMC11H1.

Figure 4-40 Connections of monitoring signal cables in a BTS3900A (Ver.C) (2 TMC11H cabinets+2 RFCs)



NOTE

The preceding figure uses the Ver.C cabinet as an example. The monitoring unit of a Ver.C cabinet is CMUE. The monitoring unit of a Ver.D cabinet is CMUEA.

Figure 4-41 Connections of monitoring signal cables in a BTS3900A (Ver.E) (2 TMC11H cabinets+2 RFCs)

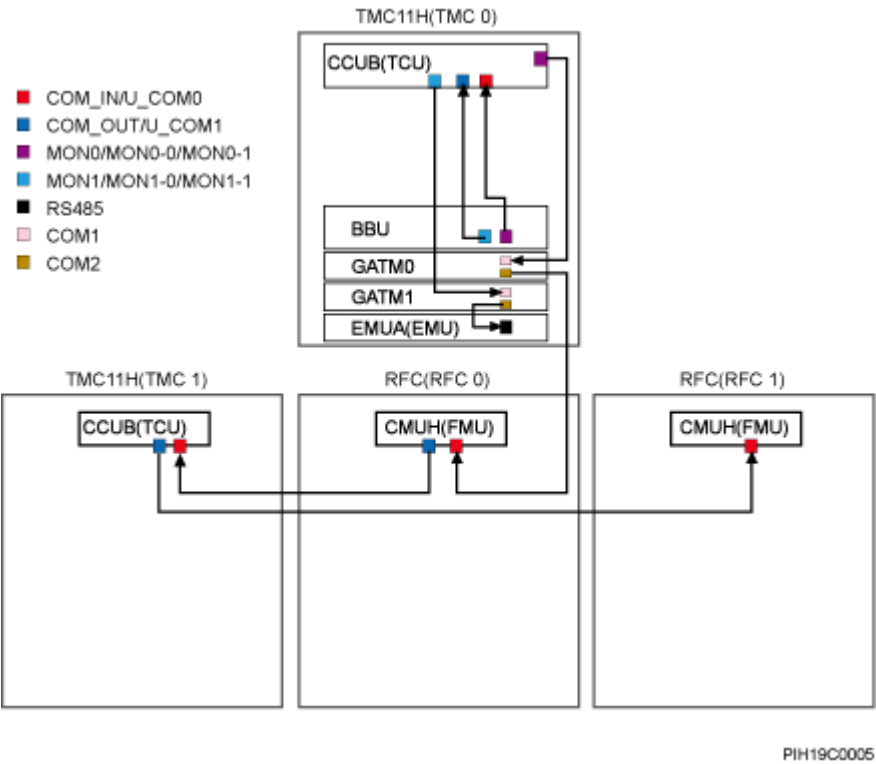


Table 4-37 describes the monitoring board configurations in this scenario.

Table 4-37 Configurations of monitoring boards in a BTS3900A (2 TMC11H cabinets+2 RFCs)

Cabin	Cabin Type	CN	SRN	SN	BT	MCN	MSR N	MPN	ADD R	Note
TMC 11H 0	TMC	0	8	0	TCU	0	0	1	7	CMU E
		0	50	0	GAT M	0	0	0	22	For GBTS only
		0	51	0	GAT M	0	0	1	22	For GBTS only
		0	40	0	EMU	0	0	1	2	-
RFC 0	RFC	1	11	0	FMU	0	0	0	14	CMU E
RFC 1	RFC	2	11	0	FMU	0	0	0	15	CMU E

Cabin	Cabin Type	CN	SRN	SN	BT	MCN	MSRN	MPN	ADDR	Note
TMC11H1	TMC	8	8	0	TCU	0	0	0	6	CMUE

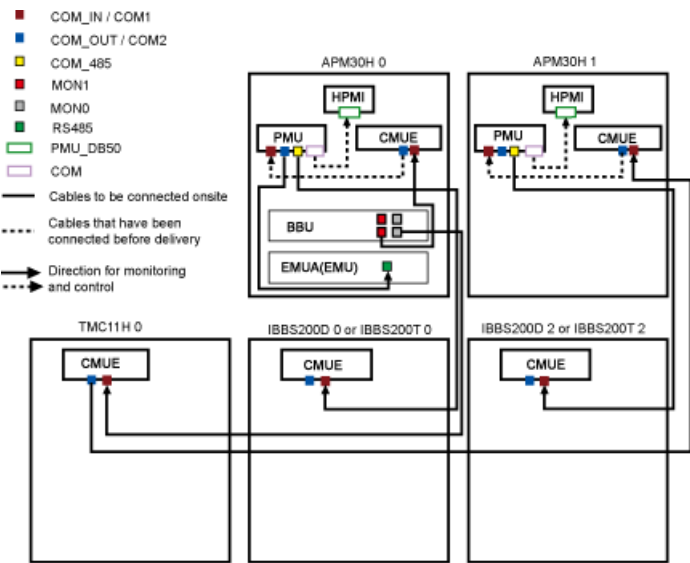
Example of Configurations in a DBS3900 (AC)

NOTICE

The HPMI board is configured in the APM30H (Ver.C), but is not configured in the APM30H (Ver.D). Therefore, the monitoring signal cable connections between the PMU and the HPMI do not need to be considered for the APM30H (Ver.D).

Figure 4-42 and Figure 4-43 show the monitoring signal cable connections in a DBS3900 (AC) (2 APM30H cabinets+2 IBBS200D/IBBS200T cabinets+1 TMC11H).

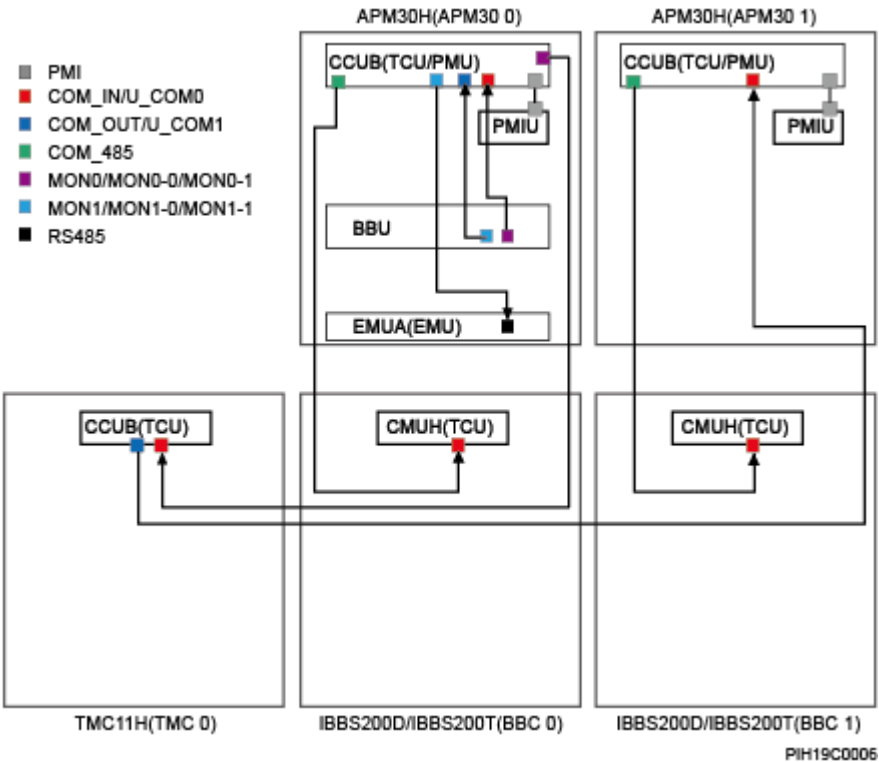
Figure 4-42 Monitoring signal cable connections in a DBS3900 (Ver.C) (2 APM30H cabinets+2 IBBS200D/IBBS200T cabinets+1 TMC11H)



NOTE

The preceding figure uses the Ver.C cabinet as an example. The TCU of a Ver.C cabinet is CMUE. The TCU of a Ver.D cabinet is CMUEA.

Figure 4-43 Monitoring signal cable connections in a DBS3900 (Ver.E) (2 APM30H cabinets+2 IBBS200D/IBBS200T cabinets+1 TMC11H)



NOTE

The preceding figure uses the Ver.E cabinet as an example. The TCU and PMU in an APM30H cabinet are connected to the same CCUB/CCUC. PMIU is the power signal transfer board and does not need to be configured.

Table 4-38 describes the monitoring board configurations in this scenario.

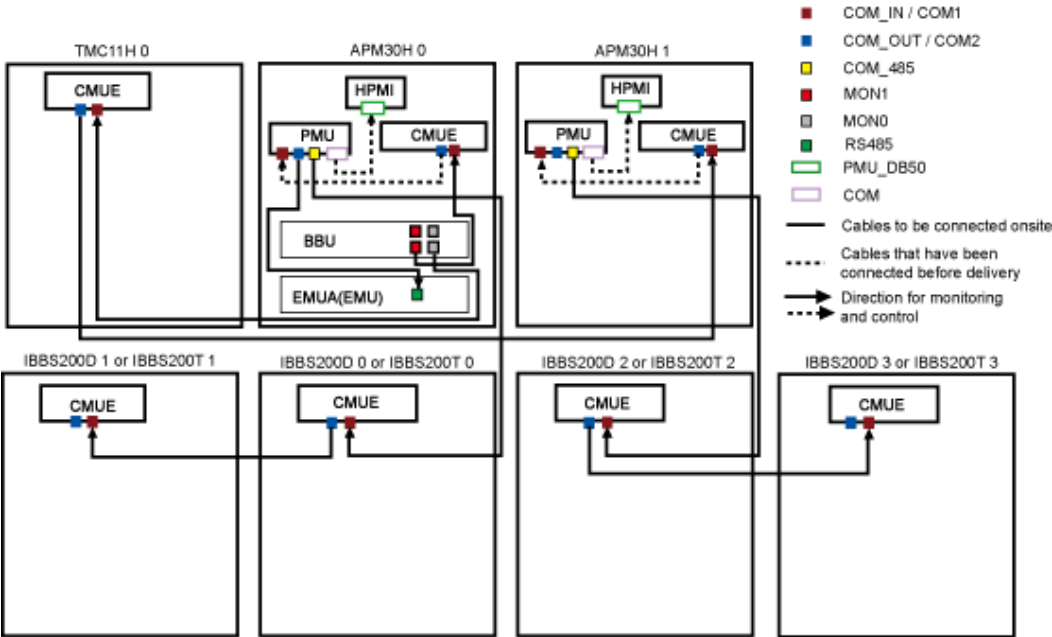
Table 4-38 Monitoring board configurations in a DBS3900 (2 APM30H cabinets+2 IBBS200D/IBBS200T cabinets+1 TMC11H)

Cabin	Cabin Type	CN	SRN	SN	BT	MCN	MSR N	MPN	ADD R	Note
APM30H 0	APM30	0	7	0	PMU	0	0	1	3	-
APM30H 0	APM30	0	8	0	TCU	0	0	1	7	CMUE
APM30H 0	APM30	0	40	0	EMU	0	0	1	2	-

Cabin et	Cabin et Type	CN	SRN	SN	BT	MCN	MSR N	MPN	ADD R	Note
APM 30H 1	APM 30	5	7	0	PMU	0	0	0	3	-
APM 30H 1	APM 30	5	8	0	TCU	0	0	0	6	CMU E
TMC 11H 0	TMC	8	8	0	TCU	0	0	0	7	CMU E
IBBS 200D / IBBS 200T 0	BBC	9	8	0	TCU	0	0	1	23	CMU E
IBBS 200D / IBBS 200T 1	BBC	11	8	0	TCU	0	0	0	23	CMU E

Figure 4-44 and Figure 4-45 show the monitoring signal cable connections in a DBS3900 (AC) (2 APM30H cabinets+4 IBBS200D/IBBS200T cabinets+1 TMC11H).

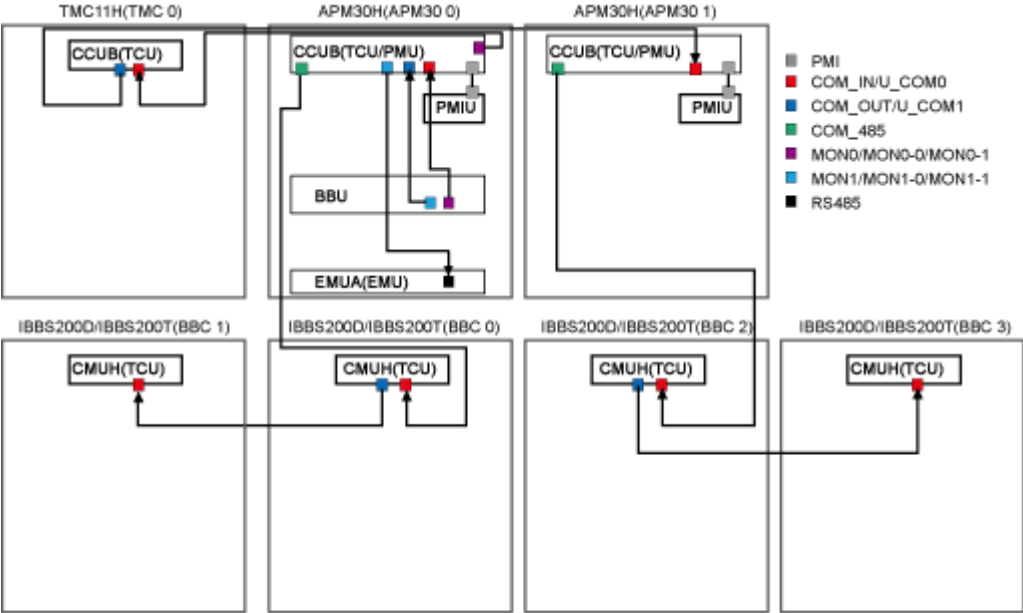
Figure 4-44 Monitoring signal cable connections in a DBS3900 (2 APM30H cabinets+4 IBBS200D/IBBS200T cabinets+1 TMC11H)



NOTE

The preceding figure uses the Ver.C cabinet as an example. The TCU of a Ver.C cabinet is CMUE. The TCU of a Ver.D cabinet is CMUEA.

Figure 4-45 Monitoring signal cable connections in a DBS3900 (Ver.E) (2 APM30H cabinets+4 IBBS200D/IBBS200T cabinets+1 TMC11H)



PIH19C0007

 NOTE

The preceding figure uses the Ver.E cabinet as an example. The TCU and PMU in an APM30H cabinet are connected to the same CCUB/CCUC. PMIU is the power signal transfer board and does not need to be configured.

Table 4-39 describes the monitoring board configurations in this scenario.

Table 4-39 Monitoring board configurations in a DBS3900 (2 APM30H cabinets+4 IBBS200D/IBBS200T cabinets+1 TMC11H)

Cabinet	Cabinet Type	CN	SRN	SN	BT	MCN	MSRN	MPN	ADDR	Note
APM30H0	APM30	0	7	0	PMU	0	0	1	3	-
APM30H0	APM30	0	8	0	TCU	0	0	1	7	CMUE
APM30H0	APM30	0	40	0	EMU	0	0	1	2	-
APM30H1	APM30	5	7	0	PMU	0	0	0	3	-
APM30H1	APM30	5	8	0	TCU	0	0	0	6	CMUE
TMC11H0	TMC	8	8	0	TCU	0	0	0	7	CMUE
IBBS200D / IBBS200T0	BBC	9	8	0	TCU	0	0	1	23	CMUE
IBBS200D / IBBS200T1	BBC	10	8	0	TCU	0	0	1	24	CMUE

Cabin	Cabin Type	CN	SRN	SN	BT	MCN	MSRN	MPN	ADDR	Note
IBBS 200D / IBBS 200T 2	BBC	11	8	0	TCU	0	0	0	23	CMUE
IBBS 200D / IBBS 200T 3	BBC	12	8	0	TCU	0	0	0	24	CMUE

Figure 4-46 and Figure 4-47 show the monitoring signal cable connections in a DBS3900 in the RRU remote monitoring scenario.

Figure 4-46 Monitoring signal cable connections in a DBS3900 in the RRU remote monitoring scenario

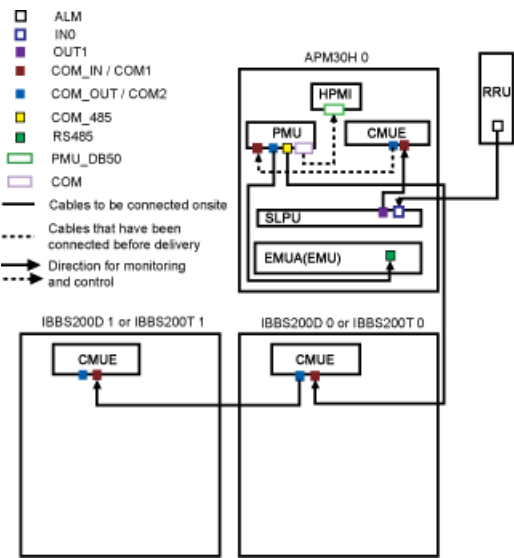


Figure 4-47 Monitoring signal cable connections in a DBS3900 (Ver.E) in the RRU remote monitoring scenario

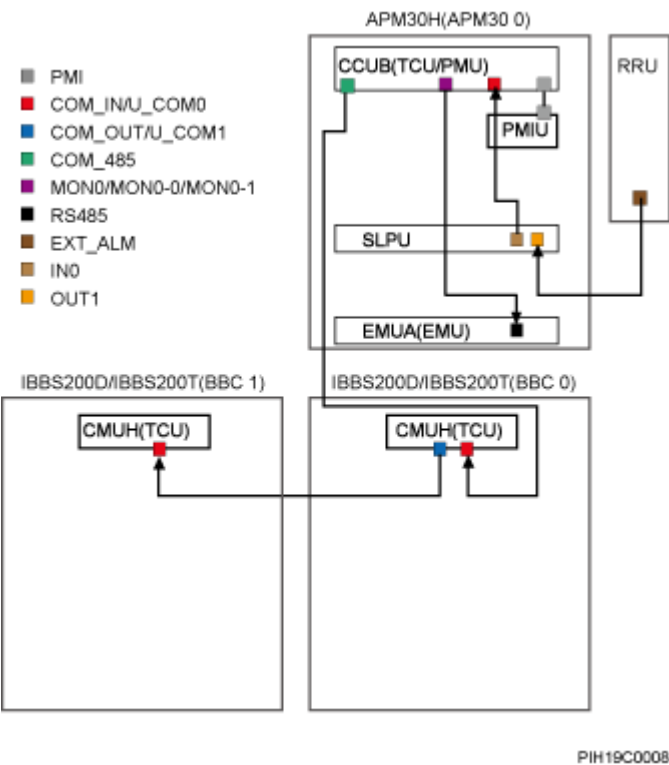


Table 4-40 describes the monitoring board configurations in this scenario.

Table 4-40 Monitoring board configurations in a DBS3900 in the RRU remote monitoring scenario

Cabin	Cabin Type	CN	SRN	SN	BT	MCN	MSR N	MPN	ADD R	Note
APM30H0	APM30	10	7	0	PMU	Cabin number of the RRU that connects to this board	Subrack number of the RRU that connects to this board	0	3	-

Cabin et	Cabin et Type	CN	SRN	SN	BT	MCN	MSR N	MPN	ADD R	Note
APM 30H 0	APM 30	10	8	0	TCU	Cabin et num ber of the RRU that conn ects to this boar d	Subr ack num ber of the RRU that conn ects to this boar d	0	6	CMU E
APM 30H 0	APM 30	10	40	0	EMU	Cabin et num ber of the RRU that conn ects to this boar d	Subr ack num ber of the RRU that conn ects to this boar d	0	2	-
IBBS 200D / IBBS 200T 0	BBC	11	8	0	TCU	Cabin et num ber of the RRU that conn ects to this boar d	Subr ack num ber of the RRU that conn ects to this boar d	0	23	CMU E

Cabin et	Cabin et Type	CN	SRN	SN	BT	MCN	MSR N	MPN	ADD R	Note
IBBS 200D / IBBS 200T 1	BBC	12	8	0	TCU	Cabin et num ber of the RRU that conn ects to this boar d	Subr ack num ber of the RRU that conn ects to this boar d	0	24	CMU E

Figure 4-48 shows the monitoring signal cable connections in a DBS3900 (AC) (1 OMB (Ver.C)+2 IBBS20D cabinets).

Figure 4-48 Monitoring signal cable connections in a DBS3900 (1 OMB (Ver.C)+2 IBBS20D cabinets)

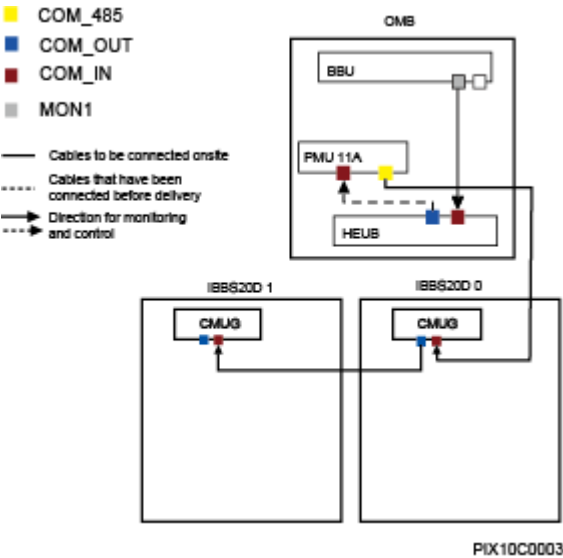


Table 4-41 describes the monitoring board configurations in this scenario.

Table 4-41 Monitoring board configurations in a DBS3900 (1 OMB (Ver.C)+2 IBBS20D cabinets)

Cabin	Cabin Type	CN	SRN	SN	BT	MCN	MSRN	MPN	ADDR	Note
OMB	OMB	0	7	0	PMU	0	0	1	3	PMU 11A (The configuration type of the PMU is EPS4815.)
		0	8	0	TCU	0	0	1	7	HEUB
IBBS20D0	BBC	11	8	0	TCU	0	0	0	23	CMUG
IBBS20D1	BBC	12	8	0	TCU	0	0	0	24	CMUG

Figure "Monitoring signal cable connections in a DBS3900 (AC) (1 OMB)" shows the monitoring signal cable connections in a DBS3900 (AC) that has the BBU installed in an OMB cabinet.

Figure 4-49 Monitoring signal cable connections in a DBS3900 (AC) (1 OMB)

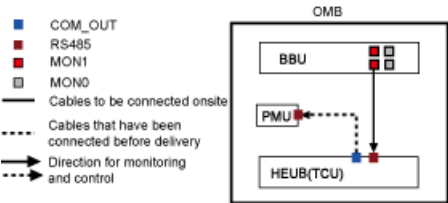


Table 4-42 describes the monitoring board configurations in this scenario.

Table 4-42 Monitoring board configurations in a DBS3900 (AC) (1 OMB)

Cabin	Cabin Type	CN	SRN	SN	BT	MCN	MSRN	MPN	ADDR	Note
OMB	OMB	0	7	0	PMU	0	0	1	3	-
		0	8	0	TCU	0	0	1	7	-

Figure "Monitoring signal cable connections in a DBS3900 (AC) (BBU+RRU+ICR)" shows the monitoring signal cable connections in a DBS3900 (110 V/220 V) deployed in BBU+RRU+ICR mode. The BBU and PMU are installed on the upper and lower racks respectively in an IMB cabinet.

Figure 4-50 Monitoring signal cable connections in a DBS3900 (AC) (BBU+RRU+ICR)

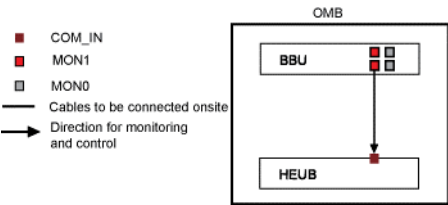


Table 4-43 describes the monitoring board configurations in this scenario.

Table 4-43 Monitoring board configurations in a DBS3900 (AC) deployed in BBU+RRU+ICR mode

Cabin	Cabin Type	CN	SRN	SN	BT	MCN	MSRN	MPN	ADDR	Note
IMB03/IMB05	Virtual	0	7	0	PMU	0	0	1	3	-

Figure 4-51 shows the monitoring signal cable connections in a DBS3900 (with AC power input) deployed in BBU+RRU+IMB mode. The BBU is installed in an IMB cabinet and the RRU is installed outdoors.

Figure 4-51 Monitoring signal cable connections in a 110 V/220 V base station deployed in BBU+RRU+IMB mode

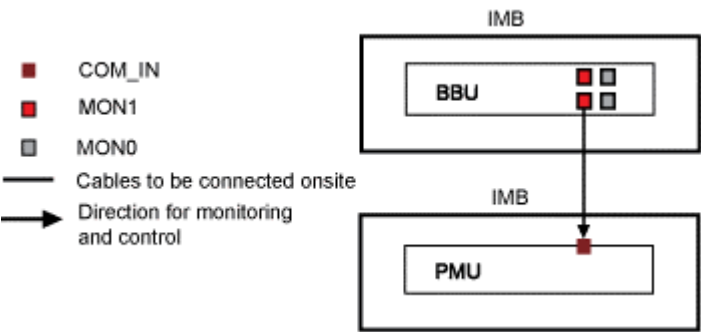


Table 4-44 describes the monitoring board configurations in this scenario.

Table 4-44 Monitoring board configurations in a DBS3900 (AC) deployed in BBU+RRU+IMB mode

Cabinet	Cabinet Type	CN	SRN	SN	BT	MCN	MSRN	MPN	ADDR	Note
IMB03	Virtual	0	7	0	PMU	0	0	1	3	-

Figure 4-52 shows the monitoring signal cable connections in an outdoor DBS3900 (AC) deployed in BBU3910A+OPM50M+IBBS20D mode based on the maximum monitoring configuration of 1 BBU3910A+3 OPM50Ms+6 IBBS20D cabinets. If only one or two OPM50M/PMU modules are configured at a site, you only need to connect the cables for the configured OPM50M/PMU modules.

Figure 4-52 Monitoring signal cable connections in a base station (1 BBU3910A+3 OPM50Ms+6 IBBS20D cabinets)

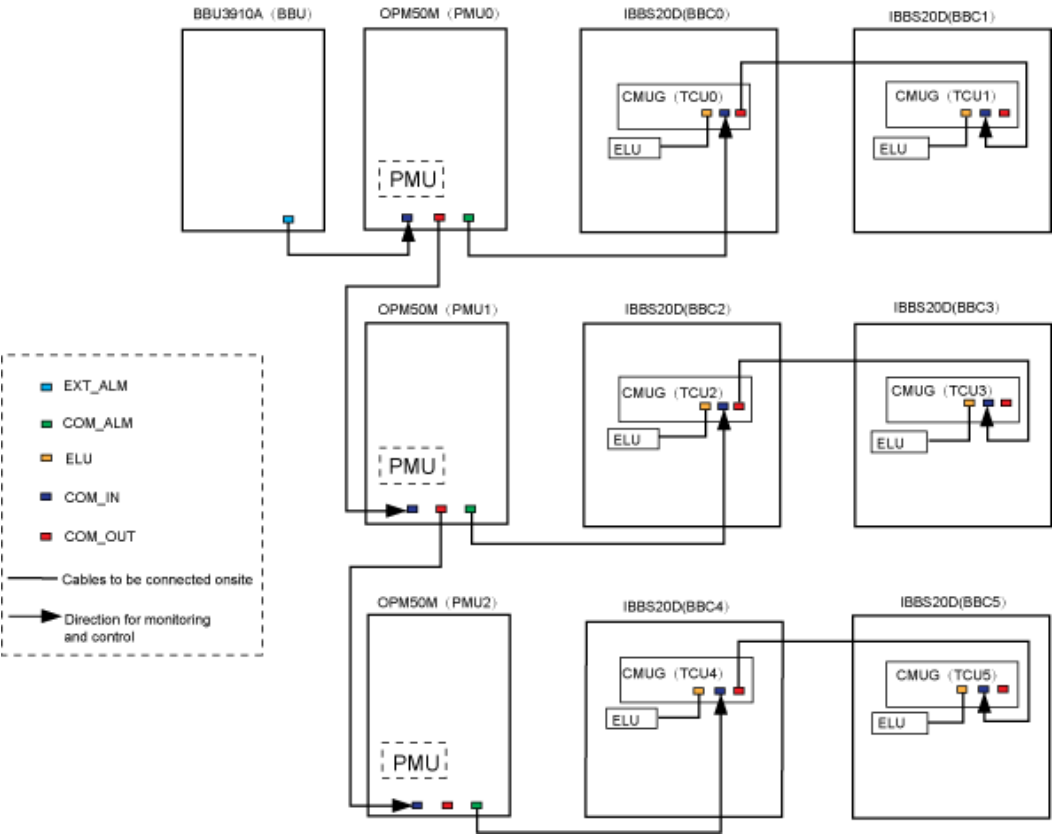


Table 4-45 describes the monitoring board configurations in this scenario.

Table 4-45 Monitoring board configurations in a base station (1 BBU3910A+3 OPM50Ms+6 IBBS20D cabinets)

Cabinet	Cabinet Type	CN	SRN	SN	BT	MCN	MSRN	MPN	ADDR	Note
OPM50M0	Virtual	0	7	0	PMU	0	0	0	3	Power System Type for the PMU is set to ETP.
Cabinet 1	BBC	9	8	0	TCU	0	0	0	23	TCU0
Cabinet 2	BBC	10	8	0	TCU	0	0	0	24	TCU1
OPM50M1 ^[1]	Virtual	5	7	0	PMU	0	0	0	4	Power System Type for the PMU is set to ETP.
Cabinet 4 ^[1]	BBC	11	8	0	TCU	0	0	0	25	TCU2
Cabinet 5 ^[1]	BBC	12	8	0	TCU	0	0	0	26	TCU3

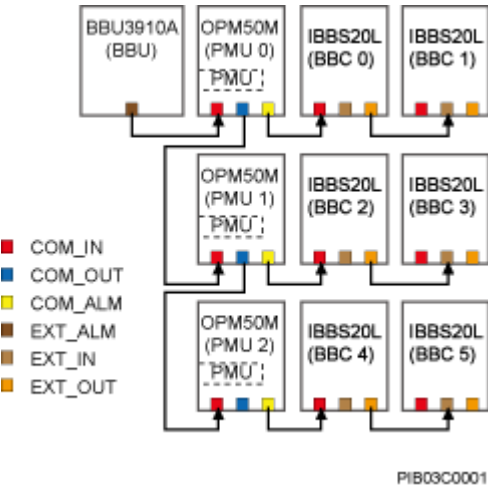
Cabinet	Cabinet Type	CN	SRN	SN	BT	MCN	MSRN	MPN	ADDR	Note
OPM50M2 ^[2]	Virtual	6	7	0	PMU	0	0	0	9	Power System Type for the PMU is set to ETP.
Cabinet 7 ^[2]	BBC	13	8	0	TCU	0	0	0	28	TCU 4
Cabinet 8 ^[2]	BBC	14	8	0	TCU	0	0	0	29	TCU 5

NOTE

- For details about monitoring board configurations, see **Operation and Maintenance (eGBTS/NodeB/eNodeB/gNodeB) > Data Configuration > Data Preparation > Deployment of Basic Power Supply Management Functions** in *Power Supply Management*.
- The subrack No. and slot No. for all OPM50M modules must be set to 7 and 0, respectively, regardless of the number of OPM50M modules configured.
- If only one OPM50M is configured, the data in the row marked ^[1]^[2] does not need to be configured. If two OPM50M modules are configured, the data in the row marked ^[2] does not need to be configured.
- If no backup power is available, you do not need to configure the parameters in the row describing the corresponding cabinet.

Figure 4-53 shows the monitoring signal cable connections in an outdoor DBS3900 (AC) deployed in BBU3910A+OPM50M+IBBS20L mode based on the maximum monitoring configuration of 1 BBU3910A+3 OPM50Ms+6 IBBS20Ls. If only one or two OPM50M/PMU modules are configured at a site, you only need to connect the cables for the configured OPM50M/PMU modules.

Figure 4-53 Monitoring signal cable connections in a base station (1 BBU3910A+3 OPM50Ms+6 IBBS20L modules)



NOTE

Blade lithium batteries do not need to be separately configured for the IBBS20L. The backup power type is lithium power.

Table 4-46 describes the monitoring board configurations in this scenario.

Table 4-46 Monitoring board configurations in a base station (1 BBU3910A+3 OPM50Ms+6 IBBS20L modules)

Cabin	Cabin Type	CN	SRN	SN	BT	MCN	MSRN	MPN	ADDR	Note
OPM50M0	Virtual	0	7	0	PMU	0	0	0	3	Power System Type for the PMU is set to ETP.

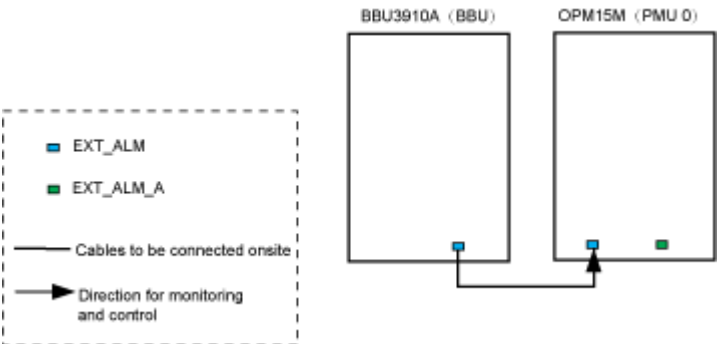
Cabinet	Cabinet Type	CN	SRN	SN	BT	MCN	MSRN	MPN	ADDR	Note
OPM50M1	Virtual	5	7	0	PMU	0	0	0	4	Power System Type for the PMU is set to ETP.
OPM50M2	Virtual	6	7	0	PMU	0	0	0	9	Power System Type for the PMU is set to ETP.

NOTE

- For details about monitoring board configurations, see **Operation and Maintenance (eGBTS/NodeB/eNodeB/gNodeB) > Data Configuration > Data Preparation > Deployment of Basic Power Supply Management Functions** in *Power Supply Management*.
- The subrack No. and slot No. for all OPM50M modules must be set to 7 and 0, respectively, regardless of the number of OPM50M modules configured.

Figure 4-54 shows the monitoring signal cable connections for the DBS3900 (1 BBU3910A+1 OPM15M).

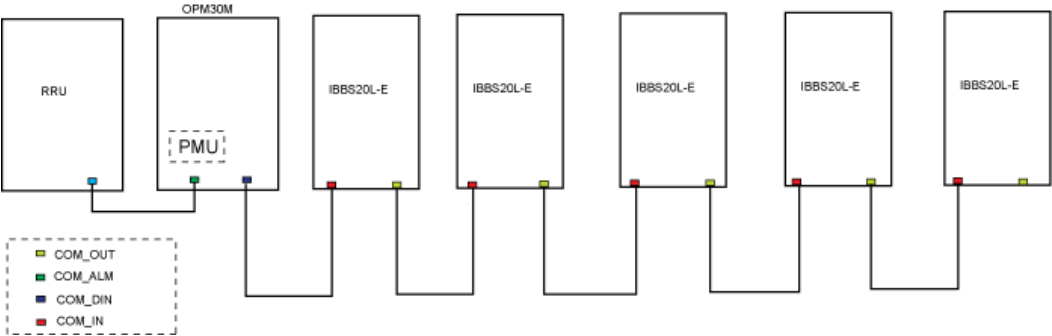
Figure 4-54 Monitoring signal cable connections for 1 BBU3910A+1 OPM15M



In this scenario, the OPM15M is automatically monitored by the BBU3910A and does not require the configuration of monitoring data.

The following figure shows the monitoring signal cable connections in a DBS3900. An OPM30M is deployed for independent power supply for the RRU. This figure uses the maximum monitoring configuration of 1 RRU+1 OPM30M+5 IBBS20L-E modules.

Figure 4-55 Monitoring signal cable connections for 1 RRU+1 OPM30M+5 IBBS20L-E modules



NOTE

Blade lithium batteries do not need to be separately configured for the IBBS20L-E. The backup power type is lithium power.

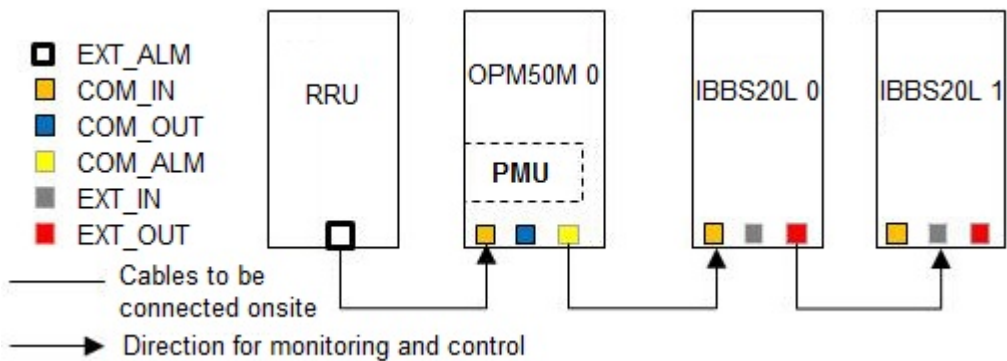
Table 4-47 describes the monitoring board configurations in this scenario.

Table 4-47 Monitoring board configurations for 1 RRU+1 OPM30M+5 IBBS20L-E modules

Cabin	Cabin Type	CN	SRN	SN	BT	MCN	MSRN	MPN	ADDR	Note
OPM30M	Virtual	0	7	0	PMU	Cabinet number of the RRU that connects to this board	Subrack number of the RRU that connects to this board	0	3	Power System Type for the PMU is set to ETP.

Figure 4-56 shows the monitoring signal cable connections in a DBS3900 (1 RRU +1 OPM50M+2 IBBS20L modules).

Figure 4-56 Monitoring signal cable connections for 1 RRU+1 OPM50M+2 IBBS20L modules



NOTE

If an RRU can receive RS485 signals and dry contact alarm signals from a DC power cable, you do not need to connect the RRU to an OPM50M (Ver.B) through a monitoring signal cable. An MEB06D/MEB06D Ver.B connects to power devices and monitored devices and carries external RS485 signals and dry contact alarm signals on a power circuit.

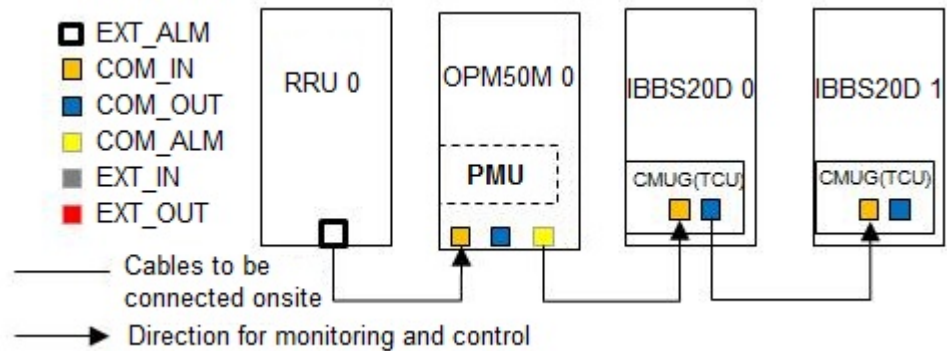
Table 4-48 describes the monitoring board configurations in this scenario.

Table 4-48 Monitoring board configurations for 1 RRU+1 OPM50M+2 IBBS20L modules

Cabin	Cabin Type	CN	SR N	SN	BT	MCN	MSRN	MPN	AD DR	Not e
OPM 50M 0	Virtu al	0	7	0	PMU	Cabine t numbe r of the RRU that connec ts to this board	Subrac k numbe r of the RRU that connec ts to this board	0	3	Power Syst em Typ e for the PMU is set to ETP .

Figure 4-57 shows the monitoring signal cable connections in a DBS3900 (1 RRU +1 OPM50M+2 IBBS20D cabinets).

Figure 4-57 Monitoring signal cable connections for 1 RRU+1 OPM50M+2 IBBS20D cabinets



NOTE

If an RRU can receive RS485 signals and dry contact alarm signals from a DC power cable, you do not need to connect the RRU to an OPM50M (Ver.B) through a monitoring signal cable. An MEB06D/MEB06D Ver.B connects to power devices and monitored devices and carries external RS485 signals and dry contact alarm signals on a power circuit.

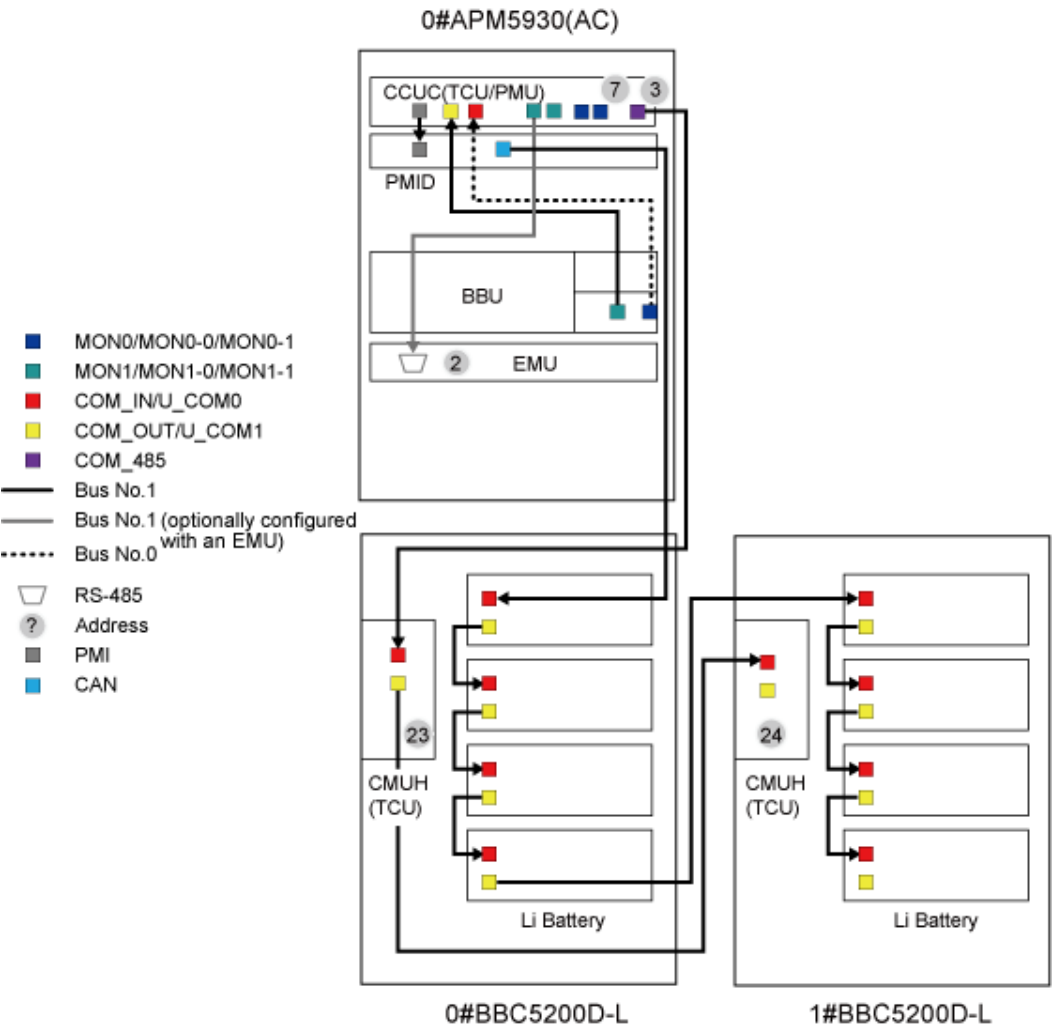
Table 4-49 describes the monitoring board configurations in this scenario.

Table 4-49 Monitoring board configurations for 1 RRU+1 OPM50M+2 IBBS20D cabinets

Cabin et	Cabin et Type	CN	SR N	SN	BT	MCN	MSRN	MPN	AD DR	Not e
OPM 50M 0	Virtu al	0	7	0	PMU	Cabine t numbe r of the RRU that connec ts to this board	Subrac k numbe r of the RRU that connec ts to this board	0	3	Power Syst em Type for the PMU is set to ETP .

Cabin et	Cabin et Type	CN	SR N	SN	BT	MCN	MSRN	MPN	AD DR	Not e
IBBS 20D 0	BBC	9	8	0	TCU	Cabin et numbe r of the RRU that connec ts to this board	Subrac k numbe r of the RRU that connec ts to this board	0	23	TCU 0
IBBS 20D 1	BBC	10	8	0	TCU	Cabin et numbe r of the RRU that connec ts to this board	Subrac k numbe r of the RRU that connec ts to this board	0	24	TCU 1

Figure 4-58 Monitoring signal cable connections in a DBS5900A (Ver.A) (1 APM5930 AC+2 BBC5200D-L cabinets) with lithium batteries as the backup power



PIH19C0051

NOTE

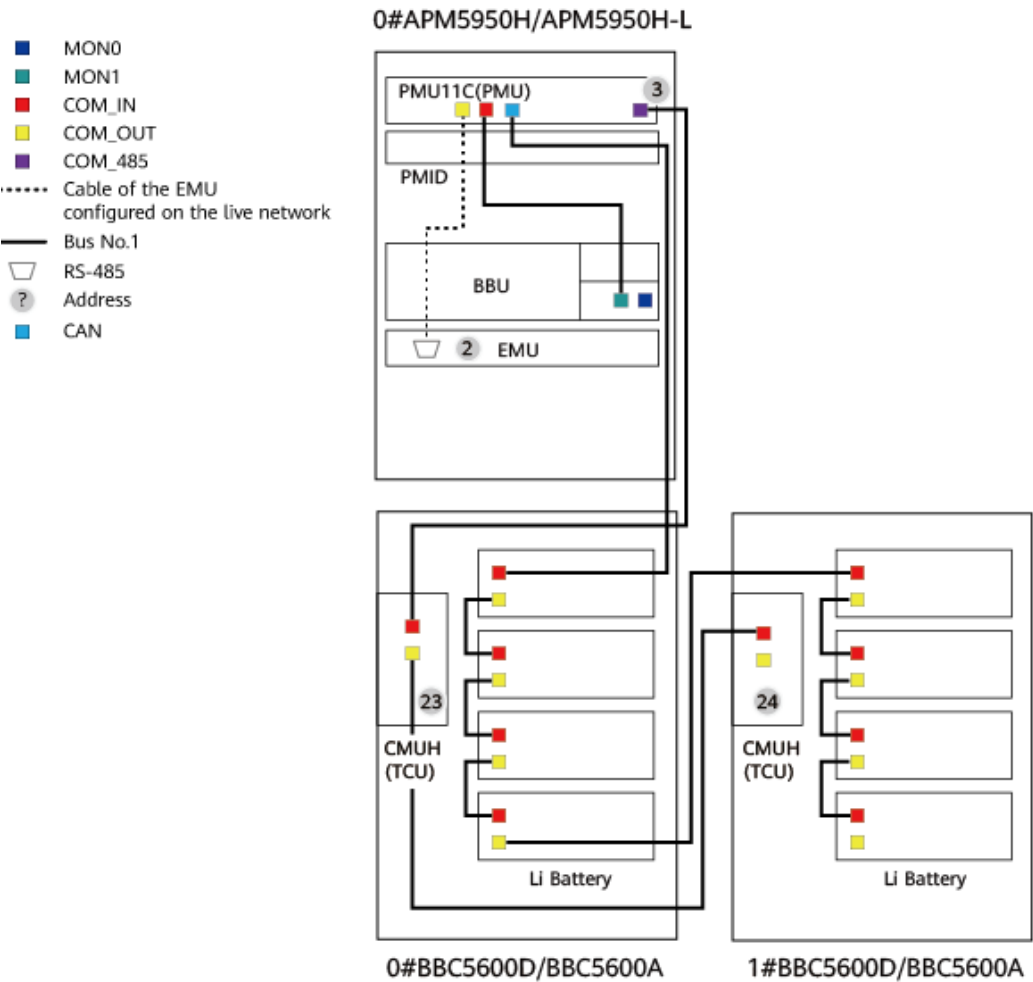
The lithium batteries are cascaded through the "COM_IN" and "COM_OUT" ports. The "COM_IN" port on the first-level lithium batteries is connected to the "CAN" port on the PMID.

Table 4-50 describes the monitoring board configurations in this scenario.

Table 4-50 Monitoring board configurations in a base station (1 APM5930 AC+2 BBC5200D-L cabinets) with lithium batteries as the backup power

Cabinet	Cabinet Type	CN	SRN	SN	BT	MCN	MSRN	MPN	ADDR	Note
APM5930(AC)0	APM5930	0	7	0	PMU	0	0	1	3	Power System Type is set to ETP .
		0	8	0	TCU	0	0	1	7	-
		0	40	0	EMU	0	0	1	2	The EMU is optional.
BBC5200-L 0	BBC	9	8	0	TCU	0	0	1	23	CMUH
BBC5200-L 1	BBC	10	8	0	TCU	0	0	1	24	CMUH

Figure 4-59 Monitoring signal cable connections for a DBS5900A (Ver.A) (1 APM5950H/APM5950H-L+2 BBC5600Ds/BBC5600As) with lithium batteries as the backup power



PIH19C0039

NOTE

- Lithium batteries in the BBC5600D/BBC5600A lithium battery cabinet are cascaded through COM_IN and COM_OUT. The COM_IN port on the first-level lithium battery is connected to the CAN port on the PMU 11C.
- When the battery cabinet is a BBC5600D-V/BBC5600A-V lead-acid battery cabinet, the lithium battery cascading cable in the preceding monitoring cables is not used.

Table 4-51 describes the monitoring board configurations in this scenario.

Table 4-51 Monitoring board configurations in the case of 1 APM5950H/
APM5950H-L+2 BBC5600Ds/BBC5600As with lithium batteries as the backup
power

Cabine t	Cabi net Typ e	C N	S R N	S N	BT	MCN	MSR N	MPN	ADD R	Note
APM59 50H/ APM59 50H-L 0	APM 595 0	0	7	0	PMU	0	0	1	3	Power System Type is set to ETP .
		0	4 0	0	EMU	0	0	1	2	The EMU is optional .
BBC56 00D ^[1] / BBC56 00A ^[2] 0	BBC	9	8	0	TCU	0	0	1	23	CMUH
BBC56 00D ^[1] / BBC56 00A ^[2] 1	BBC	1 0	8	0	TCU	0	0	1	24	CMUH
[1]: The monitoring board configurations in a BBC5600D-V are the same as those in a BBC5600D. [2]: The monitoring board configurations in a BBC5600A-V are the same as those in a BBC5600A.										

Example of Configurations in a DBS3900 (-48 V DC)

Figure 4-60, Figure 4-61, Figure 4-62, and Figure 4-63 show the monitoring
signal cable connections in a DBS3900 (-48 V DC) configured with TMC11H
cabinets.

Figure 4-60 Monitoring signal cable connections in a DBS3900 (–48 V DC) configured with TMC11H cabinets (1)

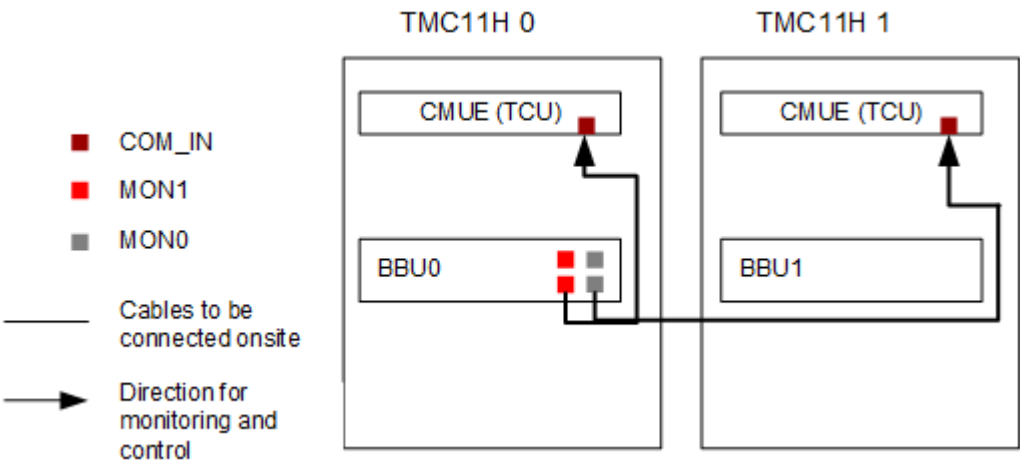


Figure 4-61 Monitoring signal cable connections in a DBS3900 (–48 V DC) configured with TMC11H cabinets (2)

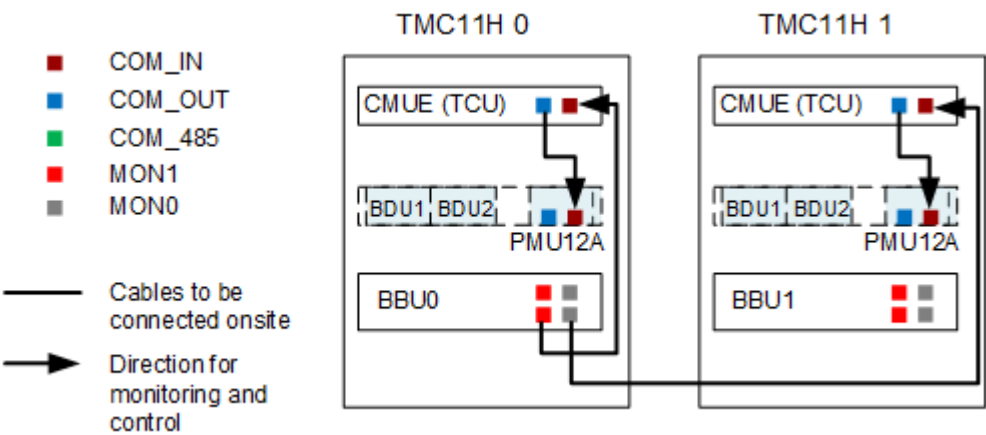


Figure 4-62 Monitoring signal cable connections in a DBS3900 (–48 V DC) configured with TMC11H (Ver.E) cabinets (1)

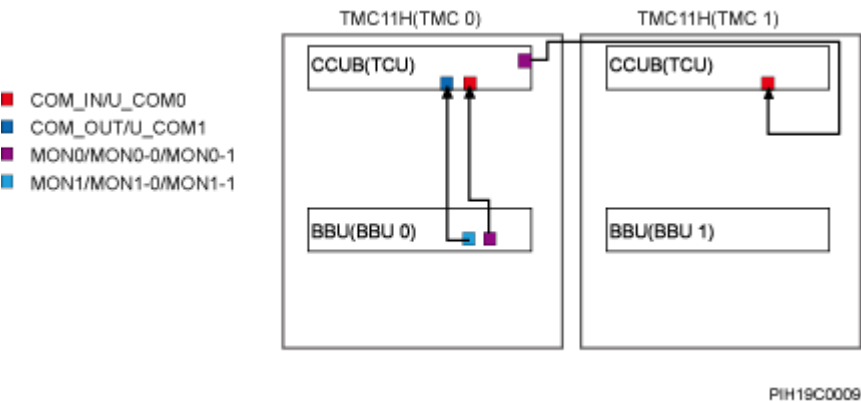


Figure 4-63 Monitoring signal cable connections in a DBS3900 (–48 V DC) configured with TMC11H (Ver.E) cabinets (2)

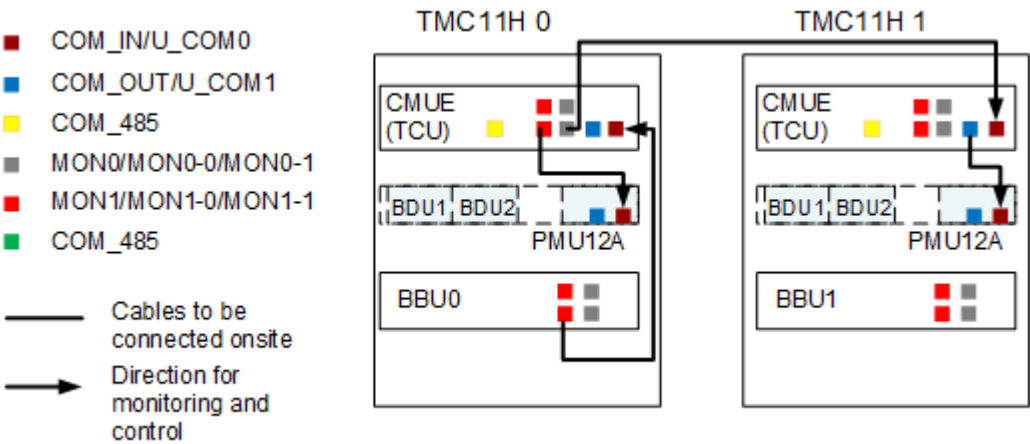


Table 4-52 describes the monitoring board configurations in this scenario.

Table 4-52 Monitoring board configurations in a DBS3900 (–48 V DC) configured with TMC11H cabinets

Cabin	Cabin Type	CN	SRN	SN	BT	MCN	MSR N	MPN	ADD R	Note
TMC 11H 0	TMC	0	8	0	TCU	0	0	1	7	CMUE
		0	17	0	PMU	0	0	1	16	PMU 12A (optional)
		0	17	1	BDU	-	-	-	-	BDU 1 (optional) [1]
		0	17	2	BDU	-	-	-	-	BDU 2 (optional) [1]
TMC 11H 1	TMC	8	8	0	TCU	0	0	0	7	CMUE
		8	17	0	PMU	0	0	0	16	PMU 12A (optional)

Cabinet	Cabinet Type	CN	SRN	SN	BT	MCN	MSRN	MPN	ADDR	Note
		8	17	1	BDU	-	-	-	-	BDU 1 (optional) [1]
		8	17	2	BDU	-	-	-	-	BDU 2 (optional) [1]

 NOTE

[1] When configuring the BDU, you can specify the type of a device. If this parameter is set to **DEFAULT**, the system does not check the BDU, HSU, and SDU types. If this parameter is set to **BDU**, **HSU**, or **SDU**, the system checks whether the type of the BDU, HSU, or SDU in use is consistent with the configuration.

Figure 4-64 Monitoring signal cable connections in a DBS5900 (-48 V DC) configured with an APM5900H cabinet

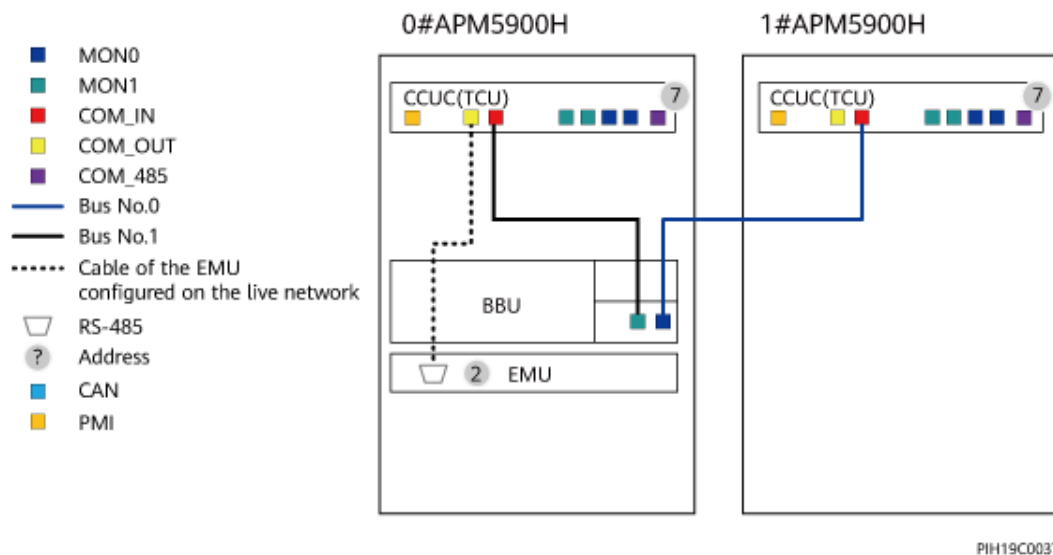


Table 4-53 describes the monitoring board configurations in this scenario.

Table 4-53 Monitoring board configurations in a DBS5900 (–48 V DC) configured with an APM5900H cabinet

Cabinet	Cabinet Type	CN	SRN	SN	BT	MCN	MSRN	MPN	ADDR	Note
APM5900H 0	APM5900	0	8	0	TCU	0	0	1	7	CCUC
		0	40	0	EMU	0	0	1	2	EMU (optional)
APM5900H 1	APM5900	8	8	0	TCU	0	0	0	7	CCUC

Figure 4-65 Monitoring signal cable connections in a DBS5900 (–48 V DC) configured with an APM5900H-L cabinet

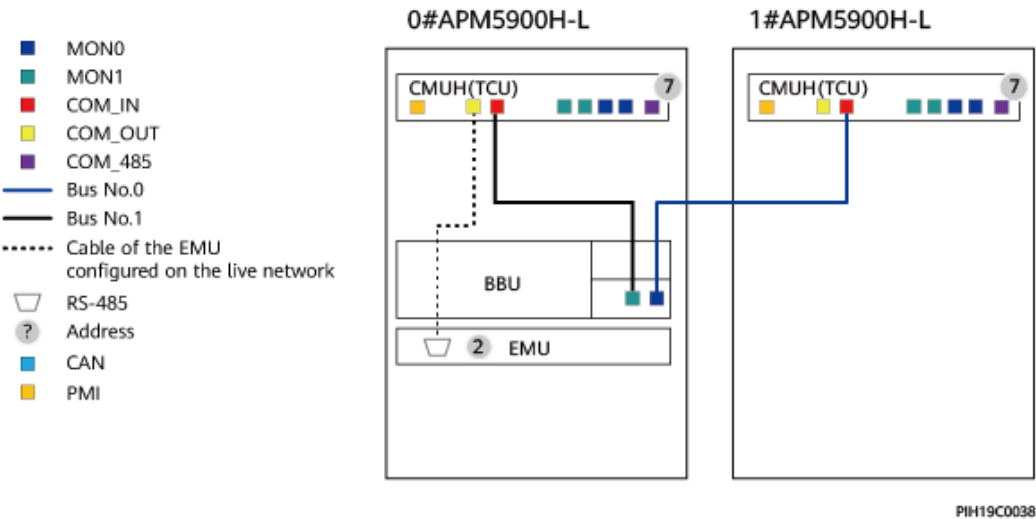


Table 4-54 describes the monitoring board configurations in this scenario.

Table 4-54 Monitoring board configurations in a DBS5900 (–48 V DC) configured with an APM5900H-L cabinet

Cabinet	Cabinet Type	CN	SRN	SN	BT	MCN	MSRN	MPN	ADDR	Note
APM5900H-L 0	APM5900	0	8	0	TCU	0	0	1	7	CMUH
		0	40	0	EMU	0	0	1	2	EMU (optional)

Cabinet	Cabinet Type	CN	SRN	SN	BT	MCN	MSRN	MPN	ADDR	Note
APM 5900 H-L 1	APM 5900	8	8	0	TCU	0	0	0	7	CMU H

Figure "Monitoring signal cable connections in a DBS3900 (–48 V DC) configured with an OMB cabinet" shows the monitoring signal cable connections in a DBS3900 (–48 V DC) that has the BBU installed in an OMB cabinet.

Figure 4-66 Monitoring signal cable connections in a DBS3900 (–48 V DC) configured with an OMB cabinet

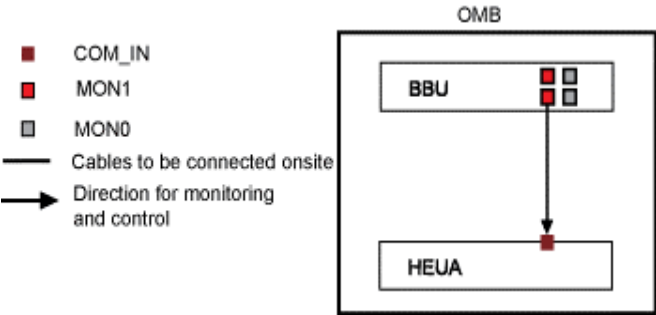


Table 4-55 describes the monitoring board configurations in this scenario.

Table 4-55 Monitoring board configurations in a DBS3900 (–48 V DC) configured with an OMB cabinet

Cabinet	Cabinet Type	CN	SRN	SN	BT	MCN	MSRN	MPN	ADDR	Note
OMB	OMB	0	8	0	TCU	0	0	1	7	-

A BBU3910C does not have any monitoring ports. If a DBS3900 supplied with –48 V power uses a BBU3910C, the power system is connected to an RRU monitoring port to implement monitoring. The RRU is connected to the BBU3910C using a CPRI optical fiber. In this way, the power system is monitored by the base station. **Figure 4-67** and **Figure 4-68** show the monitoring signal cable connections.

Figure 4-67 Example 1 of monitoring signal cable connections in a DBS3900 (–48 V DC)

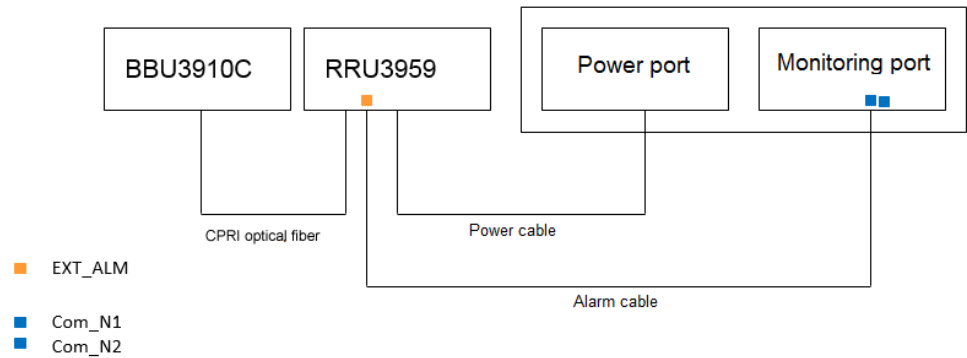
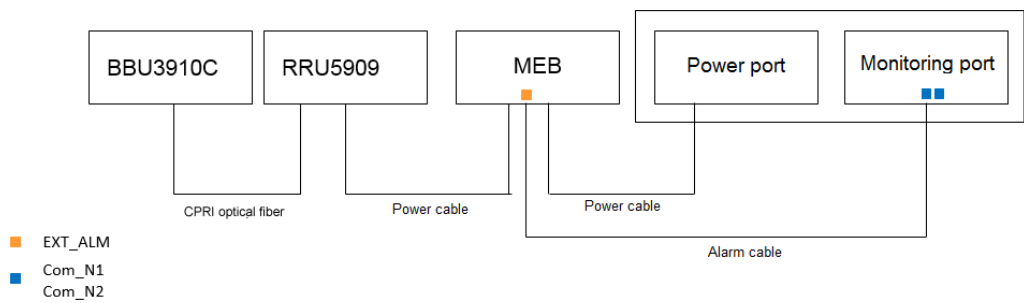


Figure 4-68 Example 2 of monitoring signal cable connections in a DBS3900 (–48 V DC)



The following table lists typical RRU monitoring port types.

Table 4-56 Typical RRU monitoring port types

Base Station Side				Monitored Equipment Side
Typical RRU Type	Port	Description	Connector	Port
RRU3936/ RRU3959	EXT_ALM port on an RRU	Alarm port, used for monitoring one RS485 input and two Boolean inputs	DB15 waterproof male connector	COM_N1/ COM_N2 port on an ICC100- N5

Base Station Side				Monitored Equipment Side
RRU5909	EXT_ALM port on an MEB06D/ MEB06D (Ver.B)	MEB alarm port, which can carry external 485 alarm signals or dry contact alarm signals on a power circuit.	DB15 waterproof male connector	

NOTE

- For details about how to connect an alarm cable to an RRU3959, see section "Installing an RRU Alarm Cable" in "Installing RRU Cables" of *RRU3959&RRU3959w Installation Guide*.
- For details about how to connect an alarm cable to an MEB, see section "Installing an MEB Alarm Cable" in *MEB06D User Guide*.
- For details about how to connect a monitoring signal cable to an ICC100-N5, see the description of ports 1 and 5 in section "Monitoring Board" of *ICC100-N5 Solar Controller User Manual*.

Take an ICC100-N5 as an example. The Com_N1 or Com_N2 port on an ICC100-N5 is connected to a base station so that the base station can monitor the ICC100-N5. The Com_N1 and Com_N2 ports are ports 1 and 5 shown in [Figure 4-69](#).

Figure 4-69 Monitoring ports on an ICC100-N5 for connections to base stations

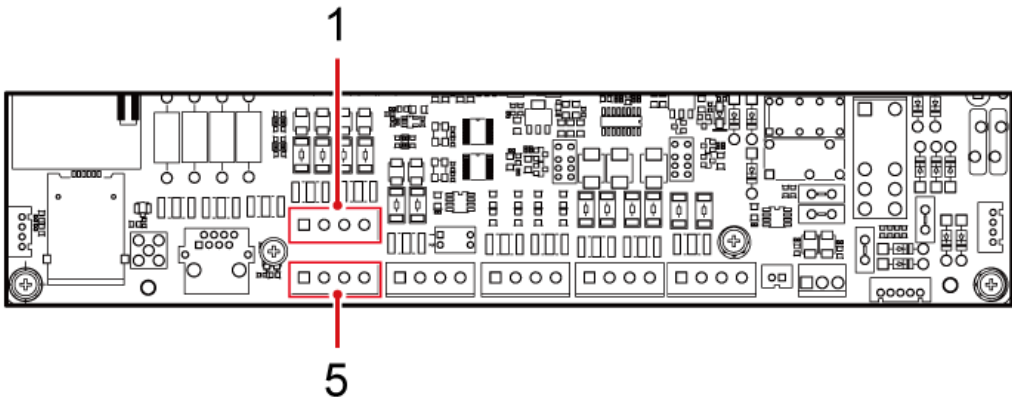


Table 4-57 Monitoring board configurations in a DBS3900 (–48 V DC) configured with a PMU cabinet

Cabin et	Cabin et Type	CN	SRN	SN	Pow er Sup ply Syst em Typ e	MC N	MS RN	MP N	AD DR	DC Volt age Alar m Low er Thr esh old	DC Volt age Alar m Upp er Thr esh old
SC4 820 0	SC4 820 0	0	7	0	SC4 820 0	Cabin et num ber of the con nect ed RRU	Subr ack num ber of the con nect ed RRU	0	3	470	585

Figure 4-70 Monitoring signal cable connections in a DBS3900 (–48 V DC) configured with an EPU02

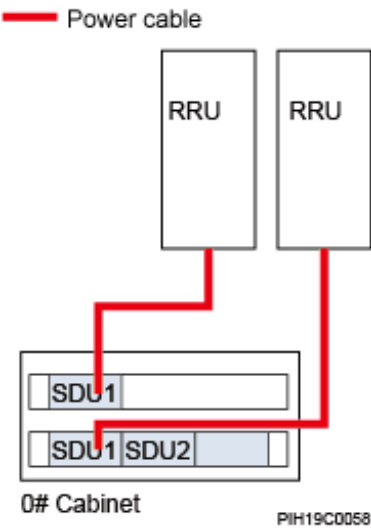


Table 4-58 describes the monitoring board configurations in this scenario.

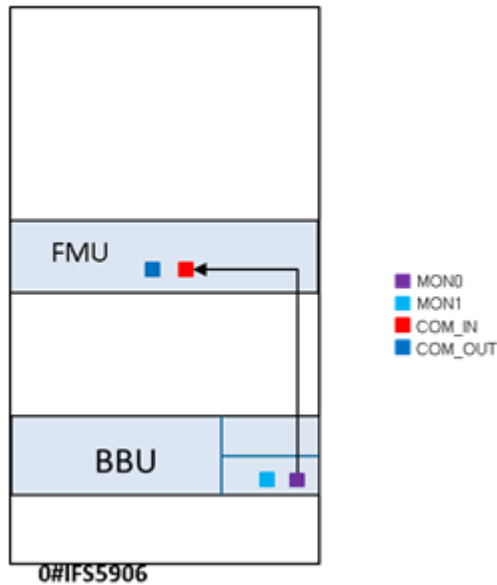
Table 4-58 Monitoring board configurations in a DBS3900 (-48 V DC) configured with an EPU02

Cab inet	Cab ine t Typ e	CN	SR N	SN	BT	MCN	MSR N	MPN	ADD R	Note
Cab inet 0	-	0	17	0	PMU	Cabin et numb er of the conn ected RRU	Subra ck num ber of the conn ected RRU	0	16	Virtual PMU on the SDU1
		0	17	1	BDU	-	-	-	-	-
		0	18	0	PMU	Cabin et numb er of the conn ected RRU	Subra ck num ber of the conn ected RRU	0	16	Virtual PMU on the SDU1
		0	18	1	BDU	-	-	-	-	-
		0	18	2	BDU	-	-	-	-	-

 NOTE

- When configuring the BDU, you can specify the type of a device. If this parameter is set to **DEFAULT**, the system does not check the BDU, HSU, and SDU types. If this parameter is set to **BDU**, **HSU**, or **SDU**, the system checks whether the type of the BDU, HSU, or SDU in use is consistent with the configuration.
- When power cables are used for monitoring, the manager of the virtual PMU on the SDU is the RRU connected to the SDU.

Figure 4-71 Monitoring signal cable connections in a DBS3900 (–48 V DC) configured with an IFS5906



The following table describes the monitoring board configurations in this scenario.

Table 4-59 Monitoring board configurations in a DBS3900 (–48 V DC) configured with an IFS5906

Cabinet	Cabinet Type	CN	SRN	SN	BT	MCN	MSRN	MPN	ADDR	Note
Cabinet 0	IFS5906	0	11	0	FMU	0	0	0	14	-

Example of Configurations in a BTS3900AL (AC)

Figure 4-72 shows the monitoring signal cable connections in a BTS3900AL housed in one AC cabinet, one DC cabinet, and one TMC11H cabinet. Ignore the HAU configuration data of BTS3900AL AC 0.

Figure 4-72 Monitoring signal cable connections in a BTS3900AL (1 AC cabinet+1 DC cabinet+1 TMC11H)

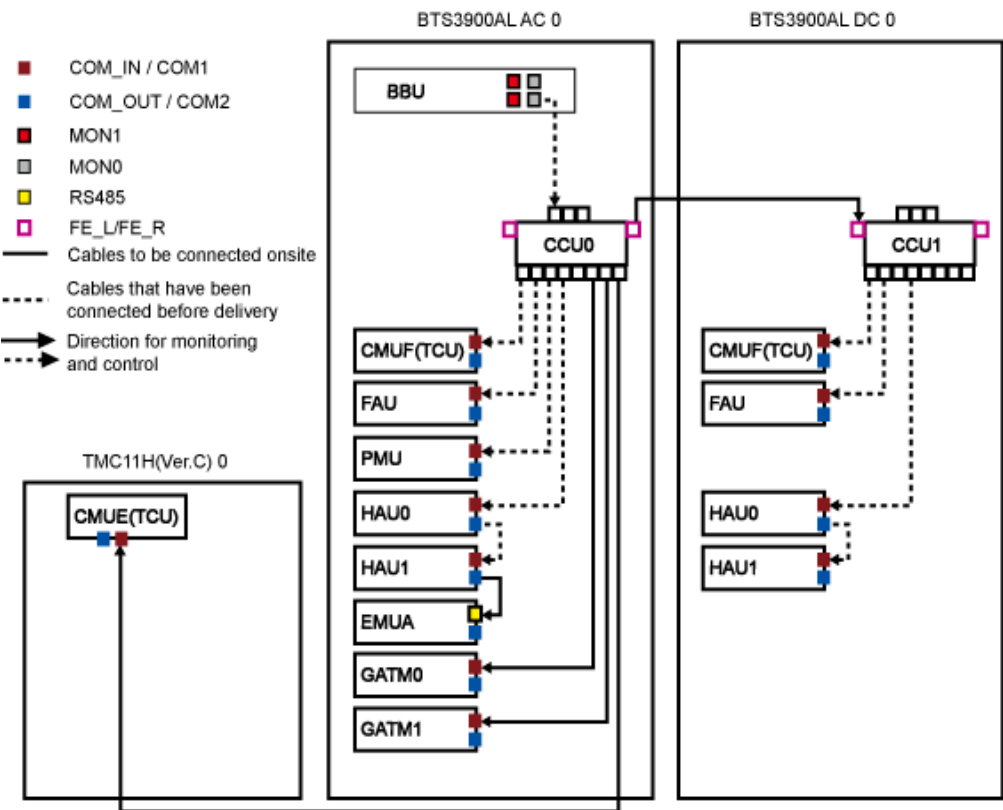


Table 4-60 describes the monitoring board configurations in this scenario.

Table 4-60 Monitoring board configurations in a BTS3900AL (1 AC cabinet+1 DC cabinet+1 TMC11H)

Cabin et	Cabin et Type	CN	SRN	SN	BT	MC N	MS RN	MP N	AD DR	Other Par ame ter	Not e
BTS 390 0AL AC 0	BTS 390 0AL	0	15	0	CCU	0	0	0	21	Level 1 whose DirectConnectFlag is set to YES. Connected RAT : Indicates the working standard of the manager unit to which the CCU is connected.	-
	BTS 390 0AL	0	8	0	TCU	0	15	0	7	-	CMUF

Cabin	Cabin Type	CN	SRN	SN	BT	MCN	MSRN	MPN	ADDR	Other Parameter	Note
	BTS 390 OAL	0	11	0	FMU	0	15	1	14	-	FAU
	BTS 390 OAL	0	7	0	PMU	0	15	2	3	-	-
	BTS 390 OAL	0	8	1	TCU	0	15	3	28	-	HAU0, optional
	BTS 390 OAL	0	8	2	TCU	0	15	3	29	-	HAU1, optional
	BTS 390 OAL	0	40	0	EMU	0	15	3	2	-	Optional
BTS 390 OAL DC0	BTS 390 OAL	1	15	0	CCU	0	0	0	21	Level 2 whose DirectConnectFlag is set to NO	-
	BTS 390 OAL	1	8	0	TCU	1	15	0	7	-	CMUF
	BTS 390 OAL	1	11	0	FMU	1	15	1	14	-	FAU
	BTS 390 OAL	1	8	1	TCU	1	15	3	28	-	HAU0, optional

Cabinet	Cabinet Type	CN	SRN	SN	BT	MCN	MSRN	MPN	ADDR	Other Parameter	Note
	BTS3900AL	1	8	2	TCU	1	15	3	29	-	HAU1, optional
TMC11H (Ver. C) 0	TMC	2	8	0	TCU	0	15	7	7	-	CMUE

Figure 4-73 shows the monitoring signal cable connections in a BTS3900AL housed in one AC cabinet, one DC cabinet, and one IBBS700D/IBBS700T cabinet.

Figure 4-73 Monitoring signal cable connections in a BTS3900AL (1 AC cabinet+1 DC cabinet+1 IBBS700D/IBBS700T)

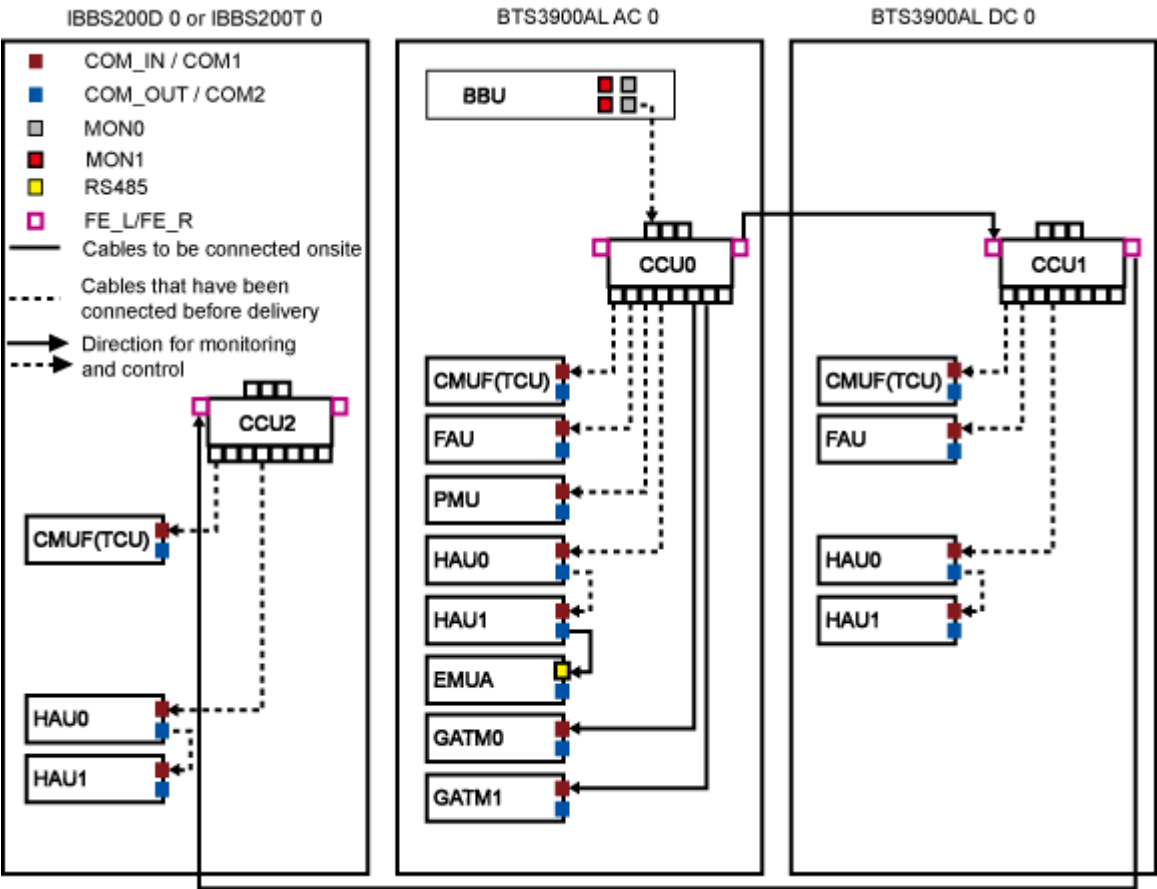


Table 4-61 describes the monitoring board configurations in this scenario.

Table 4-61 Monitoring board configurations in a BTS3900AL (1 AC cabinet+1 DC cabinet+1 IBBS700D/IBBS700T)

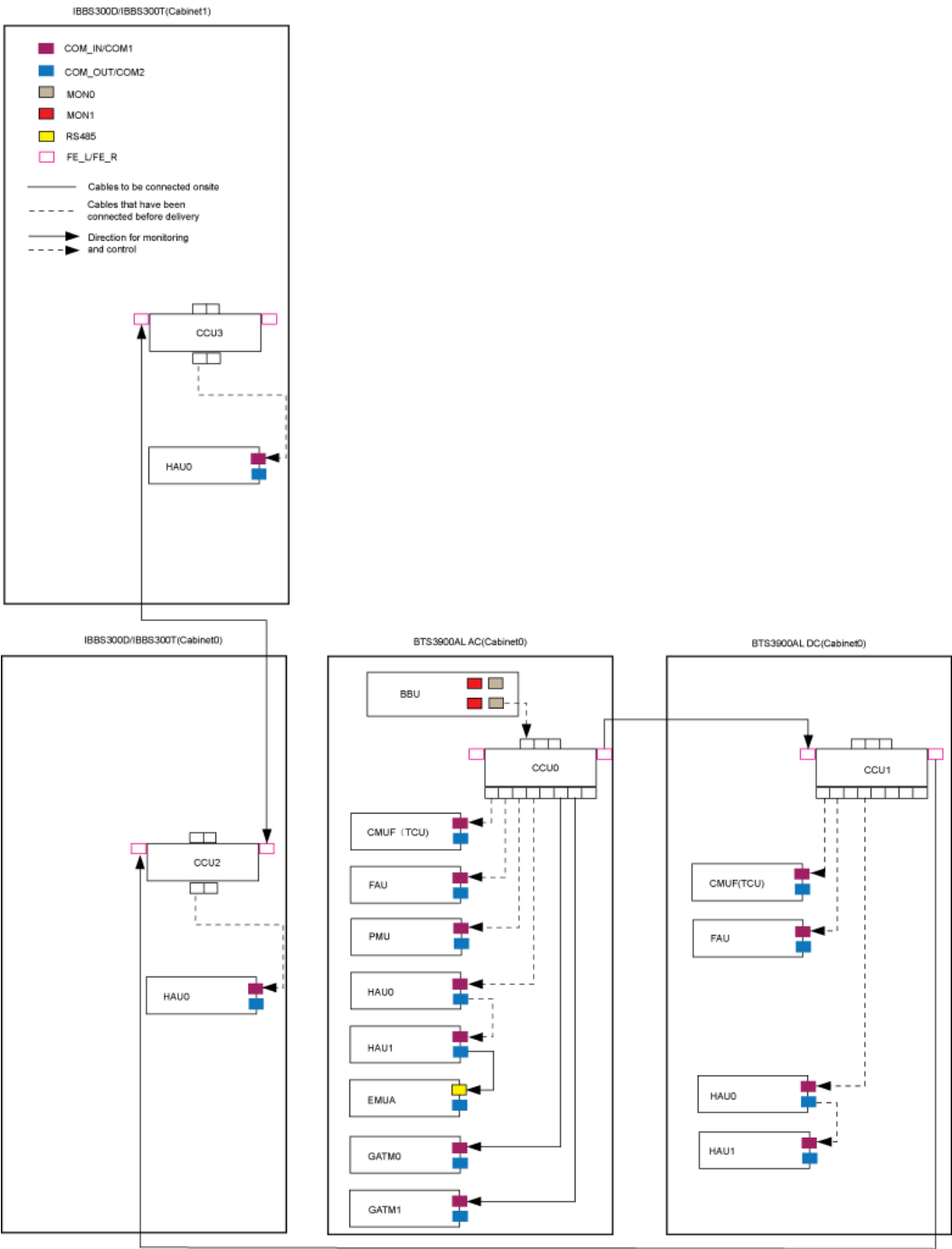
Cabin et	Cabin et Type	CN	SRN	SN	BT	MC N	MS RN	MP N	AD DR	Other Par ame ter	Not e
BTS 390 0AL AC 0	BTS 390 0AL	0	15	0	CCU	0	0	0	21	Level 1 whose DirectConnectFlag is set to YES. Connected RAT : Indicates the working standard of the manager unit to which the CCU is connected.	-
	BTS 390 0AL	0	8	0	TCU	0	15	0	7	-	CMUF

Cabin	Cabin Type	CN	SRN	SN	BT	MCN	MSRN	MPN	ADDR	Other Parameter	Note
	BTS 390 OAL	0	11	0	FMU	0	15	1	14	-	FAU
	BTS 390 OAL	0	7	0	PMU	0	15	2	3	-	-
	BTS 390 OAL	0	8	1	TCU	0	15	3	28	-	HAU0, optional
	BTS 390 OAL	0	8	2	TCU	0	15	3	29	-	HAU1, optional
	BTS 390 OAL	0	40	0	EMU	0	15	3	2	-	Optional
BTS 390 OAL DC0	BTS 390 OAL	1	15	0	CCU	0	0	0	21	Level 2 whose DirectConnectFlag is set to NO	-
	BTS 390 OAL	1	8	0	TCU	1	15	0	7	-	CMUF
	BTS 390 OAL	1	11	0	FMU	1	15	1	14	-	FAU
	BTS 390 OAL	1	8	1	TCU	1	15	3	28	-	HAU0, optional

Cabin et	Cabin et Typ e	CN	SRN	SN	BT	MC N	MS RN	MP N	AD DR	Oth er Par ame ter	Not e
	BTS 390 0AL	1	8	2	TCU	1	15	3	29	-	HA U1, opti onal
IBBS 700 D/ IBBS 700 T 0	BBC	9	15	0	CCU	0	0	0	21	Leve l 3 who se Dire ctCo nne ctFl ag is set to NO	-
	BBC	9	8	0	TCU	9	15	0	7	-	CM UF
	BBC	9	8	1	TCU	9	15	3	28	-	HA U0, opti onal
	BBC	9	8	2	TCU	9	15	3	29	-	HA U1, opti onal

Figure 4-74 shows the monitoring signal cable connections in a BTS3900AL housed in one AC cabinet, one DC cabinet, and two IBBS300D/IBBS300T cabinets.

Figure 4-74 Monitoring signal cable connections in a BTS3900AL (1 AC cabinet+1 DC cabinet+2 IBBS300D/IBBS300T cabinets)



The following table describes the monitoring board configurations in this scenario.

Table 4-62 Monitoring board configurations in a BTS3900AL (1 AC cabinet+1 DC cabinet+2 IBBS300D/IBBS300T cabinets)

Cabin et	Cabin et Type	CN	SRN	SN	BT	MC N	MS RN	MP N	AD DR	Other Par ame ter	Not e
BTS 390 0AL AC 0	BTS 390 0AL	0	15	0	CCU	0	0	0	21	Level 1 whose DirectConnectFlag is set to YES. Connected RAT : Indicates the working standard of the manager unit to which the CCU is connected.	-
	BTS 390 0AL	0	8	0	TCU	0	15	0	7	-	CMUF

Cabin	Cabin Type	CN	SRN	SN	BT	MCN	MSRN	MPN	ADDR	Other Parameter	Note
	BTS 390 OAL	0	11	0	FMU	0	15	1	14	-	FAU
	BTS 390 OAL	0	7	0	PMU	0	15	2	3	-	-
	BTS 390 OAL	0	8	1	TCU	0	15	3	28	-	HAU0, optional
	BTS 390 OAL	0	8	2	TCU	0	15	3	29	-	HAU1, optional
	BTS 390 OAL	0	40	0	EMU	0	15	3	2	-	Optional
BTS 390 OAL DC0	BTS 390 OAL	1	15	0	CCU	0	0	0	21	Level 2 whose DirectConnectFlag is set to NO	-
	BTS 390 OAL	1	8	0	TCU	1	15	0	7	-	CMUF
	BTS 390 OAL	1	11	0	FMU	1	15	1	14	-	FAU
	BTS 390 OAL	1	8	1	TCU	1	15	3	28	-	HAU0, optional

Cabin et	Cabin et Type	CN	SRN	SN	BT	MC N	MS RN	MP N	AD DR	Other Par ame ter	Not e
	BTS 390 0AL	1	8	2	TCU	1	15	3	29	-	HA U1, opti onal
IBBS 300 D/ IBBS 300 T 0	BBC	9	15	0	CCU	0	0	0	21	Level 3 whose DirectCo nne ctFlag is set to NO	-
	BBC	9	8	0	TCU	9	15	0	7	-	-
	BBC	9	8	1	TCU	9	15	0	28	-	HA U0, opti onal
IBBS 300 D/ IBBS 300 T 1	BBC	10	15	0	CCU	0	0	0	21	Level 4 whose DirectCo nne ctFlag is set to NO	-
	BBC	10	8	0	TCU	10	15	0	7	-	-
	BBC	10	8	1	TCU	10	15	0	28	-	HA U0, opti onal

Example of Configurations in a BTS5900 (-48 V DC)

The monitoring signal cable connections in a BTS5900 using -48 V DC power input are similar to the example provided in [Example of Configurations in a BTS3900 \(-48 V DC\)](#).

Example of Configurations in a BTS5900L (-48 V DC)

The monitoring signal cable connections in a BTS5900L using -48 V DC power input are similar to the example provided in [Example of Configurations in a BTS3900L \(-48 V DC\)](#).

Example of Configurations in a BTS5900A (AC)

The monitoring signal cable connections in a BTS5900A using AC power input are similar to the example where APM30H (Ver.E) is used in [Example of Configurations in a BTS3900A \(AC\)](#).

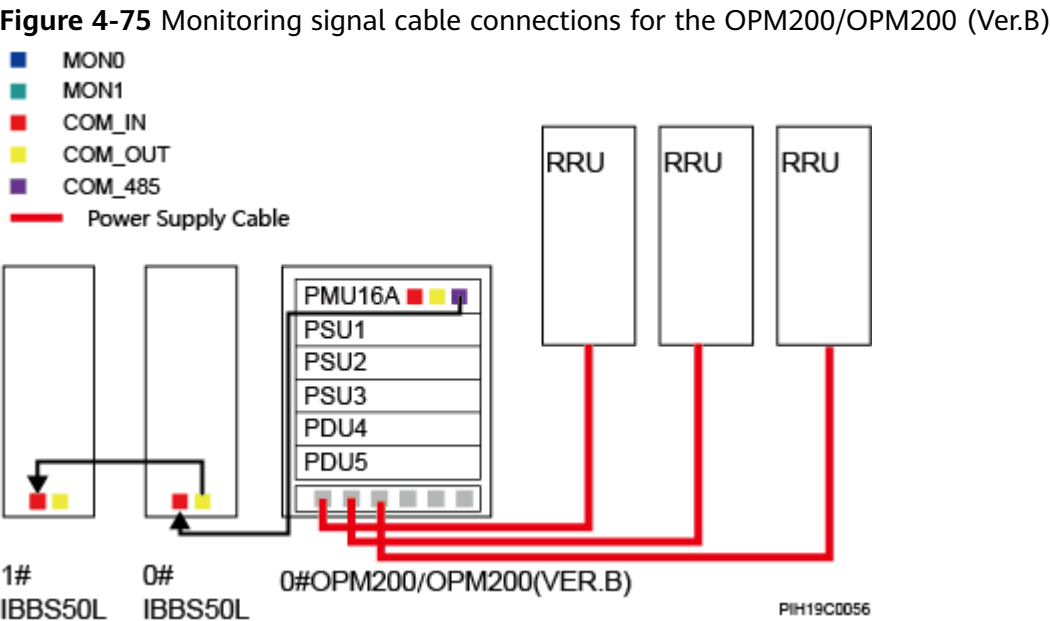
Example of Configurations in a BTS5900A (DC)

The monitoring signal cable connections in a BTS5900A using -48 V DC power input are similar to the example where the APM30H (Ver.E) is used in [Example of Configurations in a BTS5900A \(DC\)](#).

Example of Configurations in a DBS5900

- [Example of Configurations in a DBS3900 \(AC\)](#) shows the monitoring signal cable connections in a DBS5900 supplied with AC power.
- [Example of Configurations in a DBS3900 \(-48 V DC\)](#) shows the monitoring signal cable connections in a DBS5900 supplied with -48 V DC power.

The following figure shows the connections of monitoring signal cables when the OPM200/OPM200 (Ver.B) is configured and RRUs are used for remote monitoring.



The following table describes the monitoring board configurations in this scenario.

Table 4-63 Monitoring board configurations for an OPM200/OPM200 (Ver.B)

Cabine t	Ca bi ne t T y p e	CN	SR N	SN	BT	MCN	MS RN	MP N	AD DR	Oth er P ar a m e t e r	Sett ing Not es
- OPM200 - OPM200(Ver.B)	OPM	5 ^a	7	0	PMU	Cabinet number of the connected RRU	Subrack number of the connected RRU	0	3	-	Power System Type is set to ETP.

NOTE

a: The cabinet number is for reference only.

The following figure shows the connections of monitoring signal cables when the OPM200/OPM200 (Ver.B) is configured and the BBU5900A is used for monitoring.

Figure 4-76 BBU5900A monitoring signal cable connection for the OPM200/OPM200 (Ver.B)

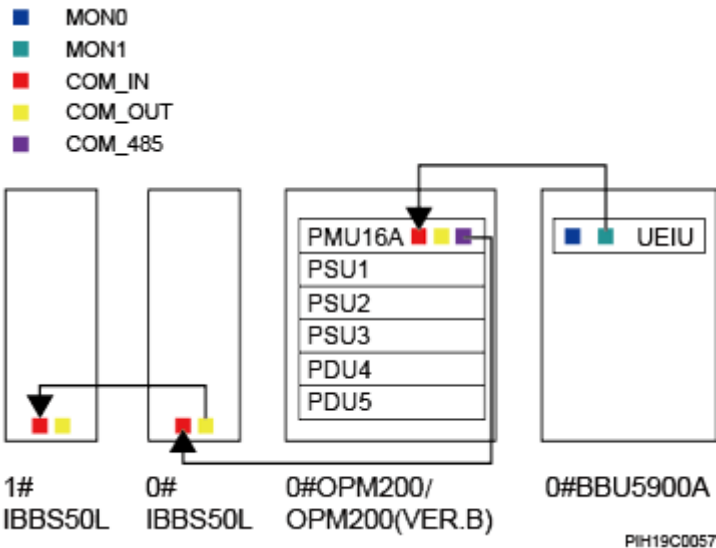


Table 4-64 Monitoring board configurations for an OPM200

Cabin et	Ca bin et Typ e	CN	SRN	SN	BT	MC N	MS RN	MP N	AD DR	Oth er Par ame ter	Sett ing Not es
OPM 200/ OPM 200 (Ver.B)	OP M	5 ^a	7	0	PM U	Cabi net num ber of the BBU that con nect s to this boar d	Subr ack num ber of the BBU that con nect s to this boar d	1	3	-	Pow er Syst em Typ e is set to ETP .

NOTE

a: The cabinet number is for reference only.

Example of Configurations in a DBS5900E

The DBS5900E is an outdoor base station and is used together with the BBU5900C/BookBBU5901/BookBBU5901a. As there is no monitoring port on the BBU5900C/BookBBU5901/BookBBU5901a, the power system is connected to the monitoring port on an RRU, and then the RRU is connected to the BBU using a CPRI optical fiber. In this way, the power system is monitored by the base station. [Figure 4-77](#) and [Figure 4-78](#) show the monitoring signal cable connections in a DBS5900E.

Figure 4-77 Example 1 of monitoring signal cable connections in a DBS5900E

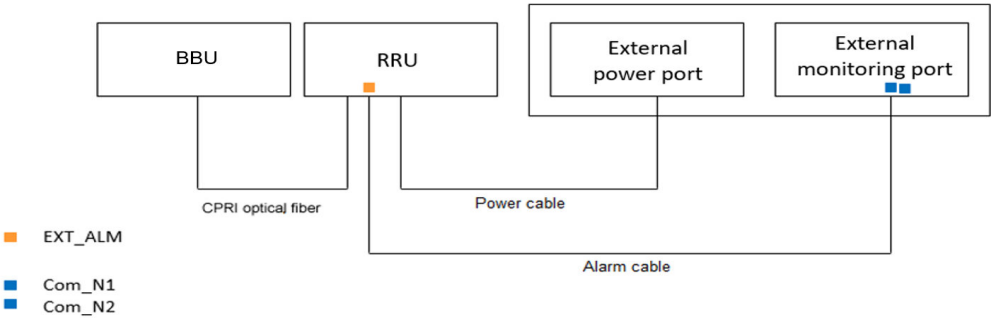
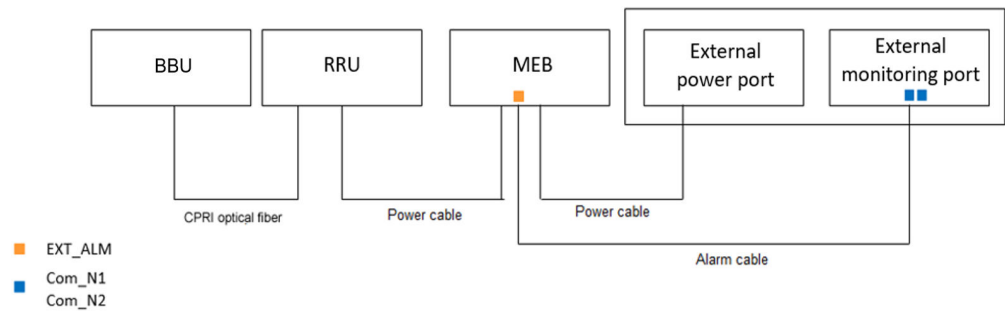


Figure 4-78 Example 2 of monitoring signal cable connections in a DBS5900E



The following table lists typical RRU monitoring port types.

Table 4-65 Typical RRU monitoring port types

Base Station Side				Monitored Equipment Side
Typical RRU Type	Port	Port Description	Connector	Typical device/Port position
RRU3936/ RRU3959	EXT_ALM (RRU)	RRU alarm port, used for monitoring one RS485 input and two Boolean inputs	DB15 waterproof male connector	OPM50M/ COM_IN
RRU5909	EXT_ALM (MEB06D/ MEB06D (Ver.B))	MEB alarm port, which can carry external RS485 alarm signals or dry contact alarm signals on a power circuit.	DB15 waterproof male connector	

NOTE

- For details about how to connect an alarm cable to an RRU, see section "Installing an RRU Alarm Cable" in "Installing RRU Cables" of the corresponding RRU installation guide in *3900 & 5900 Series Base Station Product Documentation*.
- For details about how to connect an alarm cable to an MEB, see section "Installing an MEB Alarm Cable" in *MEB06D User Guide* or *MEB06D (Ver.B) User Guide* in *3900 & 5900 Series Base Station Product Documentation*.
- For details about how to connect an alarm cable to an OPM50M, see section "Installing an OPM50M Alarm Cable" in *OPM50M User Guide* in *3900 & 5900 Series Base Station Product Documentation*.

4.4.3.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Scenario 1: Configuring the PMU

For details about how to configure the PMU, see *Power Supply Management*.

Scenario 2: Configuring the FMU

Run the **ADD FMU** command to add an FMU.

The following is an MML command example:

```
//Adding an FMU with the parameters as follows: Cabinet No.: 0; Subrack No.: 11; Manager Port No.: 0;  
Address: 14  
ADD FMU: CN=0, SN=0, SRN=11, MPN=0, ADDR=14, STC=ENABLE;
```

Scenario 3: Configuring the TCU

Run the **ADD TCU** command to add a TCU.

Scenario 4: Configuring the CCU

Run the **ADD CCU** command to add a CCU.

Scenario 5: Configuring Customized Alarms

- Step 1** Run the **ADD EMU** command to add an EMU as required. Skip this step if the EMU is not required.
- Step 2** Run the following commands to query the status of the input and output alarm ports:
- Run the **LST ALMPORT** command to query the status of the input alarm port.
 - Run the **LST OUTPORT** command to query the status of the output alarm port.
- Step 3** Based on the query results, run the following commands to enable the input alarm port and disable the unused alarm port:
- Run the **SET ALMPORT** command to set the input alarm port.
 - Run the **SET OUTPORT** command to set the output alarm port.
- Step 4** Add and bind the customized alarms on the base station to the MAE so that the alarms can be identified and reported.

Log in to the MAE. Choose **Monitor > Alarms > NE Alarm Settings** from the main menu. In the navigation tree on the left, choose **User-Defined Alarm** to add configurations, bind the alarms, and modify the alarm severity. For details, see "Operation and Maintenance > Fault Management > Alarms > Configuring Alarms or Events > Precise Monitoring > Defining an NE Alarm" in the MAE product documentation.

 NOTE

If the output alarm port is configured, you do not need to add and bind the user-defined alarms on the MAE.

----End

The following are MML command examples:

```
//Configuring an input alarm port with the parameters as follows: Cabinet No.: 0; Subrack No. (where the EMU is located): 40; Port No.: 0; Alarm ID: 65033; Alarm Voltage: HIGH  
SET ALMPORT: CN=0, SRN=40, SN=0, PN=0, SW=ON, AID=65033, PT=BOOL, AVOL=HIGH;  
//Configuring an output alarm port with the parameters as follows: Cabinet No.: 0; Subrack No. (where the EMU is located): 40; Port No.: 0; Port Name: Water Alarm  
SET OUTPORT: CN=0, SRN=40, SN=0, PN=0, NAME="Water Alarm", SW=ON;
```

The following are MML command examples for configuring customized alarms for the HAU03A-02 (an AC heater):

- In an APM5950H/APM5950H-L cabinet, the heater alarm output cable is connected to PIN4 and PIN18 (mapped to Boolean port 0) of the SIG1 port on the PMIE. The alarm level is low.
//Configuring an input alarm port with the parameters as follows: Cabinet No.: 0; Subrack No. (where the PMIE is located): 7; Port No.: 0; Alarm ID: 65033; Alarm Voltage: LOW
SET ALMPORT: CN=0, SRN=7, SN=0, PN=0, SW=ON, AID=65033, PT=BOOL, AVOL=LOW, USERLABEL="HeatingBox";
//Setting the environment alarm parameters as follows: Alarm ID: 65033; Alarm Name: Heater Fault; Alarm Level: MINOR; Event Type: ENV
SET ENVALMPARA: ALMID=65033, ANM="Heater Fault", ALVL=MINOR, ASS=ENV;
- In an APM5900H/APM5900H-L cabinet, the heater alarm output cable is connected to the IN0 port (mapped to Boolean port 0) on the FAN02I/FAN02L. The alarm level is high.
//Configuring an input alarm port with the parameters as follows: Cabinet No.: 0; Subrack No. (where the FAN02I/FAN02L is located): 8; Port No.: 0; Alarm ID: 65033; Alarm Voltage: HIGH
SET ALMPORT: CN=0, SRN=8, SN=0, PN=0, SW=ON, AID=65033, PT=BOOL, AVOL=HIGH, USERLABEL="HeatingBox";
//Setting the environment alarm parameters as follows: Alarm ID: 65033; Alarm Name: Heater Fault; Alarm Level: MINOR; Event Type: ENV
SET ENVALMPARA: ALMID=65033, ANM="Heater Fault", ALVL=MINOR, ASS=ENV;

Scenario 6: Configuring the MEB

Run the **ADD MEB** command to add an MEB.

The following is an MML command example:

```
//Adding an MEB with the parameters as follows: Cabinet No.: 0; Subrack No.: 40; Manager Subrack No.: 60  
ADD MEB: CN=0, SN=0, SRN=40, MCN=0, MSRN=60;
```

Scenario 7: Configuring the GDU

Run the **ADD GDU** command to add a GDU.

The following is an MML command example:

```
//Adding a GDU with the parameters as follows: Cabinet No.: 0; Subrack No.: 40; Manager Port No.: 0; Address: 1  
ADD GDU: CN=0, SN=0, SRN=40, MPN=0, ADDR=1, PN=AUTO;
```

4.4.3.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.4 Activation Verification

Step 1 Run the following MML commands to check whether the boards are working properly. If the system returns any command output for a board, the board is working properly. When all the boards are working properly, proceed to [Step 2](#).

- DSP PMU
- DSP FMU
- DSP TCU
- DSP CCU
- DSP EMU
- DSP MEB
- DSP GDU

Step 2 Perform the operations described in [Table 4-66](#) and check whether environment alarms are correctly reported. If alarms cannot be reported, check hardware connections and software configurations.

Table 4-66 Verifying monitoring alarms

Alarm ID	Alarm Name	Alarm Triggering Method
25652 ¹	Cabinet Temperature Unacceptable	Set the upper limit of the cabinet temperature alarm TUTHD 5°C lower than the ambient temperature or set the lower limit TLTHD 5°C higher than the ambient temperature, and wait more than 10s. Set the preceding parameters only when the PMU is configured. Select the configuration object of the board to which the sensor is connected.
25672 ²	Burglar Alarm	Open the cabinet door and wait more than 10s.
25654 ³	Battery Temperature Unacceptable	In a high temperature scenario: Set the upper limit of the battery temperature alarm TUTHD lower than the ambient temperature or set the ambient temperature higher than the upper limit, and wait more than 2s. In a low temperature scenario: Set the lower limit of the battery temperature alarm TLTHD higher than the ambient temperature or set the ambient temperature lower than the lower limit, and wait more than 2s. Set the preceding parameters only when batteries are configured. For details, see <i>Power Supply Management</i> .

Alarm ID	Alarm Name	Alarm Triggering Method
25650 ⁴	Ambient Temperature Unacceptable	In a high temperature scenario: Set the upper limit of the ambient temperature alarm TUTHD lower than the ambient temperature or set the ambient temperature higher than the upper limit, and wait more than 2s. In a low temperature scenario: Set the lower limit of the ambient temperature alarm TLTHD higher than the ambient temperature or set the ambient temperature lower than the lower limit, and wait more than 2s. Set the preceding parameters only when the EMU is configured.
25651	Ambient Humidity Unacceptable	In a high humidity scenario: Set the upper limit of the ambient humidity alarm HUTHD lower than the ambient humidity or set the ambient humidity higher than the upper limit, and wait more than 2s. In a low humidity scenario: Set the lower limit of the ambient humidity alarm HLTHD higher than the ambient humidity or set the ambient humidity lower than the lower limit, and wait more than 2s. Set the preceding parameters only when the EMU is configured.
25653	Cabinet Humidity Unacceptable	Set the upper limit of the cabinet humidity alarm HUTHD 10% RH lower than the cabinet humidity or set the lower limit of HLTHD 10% RH higher than the cabinet humidity, and wait more than 10s. Set the preceding parameters only when the EMU is configured.
25670	Water Alarm	Perform either of the following operations. If an alarm is reported, the data is properly configured. - Place the water sensor in water for more than 10s. - Disconnect the water sensor.
25671	Smoke Alarm	Perform the following operation. If an alarm is reported, the data is properly configured. Generate smoke around the smoke sensor for more than 15s.
25740 ⁵	GDU Antenna Fault	Perform the following operation. If an alarm is reported, the data is properly configured. Remove the input antenna cable for the GDU+SLPA and maintain this state for a period of time (90s by default).

Alarm ID	Alarm Name	Alarm Triggering Method
25741 ⁵	GDU Temperature Abnormal	This alarm is reported when the temperature of the GDU+SLPA is beyond the normal range. This alarm is cleared when the device temperature stays within the normal range for 3 minutes.
25742 ⁵	GDU Signal Output Abnormal	Perform either of the following operations. If an alarm is reported, the data is properly configured. - Remove the feeder between the GDU+SLPA and the GDU and keep the faulty state for 180s (by default) or longer. - Remove all the feeders between the GDU and the BBU and keep the faulty state for 180s (by default) or longer.

NOTE

1. This alarm is reported only when the temperature sensor is configured by default or is installed in the cabinet by operators themselves.

The temperature sensor is not configured by default in the BTS3900, BTS3900L, or BTS3900C cabinet.

The temperature sensor is configured by default in the BTS3900A (with RFC), DBS3900 (with RFC), and BTS3900AL cabinets.

2. This alarm is reported only when the door status sensor is configured by default or is installed in the cabinet by operators themselves.

The door status sensor is not configured by default in the BTS3900, BTS3900L, and DBS3900 (with the IMB) cabinets.

The door status sensor is configured by default in the BTS3900A, DBS3900 (with the APM30H, TMC, or TP48600A), and BTS3900AL cabinets.

3. This alarm is reported only when batteries are configured in the cabinet.

4. This alarm is reported only when the EMU is configured.

5. This alarm is reported only when the GDU is configured.

Step 3 If a customized Boolean alarm device is connected, check whether an alarm is reported by setting the alarm device to high level or low level. Verify that the alarms are correctly configured.

----End

4.4.5 Network Monitoring

None

4.4.6 Reconfiguration

When reconfiguring parameters for monitoring devices, collect information about parameters to be reconfigured based on onsite hardware configuration and environment monitoring requirements. For details, see [4.4.3.1.1 MO Setting Notes](#).

Run the **MOD PMU** command to reconfigure PMU parameters.

Run the **MOD FMU** command to reconfigure FMU parameters.

Run the **MOD TCU** command to reconfigure TCU parameters.

Run the **MOD CCU** command to reconfigure CCU parameters.

Run the **MOD EMU** command to reconfigure EMU parameters.

Run the **MOD MEB** command to reconfigure MEB parameters.

Run the **MOD GDU** command to reconfigure GDU parameters.

4.5 Operation and Maintenance (GBTS)

4.5.1 When to Use

It is recommended that the monitoring management feature be enabled when you intend to monitor power and fan units in a base station. If auxiliary devices with monitoring functions are configured for a base station, it is recommended that customized alarm function be enabled.

4.5.2 Precautions

Single-side configuration is recommended for monitoring devices in a multimode base station. The monitoring devices can also be configured in dual-side configuration mode. For details about the dual-side configuration, see section "[4.6 Operation and Maintenance \(Multimode Base Station\)](#)".

4.5.3 Data Configuration

The scenarios for configuring monitoring management can be used together. If they are used together, data for all these scenarios must be configured. There is no fixed sequence between the configuration scenarios. When the EMU/EMUB is installed for configuring customized alarms, configure the EMU, and then the **ALMPORT** or **OUTPORT MO**.

4.5.3.1 Data Preparation

4.5.3.1.1 MO Setting Notes

The following scenarios can co-exist. If they are used together, data for all these scenarios must be prepared.

Scenario 1: Configuring the PMU

For the parameter settings for the PMU, see *Power Supply Management*.

Scenario 2: Configuring the FMU

The following table describes the parameter settings for the FMU.

Table 4-67 Key parameters to be set for the FMU

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	BTSFMUABP.CN	If the number of the RFC housing the FMU is 1, set this parameter to 1 .
Subrack No.	BTSFMUABP.SRN	This parameter specifies the number of the subrack housing the FMU. Set this parameter to 11 or 12 .
Slot No.	BTSFMUABP.SN	This parameter specifies the number of the slot housing the FMU. Set this parameter to the default value 0 .
Board Parameter Configuration Enabled	BTSFMUABP.CFGFLAG	Set this parameter to Yes .
Smart Temperature Control	BTSFMUABP.STC	This parameter specifies whether to enable smart temperature control of the FMU. ENABLED is recommended.
Special Boolean Alarm Flag	BTSFMUABP.SBAF	It is recommended that the FMU not be connected to external sensors and all sensors be disabled by default. If you want to use the sensors, contact Huawei technical support.
Special Alarm Valid Level Configuration	BTSFMUABP.SPECIALSWITCHERLEVEL	This parameter specifies the level specifications for the dedicated Boolean alarm. There are two specifications: high-level and low-level alarms. It is recommended that the dedicated Boolean alarm function be disabled on the FMU. If all sensors in the SBAF parameter are disabled, you do not need to configure this parameter.
Manager Cabinet No.	BTSFMUABP.MCN	The manager of the board is generally a TMU or CCU. TMU is the logical name of the GTMU. Set this parameter to the cabinet number of the TMU or CCU.
Manager Subrack No.	BTSFMUABP.MSRN	The manager of the board is generally a TMU or CCU. TMU is the logical name of the GTMU. Set this parameter to the subrack number of the TMU or CCU.

Parameter Name	Parameter ID	Setting Notes
Manager Port No.	BTSFMUABP.MPN	This parameter specifies the number of the serial port on the manager to which the board is connected. If the FMU is connected to the BBU: • UPEU/UEIU (MON1): 1 • UPEU/UEIU (MON0): 0 If the FMU is connected to the CCU, it must be connected to port D_COM1. Set this parameter to 1.
Address	BTSFMUABP.ADDR	This parameter specifies the communication address of the FMU. A maximum of two FMUs can be configured on an RS485 bus. Set this parameter as follows: • Upper-level FMU in two cascaded FMUs: 14 • Lower-level FMU in two cascaded FMUs: 15
FMU Board Type	BTSFMUABP.FMUTYPE	This parameter specifies the FMUA board type. • Set this parameter to OUTDOOR when the board is installed in an outdoor cabinet. • Set this parameter to INDOOR when the board is installed in an indoor cabinet.
Temperature Control Mode	BTSFMUABP.TCMODE	This parameter specifies which mode is adopted to adjust the fan speed. The value range is 0 to 65535. 0: default mode. This mode achieves a balance among system power consumption, noise, and temperature control. It applies to most scenarios. 1: low noise mode. This mode brings low noise and applies to scenarios with strict requirements for noise control. 2: low temperature rise mode. This mode applies to high-temperature areas. When this mode is used, power consumption and noise are higher than those in the default mode but heat dissipation effect is better than that in the default mode. Other values: Before using other modes, run the DSP BTSTCMODE command to query the temperature control modes and features that the base station supports. If the query result does not contain a certain mode, do not configure this mode. This parameter is valid only when CFGFLAG is set to YES.

Scenario 3: Configuring the TCU

The following table describes the parameter settings for the TCU.

Table 4-68 Key parameters to be set for the TCU

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	BTSDHEUBP.CN	This parameter specifies the number of the cabinet housing the TCU. If the number of the TMC housing the TCU is 1, set this parameter to 1 .
Subrack No.	BTSDHEUBP.SRN	This parameter specifies the number of the subrack housing the TCU. Set this parameter to 8 or 14 .
Slot No.	BTSDHEUBP.SN	This parameter specifies the number of the slot housing the TCU. Value range: 0 to 3 If the TCU is installed in the BTS3900AL AC, BTS3900AL DC, IBBS700D, or IBBS700T cabinet, set this parameter as follows: - Set the slot number of the CMUF/CMUEA to 0. - Set the slot number of HAU0 to 1. - Set the slot number of HAU1 to 2. If the TCU is installed in the TP48600A cabinet, set this parameter as follows: - Set the slot number of the CMUF for cabinet temperature control to 0. - Set the slot number of the CMUF for battery cabinet temperature control to 1. - Set the slot number of HAU0 to 2. - Set the slot number of HAU1 to 3. If the TCU is installed in other cabinets, set this parameter to 0 .
Board Parameter Configuration Enabled	BTSDHEUBP.CFGFLAG	This parameter specifies whether the board configuration is allowed. Set this parameter to Yes .
Temperature Alarm Upper Threshold	BTSDHEUBP.TUTHD	This parameter specifies the upper temperature limit for the cabinet. If the temperature exceeds the upper limit, ALM-25652 Cabinet Temperature Unacceptable is reported. It is recommended that this parameter be set to 50 for the TCU in the BBC and 68 for the TCU in other cabinets.

Parameter Name	Parameter ID	Setting Notes
Temperature Alarm Lower Threshold	BTSDHEUBP.TLTHD	This parameter specifies the lower temperature limit for the cabinet. If the temperature is below the lower limit, ALM-25652 Cabinet Temperature Abnormal is reported. It is recommended that this parameter be set to -33 for the TCU in the OMB and -20 for the TCU in other cabinets.
Manager Cabinet No.	BTSDHEUBP.MCN	The manager of the board is generally a TMU, a CCU, or an RXU. TMU, CCU, and RXU are the logical names of the GTMU, CCU, and RRU respectively. Set this parameter to the cabinet number of the TMU, CCU, or RXU.
Manager Subrack No.	BTSDHEUBP.MSRN	The manager of the board is generally a TMU, a CCU, or an RXU. TMU, CCU, and RXU are the logical names of the GTMU, CCU, and RRU respectively. Set this parameter to the subrack number of the TMU, CCU, or RXU.
Manager Port No.	BTSDHEUBP.MPN	This parameter specifies the number of the serial port on the manager to which the board is connected. If the TCU is connected to the BBU: • UPEU/UEIU (MON1): 1 • UPEU/UEIU (MON0): 0 If the TCU is connected to the CCU: • The CMUF must be connected to port D_COM0 on the CCU. In this case, set this parameter to 0. • The FAU0 or FAU1 must be connected to port D_COM3 on the CCU. In this case, set this parameter to 3. If the module is connected to the RRU, set this parameter to 0.

Parameter Name	Parameter ID	Setting Notes
Address	BTSDHEUBP.ADDR	In the APM30H or TMC11H cabinet, a maximum of two TCUs can be configured on an RS485 bus. Set this parameter as follows: • Upper-level TCU in two cascaded TCUs: 7 • Lower-level TCU in two cascaded TCUs: 6 In the IBBS200D/IBBS200T cabinet, a maximum of four TCUs can be configured on an RS485 bus. • If the upper-level TCU is managed by the PMU whose communication address is 3, set this parameter to 23. If the lower-level TCU is managed by this PMU, set this parameter to 24. • If the upper-level TCU is managed by the PMU whose communication address is 4, set this parameter to 25. If the lower-level TCU is managed by this PMU, set this parameter to 26. If the HAU is installed, a maximum of two TCUs can be configured on an RS485 bus. Set this parameter as follows: • Upper-level TCU in two cascaded TCUs: 28 • Lower-level TCU in two cascaded TCUs: 29
Alarm Parameter Configuration Enabled	BTSDHEUBP.ALMPARACFGFLAG	This parameter specifies whether SBAF configuration is allowed. It is recommended that this parameter be set to No, which means that SBAF is disabled.
Special Boolean Alarm Flag	BTSDHEUBP.SBAF	This parameter specifies whether to report a dedicated Boolean alarm. If the shield flag for a Boolean alarm is selected, the Boolean alarm cannot be reported. It is recommended that the TCU not be connected to external sensors and all sensors be disabled. If you want to use the sensors, contact Huawei technical support. Retain the default settings, which mean that all sensors are disabled.

Parameter Name	Parameter ID	Setting Notes
Temperature Control Mode	BTSDHEUBP.TCMODE	This parameter specifies which mode is adopted to adjust the fan speed. The value range is 0 to 65535. 0: default mode. This mode achieves a balance among system power consumption, noise, and temperature control. It applies to most scenarios. 1: low noise mode. This mode brings low noise and applies to scenarios with strict requirements for noise control. 2: low temperature rise mode. This mode applies to high-temperature areas. When this mode is used, power consumption and noise are higher than those in the default mode but heat dissipation effect is better than that in the default mode. Other values: Before using other modes, run the DSP BTSTCMODE command to query the temperature control modes and features that the base station supports. If the query result does not contain a certain mode, do not configure this mode. This parameter is valid only when CFGFLAG is set to YES.

Scenario 4: Configuring the CCU

The following table describes the parameter settings for the CCU.

Table 4-69 Key parameters to be set for the CCU

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	BTSCCUBP.CM	This parameter specifies the number of the cabinet housing the CCU. If the number of the BTS3900AL cabinet housing the CCU is 0, set this parameter to 0.
Subrack No.	BTSCCUBP.SRN	This parameter specifies the number of the subrack housing the CCU. Set this parameter to the default value 15.
Slot No.	BTSCCUBP.SN	This parameter specifies the number of the slot housing the CCU. Set this parameter to the default value 0.

Parameter Name	Parameter ID	Setting Notes
Board Parameter Configuration Enabled	BTSCCUBP.CFGFLAG	This parameter specifies whether the board configuration is allowed. Set this parameter to Yes .
Manager Cabinet No.	BTSCCUBP.MCN	The manager of the board is generally a TMU, a CCU, or an RXU. TMU, CCU, and RXU are the logical names of the GTMU, CCU, and RRU respectively. Set this parameter to the cabinet number of the TMU, CCU, or RXU. The manager of the CCU is a TMU or an RXU.
Manager Subrack No.	BTSCCUBP.MSRN	The manager of the board is generally a TMU, a CCU, or an RXU. TMU, CCU, and RXU are the logical names of the GTMU, CCU, and RRU respectively. Set this parameter to the subrack number of the TMU, CCU, or RXU. The manager of the CCU is a TMU or an RXU.
Manager Port No.	BTSCCUBP.MPN	This parameter specifies the number of the serial port on the manager to which the board is connected. The CCU must be connected to port MON0 on the BBU. In this case, this parameter must be set to 0 . The CCU must be connected to port EXT_ALM on the RRU. In this case, this parameter must be set to 0 .
Direct Connection Flag	BTSCCUBP.DCF	This parameter specifies whether the CCU is directly connected to its manager. If the CCU is directly connected to its manager, set this parameter to YES . If the CCU is connected to its manager over a cascaded CCU, set this parameter to NO .
Level	BTSCCUBP.CCN	This parameter specifies the position of the CCU on a CCU chain. Set this parameter to 1 for the leftmost CCU on the chain (the CCU whose port FE_L is not connected). Set this parameter to an integer in ascending order for the other CCUs on the chain.
Connected Standard	BTSCCUBP.CS	This parameter specifies the working standard of the BBU that is directly connected to the CCU. Set this parameter to GSM . The parameter is valid only when DCF is set to Yes .

Parameter Name	Parameter ID	Setting Notes
Special Boolean Alarm Enable Flag	BTSCCUBP.SBAF	This parameter specifies whether to report a dedicated Boolean alarm. If the shield flag for a Boolean alarm is selected, the Boolean alarm cannot be reported. Value range: WS_DISABLE(Water-Immersed Sensor Disabled), SS_DISABLE(Smog Sensor Disabled), and GS_DISABLE(Gating Sensor Disabled) It is recommended that the CCU not be connected to external sensors and all sensors be disabled. If you want to use the sensors, contact Huawei technical support. Retain the default settings, which mean that all sensors are disabled.

Scenario 5: Configuring the GATM

The following table describes the parameter settings for the GATM.

Table 4-70 Key parameters to be set for the GATM

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	BTSDATUBP.CN	This parameter specifies the number of the cabinet housing the GATM. If the number of the APM30H housing the GATM is 1, set this parameter to 1.
Subrack No.	BTSDATUBP.SRN	This parameter specifies the number of the subrack housing the GATM. It can be set to a value in the range of 40 to 59.
Slot No.	BTSDATUBP.SN	This parameter specifies the number of the slot housing the GATM. Set this parameter to the default value 0.
Board Parameter Configuration Enabled	BTSDATUBP.CFGFLAG	This parameter specifies whether the board configuration is allowed. Set this parameter to Yes.

Parameter Name	Parameter ID	Setting Notes
ANT0 ALD Power Switch to ANT5 ALD Power Switch	BTSDATUBP.AMPC0 to BTSDATUBP.AMPC5	This parameter specifies whether to provide power to tower-mounted amplifiers (TMAs) of branches 0 through 5. If a branch is connected to a TMA, set this parameter to ON. If a branch is not connected to a TMA, set this parameter to OFF.
ANT0 Alarm Mode to ANT5 Alarm Mode	BTSDATUBP.MODE0 to BTSDATUBP.MODE5	This parameter specifies the mode for reporting the feeding current alarms of TMAs of branches 0 through 5. When the feeding current exceeds the threshold, an alarm is reported. Value range: MODE1, MODE2, and MODE3 • MODE1: Only critical alarms are reported. • MODE2: Warning and critical alarms are reported. In this mode, both warning and critical alarms are reported as fixed-value current. • MODE3: Warning and critical alarms are reported. In this mode, warning alarms are reported as periodic pulse current, and critical alarms are reported as fixed-value current. Periodic pulse current is reported when the alarm reporting requirement is met within a preset period. Compared with fixed-value current, periodic pulse current is easily reported. The value MODE1 is recommended.
ANT0 ALD Over Critical Alarm Threshold(mA) to ANT5 ALD Over Critical Alarm Threshold(mA)	BTSDATUBP.MAJORALMU P0 to BTSDATUBP.MAJORALMU P5	These parameters specify the thresholds for reporting the critical alarms related to the feeding current of the TMA. When the feeding current of the TMA onsite exceeds the threshold, ALM-28002 GATM ALD Current Out of Range is reported.
ANT0 ALD Over Warning Alarm Threshold(mA) to ANT5 ALD Over Warning Alarm Threshold(mA)	BTSDATUBP.MINORALMU P0 to BTSDATUBP.MINORALMU P5	These parameters specify the thresholds for reporting the warning alarms related to the feeding current of the TMA. When the feeding current of the TMA onsite exceeds the threshold, ALM-28002 GATM ALD Current Out of Range is reported.

Parameter Name	Parameter ID	Setting Notes
ANT0 ALD Low Current Alarm Threshold(mA) to ANT5 ALD Low Current Alarm Threshold(mA)	BTSDATUBP.ALMD0 to BTSDATUBP.ALMD5	These parameters specify the thresholds for reporting the warning alarms related to the feeding current of the TMA. When the feeding current of the TMA onsite is less than the threshold, ALM-28002 GATM ALD Current Out of Range is reported.
Manager Cabinet No.	BTSDATUBP.MCN	The manager of the board is generally a TMU, a CCU, or an RXU. TMU, CCU, and RXU are the logical names of the GTMU, CCU, and RRU respectively. Set this parameter to the cabinet number of the TMU, CCU, or RXU.
Manager Subrack No.	BTSDATUBP.MSRN	The manager of the board is generally a TMU, a CCU, or an RXU. TMU, CCU, and RXU are the logical names of the GTMU, CCU, and RRU respectively. Set this parameter to the subrack number of the TMU, CCU, or RXU.
Manager Port No.	BTSDATUBP.MPN	This parameter specifies the number of the serial port on the manager to which the board is connected. If the GATM is connected to the BBU: - GATM0 is connected to MON0 on the UPEU. Set this parameter to 0. - GATM1 is connected to MON1 on the UPEU. Set this parameter to 1. If the GATM is connected to CCU: - The GATM0 must be connected to port D_COM5 on the CCU. In this case, set this parameter to 5. - The GATM1 must be connected to port D_COM6 on the CCU. In this case, set this parameter to 6.
Address	BTSDATUBP.ADDR	This parameter specifies the communication address of the GATM. Set this parameter to 22 .

Scenario 6: Configuring Customized Alarms

To prepare data for customized alarm configuration, perform the following steps:

- Step 1** Prepare EMU/MEB configuration data. Before data preparation, check whether the EMU/MEB needs to be configured by referring to [Table 4-20](#) and [Table 4-21](#). If the EMU/MEB is required, prepare configuration data by referring to the following table. Otherwise, skip this step.

Step 2 Prepare configuration data of the alarm signal input and output ports based on the input and output requirements of customized alarms.

----End

- Prepare the configuration data for the **ALMPORT** MO by referring to [Table 4-72](#).
- Prepare the configuration data for the **OUTPORT** MO by referring to [Table 4-73](#).

Table 4-71 Key parameters to be set for the EMU

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	BTSDEMUBP.CN	This parameter specifies the number of the cabinet housing the EMU. If the number of the BTS3900 cabinet housing the EMU is 0, set this parameter to 0 .
Subrack No.	BTSDEMUBP.SRN	This parameter specifies the number of the subrack housing the EMU. Set this parameter to the default value.
Slot No.	BTSDEMUBP.SN	This parameter specifies the number of the slot housing the EMU. Set this parameter to the default value.
Board Parameter Configuration Enabled	BTSDEMUBP.CFGFLAG	This parameter specifies whether the board configuration is allowed. Set this parameter to Yes .
Special Analog Alarm Flag	BTSDEMUBP.SAAF	This parameter specifies whether to report a dedicated Boolean alarm. If the shield flag for a Boolean alarm is selected, the Boolean alarm cannot be reported. Set this parameter as follows: • The -48 V alarm is used to check whether the input voltage for the cabinet is normal. When the input voltage for the cabinet is -48 V, deselect the 48V_DISABLE(-48 Voltage Disabled) option. • Set this parameter based on the types of sensors connected to the EMU.
Special Boolean Alarm Flag	BTSDEMUBP.SBAF	This parameter specifies whether to report a dedicated Boolean alarm. If the shield flag for a Boolean alarm is selected, the Boolean alarm cannot be reported. Set this parameter according to the type of the sensor or external device connected to the EMU.

Parameter Name	Parameter ID	Setting Notes
Special Alarm Switch Valid Vol. Level	BTSDMUBP.SASV	This parameter specifies the level specifications for the dedicated Boolean alarm. There are two specifications: high-level and low-level alarms. Set this parameter according to the type of the sensor or external device connected to the EMU.
Manager Cabinet No.	BTSDMUBP.MCN	The manager of the board is generally a TMU, a CCU, or an RXU. TMU, CCU, and RXU are the logical names of the GTMU, CCU, and RRU respectively. Set this parameter to the cabinet number of the TMU, CCU, or RXU.
Manager Subrack No.	BTSDMUBP.MSRN	The manager of the board is generally a TMU, a CCU, or an RXU. TMU, CCU, and RXU are the logical names of the GTMU, CCU, and RRU respectively. Set this parameter to the subrack number of the TMU, CCU, or RXU.
Manager Port No.	BTSDMUBP.MPN	This parameter specifies the number of the serial port on the manager to which the board is connected. If the EMU is connected to the BBU: • UPEU/UEIU (MON1): 1 • UPEU/UEIU (MON0): 0 If the EMU is connected to the CCU, it must be connected to port D_COM3. Set this parameter to 3 . If the module is connected to the RRU, set this parameter to 0 .
Address	BTSDMUBP.ADDR	This parameter specifies the communication address of the EMU. Set this parameter to 2 .
Temperature Alarm Lower Threshold	BTSEXD.TLTHD	This parameter specifies the lower ambient temperature limit. If the ambient temperature is below the lower limit, the base station reports ALM-25650 Ambient Temperature Unacceptable. This parameter needs to be configured only when TS_DISABLE in SAAF is cleared. The default value is recommended for this parameter.

Parameter Name	Parameter ID	Setting Notes
Temperature Alarm Upper Threshold	BTSEXD.TUTHD	This parameter specifies the upper ambient temperature limit. If the ambient temperature exceeds the upper limit, the base station reports ALM-25650 Ambient Temperature Unacceptable. This parameter needs to be configured only when TS_DISABLE in SAAF is cleared. The default value is recommended for this parameter.
Humidity Alarm Lower Threshold	BTSEXD.HLTHD	This parameter specifies the lower ambient humidity limit. If the ambient humidity is below the lower limit, ALM-25651 Ambient Humidity Unacceptable is reported. This parameter needs to be configured only when HS_DISABLE in SAAF is cleared. The default value is recommended for this parameter.
Humidity Alarm Upper Threshold	BTSEXD.HUTHD	This parameter specifies the upper ambient humidity limit. If the ambient humidity exceeds the upper limit, ALM-25651 Ambient Humidity Unacceptable is reported. This parameter needs to be configured only when HS_DISABLE in SAAF is cleared. The default value is recommended for this parameter.

Table 4-72 Key parameters to be set in the **ALMPORT MO**

Parameter Name	Parameter ID	Setting Notes
Subrack No.	ALMPORT.SRN	This parameter specifies the subrack number of the board where the alarm port is located.
Port No.	ALMPORT.PN	This parameter specifies the number of the alarm port on each monitoring board as follows: UPEU/UEIU: 0 to 7 (for Boolean alarms) RRU: 0 to 3 (for Boolean alarms) EMU/EMUB: 0 to 31 (for Boolean alarms) EMU/EMUB: 32 to 35 (for analog alarms) BBU3910A: 0 to 5 (for Boolean alarms)

Parameter Name	Parameter ID	Setting Notes
Switch	ALMPORT.SW	Turn on this switch for the port configured with a customized alarm. This parameter specifies the state of the port switch and can be set as follows: OFF(Off): indicates that the port is disabled and alarms cannot be detected. This is the default value for this parameter. ON(On): indicates that the port is enabled and alarms can be reported through the port according to the customized alarm IDs. In addition, the alarm voltage level (specified by the AVOL parameter) for Boolean alarms or the threshold for alarm generation can be set.
Alarm ID	ALMPORT.AID	This parameter specifies the alarm ID pertaining to the alarm port. If an alarm is detected through the port, the value of this parameter is used as the alarm ID and is reported with the alarm. The value range of this parameter is 65033 to 65233. The alarm name, alarm severity, and OSS type corresponding to the alarm ID are configured and saved on the MAE client.
Port Type	ALMPORT.PT	This parameter specifies the type of an alarm port. Set this parameter to BOOL or VALUE. An analog port, provided by the EMU, is used to monitor the external environment that changes continuously. A digital port, provided by the UPEU, UEIU, RRU, or EMU, is used to monitor the external environment with Boolean or discrete values.
Alarm Voltage	ALMPORT.AVOL	This parameter is valid only when PT is set to BOOL. This parameter specifies whether the sensor sends the port a high-level or low-level alarm. Set this parameter based on the alarm cable connection. For details about the pin assignment for the wires of alarm cables, see the following sections: • "BBU Alarm Cable" in <i>3900 & 5900 Series Base Station Cables</i> • "RRU Alarm Cable" in the RRU-specific hardware description • Section "RS485 Serial Port" in <i>EMU User Guide</i> and <i>EMUB User Guide</i>

Parameter Name	Parameter ID	Setting Notes
Upper Limit	ALMPORT.UL	This parameter specifies the alarm upper limit. If the environment measurement value is greater than the alarm upper limit of the sensor, an alarm is reported. Environment measurement value = Measurement lower limit + (Measurement upper limit - Measurement lower limit) x (Observed current or voltage - Output lower limit)/(Output upper limit - Output lower limit). UL must be less than or equal to SMUL and greater than LL.
Lower Limit	ALMPORT.LL	This parameter specifies the alarm lower limit. If the environment measurement value is below the alarm lower limit of the sensor, the alarm is reported. Environment measurement value = Measurement lower limit + (Measurement upper limit - Measurement lower limit) x (Observed current or voltage - Output lower limit)/(Output upper limit - Output lower limit). LL must be greater than or equal to SMLL and less than UL.
Sensor Type	BTSINVALMPORT.ST	This parameter specifies the type of a sensor. Set this parameter to VOLTAGE or CURRENT based on the external sensor specifications.
Sensor Measurement Upper Limit	BTSINVALMPORT.SMUL	This parameter is valid only when PT is set to VALUE. This parameter specifies the measurement upper limit of the analog port. Set this parameter based on the site conditions, such as the temperature and humidity. SMUL must be greater than or equal to UL. SMUL must be greater than SMLL.
Sensor Measurement Lower Limit	BTSINVALMPORT.SMLL	This parameter is valid only when PT is set to VALUE. This parameter specifies the measurement lower limit of the analog port. Set this parameter based on the site conditions, such as the temperature and humidity. SMLL must be less than or equal to LL. SMLL must be less than SMUL.

Parameter Name	Parameter ID	Setting Notes
Sensor Output Signal Upper Limit	BTSENVALMPORT.SOUL	This parameter is valid only when PT is set to VALUE. This parameter specifies the output signal upper limit of the analog port. For a sensor of either the current type or the voltage type, set this parameter based on the external sensor specifications. SOUL must be greater than SOLL.
Sensor Output Signal Lower Limit	BTSENVALMPORT.SOLL	This parameter is valid only when PT is set to VALUE. This parameter specifies the output signal lower limit of the analog port. For a sensor of either the current type or the voltage type, set this parameter based on the external sensor specifications. SOLL must be less than SOUL.

Table 4-73 Key parameters to be set in the **OUTPORT MO**

Parameter Name	Parameter ID	Setting Notes
Refrigeration Equipment	BTSOUTPUT.COLDSET	This parameter specifies whether to start the refrigeration equipment of the base station. Set this parameter to channel 0 output of the EMU.
Heating Equipment	BTSOUTPUT.WARMSET	This parameter specifies whether to start a heating device of the base station. Set this parameter to channel 1 output of the EMU.
Dehumidification Equipment	BTSOUTPUT.DRYSET	This parameter specifies whether to start a dehumidifier of the base station. Set this parameter to channel 2 output of the EMU.
Humidification Equipment	BTSOUTPUT.WETSET	This parameter specifies whether to start a humidifier of the base station. Set this parameter to channel 3 output of the EMU.
Fire-extinguishing Equipment	BTSOUTPUT.FIRESET	This parameter specifies whether to start a fire-extinguishing device of the base station. Set this parameter to channel 4 output of the EMU.

Parameter Name	Parameter ID	Setting Notes
Anti-theft Equipment	BTSOUTPUT.DESTEALSET	This parameter specifies whether to start an anti-theft device of the base station. Set this parameter to channel 5 output of the EMU.

Scenario 7: Configuring the MEB

If the software version of the base station does not support MEB configuration, the MEB provides physical links only and does not require configuration. The RS485 and dry contact functions are configured on the RRU connecting to the MEB. In this case, the MEB does not support the alarm, log, inventory, and upgrade functions.

If the software version of the base station supports MEB configuration, the MEB must be configured in order to support the alarm, log, inventory, and upgrade functions. The MEB can be configured for only one RAT. When an MEB is configured, the dry contacts configured on the manager RRU do not take effect.

Configure an MEB by referring to [Table 4-74](#).

NOTE

If devices to be monitored connect to an MEB, the manager of the boards is the RRU which also connects to the MEB.

Table 4-74 Key parameters to be set for the MEB

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	BTSDEMUBP.CN	This parameter specifies the number of the cabinet housing the MEB.
Subrack No.	BTSDEMUBP.SRN	This parameter specifies the number of the subrack housing the MEB.
Slot No.	BTSDEMUBP.SN	This parameter specifies the number of the slot housing the MEB.
Manager Cabinet No.	BTSDEMUBP.MCN	The manager is the RRU which connects to the board using a power cable. Set this parameter to the cabinet number of the manager.
Manager Subrack No.	BTSDEMUBP.MSRN	The manager is the RRU which connects to the board using a power cable. Set this parameter to the subrack number of the manager.

4.5.3.1.2 Reference for Configuring the Monitoring Function

For details, see [4.4.3.1.2 Reference for Configuring the Monitoring Function](#).

4.5.3.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

NOTE

Before configuring the PMU, FMU, TCU, or CCU, run the **ADD CABINET** command to configure the cabinet where the PMU, FMU, TCU, or CCU is located if the cabinet is not configured.

Scenario 1: Configuring the PMU

For details about how to configure the PMU, see *Power Supply Management*.

Scenario 2: Configuring the FMU

- Step 1** Run the **ADD BTSBRD** command to add an FMU.
- Step 2** Run the **SET BTSFMUABP** command to set FMU parameters.
- End

Scenario 3: Configuring the TCU

- Step 1** Run the **ADD BTSBRD** command to add a TCU.
- Step 2** Run the **SET BTSDHEUBP** command to set TCU parameters.
- End

Scenario 4: Configuring the CCU

- Step 1** Run the **ADD BTSBRD** command to add a CCU.
- Step 2** Run the **SET BTSCCUBP** command to set CCU parameters.
- End

Scenario 5: Configuring the GATM

- Step 1** Run the **ADD BTSBRD** command to add a GATM.
- Step 2** Run the **SET BTSDATUBP** command to set GATM parameters.
- End

Scenario 6: Configuring Customized Alarms

- Step 1** (Optional) Configure devices and parameters related to the EMU.
1. Run the **ADD BTSBRD** command to add an EMU.
 2. Run the **SET BTSDMUBP** command to set EMU parameters.

3. If the EMU is connected to a humidity or temperature sensor, run the **SET BTSEXD** command to set EMU extension parameters.

Step 2 Run the following commands to query the status of the input and output alarm ports:

- Run the **LST BTSENVALMPORT** command to query the status of the input alarm port.
- Run the **LST BTSOUTPUT** command to query the status of the output alarm port.

Step 3 Based on the query results, run the following commands to enable the input alarm port and disable the unused alarm port:

- Run the **SET BTSENVALMPORT** command to set the input alarm port.
- Run the **SET BTSOUTPUT** command to set the output alarm port.

Step 4 Add and bind the customized alarms on the base station to the MAE so that the alarms can be identified and reported.

Log in to the MAE. Choose **Monitor > Alarms > NE Alarm Settings** from the main menu. In the navigation tree on the left, choose **User-Defined Alarm** to add configurations, bind the alarms, and modify the alarm severity. For details, see "Operation and Maintenance > Fault Management > Alarms > Configuring Alarms or Events > Precise Monitoring > Defining an NE Alarm" in the MAE product documentation.

 NOTE

If the output alarm port is configured, you do not need to add and bind the user-defined alarms on the MAE.

----End

4.5.3.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.5.4 Activation Verification

Step 1 Run the **DSP BTSBRD** command to check whether boards are working properly. If the system returns any command output for a board, the board is working properly. When all the boards are working properly, proceed to [Step 2](#).

Step 2 Perform the operations described in [Table 4-66](#) and check whether environment alarms are correctly reported. If alarms cannot be reported, check hardware connections and software configurations.

Step 3 If a customized Boolean alarm device is connected and the **AVOL** parameter is configured, check whether an alarm is reported to verify that the alarm configurations are correct.

----End

4.5.5 Network Monitoring

None

4.6 Operation and Maintenance (Multimode Base Station)

The multimode base station has multiple RAT-specific monitoring devices. The component management depends on the configuration mode of the base station. The following multimode base stations support both single-side configuration and dual-side configuration: BTS3900, BTS3900L, BTS3900A, and DBS3900 without the TP48600A. Single-side configuration is recommended for these multimode base stations.

NOTE

Single-side configuration has been supported since SRAN6.0. Versions earlier than SRAN6.0 support only dual-side configuration. RRU remote monitoring supports single-side configuration only.

Only single-side configuration can be applied to the following multimode base stations: DBS3900 with the TP48600A and BTS3900AL. Monitoring a power device by multiple RRUs is not supported for these base stations. If these base stations are configured in dual-side mode, an ALM-26271 Inter-System Monitoring Device Parameter Settings Conflict is reported even when the parameter settings for RATs on the two sides are consistent.

The following sections provide the precautions for configuring multimode base stations in single-side and dual-side modes.

4.6.1 Precautions for Single-Side Configuration

When the single-side configuration is performed in a multimode base station, select the RAT according to the following precautions, and then configure data according to [4.4 Operation and Maintenance \(eGBTS/NodeB/eNodeB/gNodeB\)](#) or [4.5 Operation and Maintenance \(GBTS\)](#).

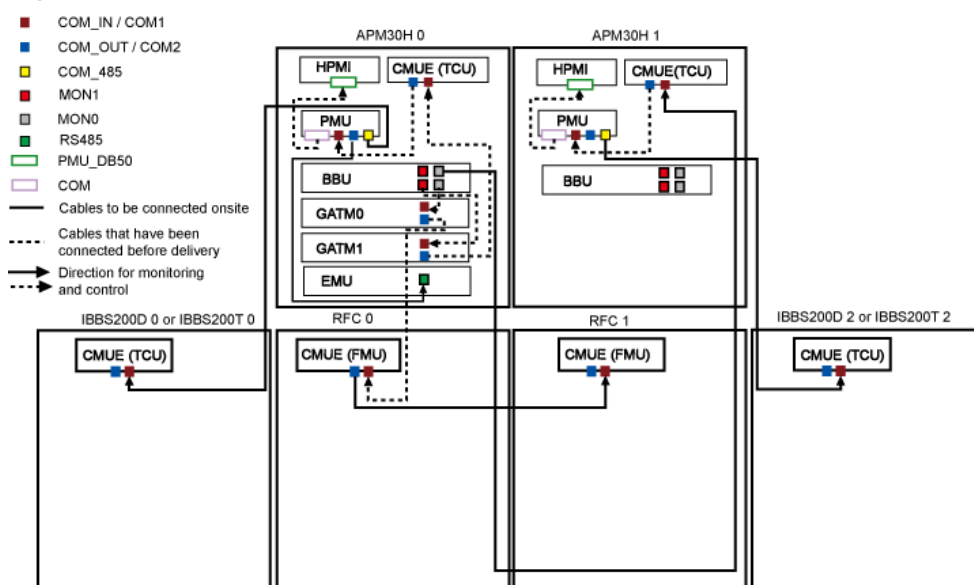
- The monitoring devices of the same base station must be configured on the same RAT.
- The configuration must be consistent with actual cable connections. If a monitoring device connects to a BBU, the monitoring device must be configured on one RAT of the BBU.
- For a separate-MPT dual-mode base station, the monitoring devices are configured in the following priority order: GBTS/eGBTS > NodeB > eNodeB (FDD) > eNodeB (TDD) > gNodeB.

For example, assume that a UO+LO BTS3900A has the following configurations: 2 APM30Hs+2 RFCs+2 IBBS200Ds/IBBS200Ts. In this example, the monitoring devices of the BTS3900A should be configured on the NodeB.

- For a separate-MPT triple-mode base station, the monitoring devices should be configured according to the monitoring signal cable connections in the cabinet. If the monitoring signal cables are connected to the dual-mode BBU, configure the monitoring devices in the following priority order: GBTS/eGBTS > NodeB > eNodeB (FDD) > eNodeB (TDD) > gNodeB.

For example, as shown in the following figure, a separate-MPT GU+LO BTS3900A has the following configurations: 2 APM30Hs+2 RFCs+2 IBBS200Ds/IBBS200Ts. APM30H 0 and RFC 0 house GU devices (BBUs and RFUs), and APM30H 1 and RFC 1 house LTE devices. The monitoring devices and their cascaded monitoring devices of APM30H 1 are connected to APM30H 0. The BBU in APM30H 0 performs monitoring management. In this example, the monitoring devices should be configured on the GSM base station.

Figure 4-79 2 APM30Hs+2 RFCs+2 IBBS200Ds/IBBS200Ts



NOTICE

The HPMI board is configured in the APM30H (Ver.C), but is not configured in the APM30H (Ver.D). Therefore, the monitoring signal cable connections between the PMU and the HPMI do not need to be considered for the APM30H (Ver.D).

- For a co-MPT multimode base station, the monitoring devices are configured in the following priority order: GBTS/eGBTS > NodeB > eNodeB (FDD) > eNodeB (TDD) > gNodeB.

For example, a co-MPT GUL BTS3900A (BBUs and RFUs) has the following configurations: 1 APM30H+1 RFC+1 IBBS200D/IBBS200T. In this example, the monitoring devices of the BTS3900A are configured on the GSM base station.

- In the evolution scenario, it is recommended that new monitoring devices be configured on an existing RAT before the evolution to prevent reconstruction on legacy devices. If the RAT existing before the evolution cannot be selected, configure the monitoring devices in the following priority order: GBTS/eGBTS > NodeB > eNodeB (FDD) > eNodeB (TDD) > gNodeB. Then, reconstruct legacy devices.

For example, if a GBTS (BTS3900) has evolved into a GU dual-mode base station, configure the monitoring devices of the evolved base station on the GSM mode.

- In the RRU remote monitoring scenario, configure the monitoring devices of the single-mode RRU on the RAT of the RRU. It is recommended that you configure the monitoring devices of the GU, GL, or UL dual-mode RRU in the following priority order: GBTS/eGBTS > NodeB > eNodeB (FDD) > eNodeB (TDD) > gNodeB.
- In the RRU remote monitoring scenario, a BDU or SDU supporting the virtual PMU in the voltage boosting power distribution subrack EPU02B/EPU02S-02/EPU02S can only be configured as the PMU for one RRU.

4.6.2 Precautions for Dual-Side Configuration

Dual-side configuration applies to separate-MPT multimode base stations. When dual-side configuration is performed in a multimode base station, the configuration sequence is not critical. You only need to configure the data on both sides according to the operations in [4.4 Operation and Maintenance \(eGBTS/NodeB/eNodeB/gNodeB\)](#) or [4.5 Operation and Maintenance \(GBTS\)](#). In addition, the following precautions must be observed:

- If all monitoring devices are monitored by multiple RATs of the connected BBUs, ensure that the number of physical devices and the settings of the parameters listed in the following table are consistent on the two sides. The following tables show the parameter settings. The monitoring device SMU02C only supports single-side configuration.
- RRU remote monitoring only supports single-side configuration of monitoring devices.
 - After a monitoring device is configured on NE A, if NE B is also configured with the monitoring device under a common monitoring manager (RRU), the monitoring device on NE A cannot be managed properly. The configuration of the monitoring device from NEs A and B, and then configure it again on NE A to restore the management function.
 - After a monitoring device is configured on NE A, if NE B runs the **SCN RS485** command to scan the RS485 bus for the common monitoring manager (RRU), NE A reports ALM-25600 Monitoring Device Maintenance Link Failure.

Table 4-75 Common GSM parameters of the multimode base stations (1)

Object	Parameter Name	Parameter ID	MML Command	Setting Notes
FMU	Special Boolean Alarm Flag	SBAF	SET BTSFMUABP	Set this parameter to the default value. Essentially, select the WS_DISABLE(Water-Immersed Sensor Disabled) and SS_DISABLE(Smog Sensor Disabled) options.
	Smart Temperature Control	STC	SET BTSFMUABP	ENABLE(Enable)
TCU	Temperature Alarm Lower Threshold	TLTHD	SET BTSDHEUBP	Set this parameter to the default value.
	Temperature Alarm Upper Threshold	TUTHD	SET BTSDHEUBP	- Set this parameter to 50 for the TCU in the BBC. - Set this parameter to 68 for the TCU in other cabinets.
	Special Boolean Alarm Flag	SBAF	SET BTSDHEUBP	Set this parameter to the default value. Essentially, select the WS_DISABLE(Water-Immersed Sensor Disabled) and SS_DISABLE(Smog Sensor Disabled) options.

Object	Parameter Name	Parameter ID	MML Command	Setting Notes
CCU	Special Boolean Alarm Flag	SBAF	SET BTSCCUBP	Set this parameter to the default value. Essentially, select the WS_DISABLE(Water-Immersed Sensor Disabled) and SS_DISABLE(Smog Sensor Disabled) options.

Table 4-76 Common GSM parameters of the multimode base stations (2)

Object	Parameter Name	Parameter ID	MML Command	Setting Notes
EMU	Temperature Alarm Lower Threshold	TLTHD	SET BTSEXD	Set this parameter to the default value.
	Temperature Alarm Upper Threshold	TUTHD	SET BTSEXD	Set this parameter to the default value.
	Humidity Alarm Lower Threshold	HLTHD	SET BTSEXD	Set this parameter to the default value.
	Humidity Alarm Upper Threshold	HUTHD	SET BTSEXD	Set this parameter to the default value.

Object	Parameter Name	Parameter ID	MML Command	Setting Notes
	Special Analog Alarm Flag	SAAF	SET BTSDEMUBP	- The -48 V alarm is used to check whether the input voltage for the cabinet is normal. When the input voltage for the cabinet is -48 V, deselect the 48V_DISABLE(-48 Voltage Disabled) option. - Set this parameter based on the types of sensors connected to the EMU/EMUB.
	Special Boolean Alarm Flag	SBAF	SET BTSDEMUBP	Set this parameter based on the types of sensors connected to the EMU/EMUB.

Table 4-77 Common GSM parameters of the multimode base stations (3)

Object	Parameter Name	Parameter ID	MML Command	Setting Notes
ALMPORT	Port No.	PN	SET BTSENVALMPOR T	Set this parameter based on cable connections.
	Switch	SW	SET BTSENVALMPOR T	Turn on this switch only for the port configured with a customized alarm.

Object	Parameter Name	Parameter ID	MML Command	Setting Notes
	Alarm ID	AID	SET BTSENVAMPOR T	Set this parameter to a value ranging from 65033 to 65233 .
	Port Type	PT	SET BTSENVAMPOR T	Set this parameter to BOOL(Digital Port) or VALUE(Analog Port) .
	Alarm VOL.	AVOL	SET BTSENVAMPOR T	Set this parameter to HIGH or LOW based on cable connections.
	Upper Limit	UL	SET BTSENVAMPOR T	Set this parameter based on the sensor types.
	Lower Limit	LL	SET BTSENVAMPOR T	Set this parameter based on the sensor types.
	Sensor Type	ST	SET BTSENVAMPOR T	Set this parameter based on the sensor types.
	Sensor Measurement Upper Limit	SMUL	SET BTSENVAMPOR T	Set this parameter based on the sensor types.
	Sensor Measurement Lower Limit	SMLL	SET BTSENVAMPOR T	Set this parameter based on the sensor types.
	Sensor Output Signal Upper Limit	SOUL	SET BTSENVAMPOR T	Set this parameter based on the sensor types.

Object	Parameter Name	Parameter ID	MML Command	Setting Notes
	Sensor Output Signal Lower Limit	SOLL	SET BTSENVAMPOR T	Set this parameter based on the sensor types.

Table 4-78 Common UMTS, LTE, and NR parameters of the multimode base stations (4)

Object	Parameter Name	Parameter ID	MML Command	Setting Notes
FMU	Special Boolean Alarm Flag	SBAF	ADD FMU	Set this parameter to the default value. Essentially, select the WS_DISABLE(Water-Immersed Sensor Disabled) and SS_DISABLE(Smog Sensor Disabled) options.
	Smart Temperature Control	STC	ADD FMU	ENABLE(Enable)
TCU	Temperature Alarm Lower Threshold	TLTHD	ADD TCU	Set this parameter to the default value.
	Temperature Alarm Upper Threshold	TUTHD	ADD TCU	- Set this parameter to 50 for the TCU in the BBC. - Set this parameter to 68 for the TCU in other cabinets.

Object	Parameter Name	Parameter ID	MML Command	Setting Notes
	Special Boolean Alarm Flag	SBAF	ADD TCU	Set this parameter to the default value. Essentially, select the WS_DISABLE(Water-Immersed Sensor Disabled) and SS_DISABLE(Smog Sensor Disabled) options.
CCU	Special Boolean Alarm Flag	SBAF	ADD CCU	Set this parameter to the default value. Essentially, select the WS_DISABLE(Water-Immersed Sensor Disabled) and SS_DISABLE(Smog Sensor Disabled) options.

Table 4-79 Common UMTS, LTE, and NR parameters of the multimode base stations (5)

Object	Parameter Name	Parameter ID	MML Command	Setting Notes
EMU	Temperature Alarm Lower Threshold	TLTHD	ADD EMU	Set this parameter to the default value.
	Temperature Alarm Upper Threshold	TUTHD	ADD EMU	Set this parameter to the default value.
	Humidity Alarm Lower Threshold	HLTHD	ADD EMU	Set this parameter to the default value.
	Humidity Alarm Upper Threshold	HUTHD	ADD EMU	Set this parameter to the default value.

Object	Parameter Name	Parameter ID	MML Command	Setting Notes
	Special Analog Alarm Flag	SAAF	ADD EMU	- The -48 V alarm is used to check whether the input voltage for the cabinet is normal. When the input voltage for the cabinet is -48 V, deselect the 48V_DISABLE(-48 Voltage Disabled) option. - Set this parameter based on the types of sensors connected to the EMU/EMUB.
	Special Boolean Alarm Flag	SBAF	ADD EMU	Set this parameter based on the types of sensors connected to the EMU/EMUB.

Table 4-80 Common UMTS, LTE, and NR parameters of the multimode base stations (6)

Object	Parameter Name	Parameter ID	MML Command	Setting Notes
ALMPORT	Port No.	PN	SET ALMPORT	Set this parameter based on cable connections.
	Switch	SW	SET ALMPORT	Turn on this switch only for the port configured with a customized alarm.

Object	Parameter Name	Parameter ID	MML Command	Setting Notes
	Alarm ID	AID	SET ALMPORT	Set this parameter to a value ranging from 65033 to 65233 .
	Port Type	PT	SET ALMPORT	Set this parameter to BOOL(Digital Port) or VALUE(Analog Port) .
	Alarm VOL.	AVOL	SET ALMPORT	Set this parameter to HIGH(High Voltage) or LOW(Low Voltage) based on cable connections.
	Upper Limit	UL	SET ALMPORT	Set this parameter based on the sensor types.
	Lower Limit	LL	SET ALMPORT	Set this parameter based on the sensor types.
	Sensor Type	ST	SET ALMPORT	Set this parameter based on the sensor types.
	Sensor Measurement Upper Limit	SMUL	SET ALMPORT	Set this parameter based on the sensor types.
	Sensor Measurement Lower Limit	SMLL	SET ALMPORT	Set this parameter based on the sensor types.
	Sensor Output Signal Upper Limit	SOUL	SET ALMPORT	Set this parameter based on the sensor types.

Object	Parameter Name	Parameter ID	MML Command	Setting Notes
	Sensor Output Signal Lower Limit	SOLL	SET ALMPORT	Set this parameter based on the sensor types.

- The alarm parameter settings related to the ambient environment (such as settings of parameters related to the water sensor and smoke sensor) must be consistent between RATs.

If the base station detects any inconsistency, ALM-26271 Inter-System Monitoring Device Parameter Settings Conflict is reported in both RATs. To avoid inconsistency, you can perform consistency check for the multimode base station on the MAE-Deployment after configuring parameters. To perform a consistency check on the MAE-Deployment, choose **SRAN Application > Data Management > MBTS Consistency Check**. For detailed operations, press  on the MAE-Deployment to see section "Checking MBTS Data Consistency" in the online help.

5 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

- Step 2** On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.
- End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

This document uses the following references:

1. Power Supply Management Feature Parameter Description
2. 3900 & 5900 Series Base Station Component Description
3. 3900 & 5900 Series Base Station Cabinet Description
4. 3900 & 5900 Series Base Station Cables
5. TP48600A-H17B1 User Manual
6. EMU User Guide
7. EMUB User Guide
8. RRU3959 & RRU3959w Installation Guide
9. MEB06D User Guide
10. ICC100-N5 Solar Controller User Manual